

Following Evidence About the Same Part: Supplementary Protocols and Complete Question Index

Anonymous submission

1 Scope and Reading Guide

Analysis: `ldraw2-evidence-v1`. Source release: `brickatlas-ldraw-2`. All 617 original questions, 24 sources and 15,334 placed instances are retained. The main paper reports a method/resource contribution and measured algorithmic controls. No validated learned-model or human study results are available. “–” denotes not run and is distinct from measured zero.

Sections 2–9 document measurement, comparisons, independent checks, related protocols and reproducibility. The subsequent source dossiers reproduce every original English question, option and reference output, with source attribution, operand mapping, derivation and review status. They are evaluator material. They must not be provided to an inference model. The compact public-evidence summaries in a dossier do not replace exact model request bodies.

2 Reporting Roles and Full Denominators

Table 1: The complete disjoint family partition. A source missing a family is absent from its macro denominator, not assigned zero.

Family	<i>N</i>	Sources	Report role
Color	73	24	core visual
Part type	67	24	core visual
Distance	73	24	nonconstant controls
Interface	87	24	nonconstant controls
Neighbors	73	24	structural evidence
Graph deletion	73	24	structural evidence
Coverage	17	17	nonconstant controls
Evidence limit	24	24	constant controls
Step lookup	55	17	nonconstant controls
Step sequence	17	17	constant controls
Restoration	58	24	nonconstant controls

The union is 140 visual, 146 structural, 290 nonconstant controls and 41 constant controls. The first two roles contain candidate core measurements; the grouping does not certify visual answerability or nontrivial graph difficulty. Source-size bands D1–D4 and atomic/metacognitive/procedural labels are descriptive organization, not empirically validated difficulty or diagnostic localization.

Band	Sources	V	S	G	E	Total
D1	5	17	21	33	12	83
D2	9	34	47	69	25	175
D3	2	14	15	23	7	59
D4	8	75	86	108	31	300

2.1 State-driven Family and Control Tables

All tables below are generated from the selected dataset manifest and versioned result index. Completed entries must trace to raw responses and recomputed analyses; unrun entries have null metrics. There is no fixed assertion that future results must remain null. The current lack of results is a data state, not a publishing rule.

Table 2: Core visual families; condition-specific answerability pending.

Family / model	N	Sources	Source macro (%)
Color	73	24	–
Part-type	67	24	–

Table 3: Structural evidence families; deletion results require the node-count baseline and paired comparisons.

Family / model	N	Sources	Source macro (%)
Neighbors	73	24	–
Graph deletion	73	24	–

Table 4: Nonconstant controls, independently reported.

Family / model	N	Sources	Source macro (%)
Distance	73	24	–
Interface	87	24	–
Coverage	17	17	–
Step lookup	55	17	–
Restoration	58	24	–

Table 5: Constant controls. They measure scope or interface compliance and cannot decide a primary capability ranking.

Family / model	N	Sources	Source macro (%)
Evidence limit	24	24	–
Step sequence	17	17	–

2.2 Descriptive All-task Results and Planned Conditions

Table 6: All-task descriptive scores, not an overall assembly capability ranking. Configurations are planned, not confirmed service availability. Exact checkpoint/API revisions must be frozen before live inference.

Model	N	Micro (%)	Source macro (%)
claude-sonnet-4-20250514	617	–	–
gemini-2.5-pro	617	–	–
gpt-4.1-2025-04-14	617	–	–
meta-llama/Llama-4-Scout-17B-16E-Instruct	617	–	–
Qwen/Qwen3-VL-8B-Instruct	617	–	–
HuggingFaceTB/SmolVLM-256M-Instruct	617	–	–

Config	Exact model ID	Revision
Claude	claude-sonnet-4-20250514	2025-05-14
Gemini	gemini-2.5-pro	Freeze before live use
GPT-4.1	gpt-4.1-2025-04-14	2025-04-14
Llama 4 Scout	meta-llama/Llama-4-Scout-17B-16E-Instruct	Freeze before live use
Qwen3-VL	Qwen/Qwen3-VL-8B-Instruct	Freeze before live use
SmolVLM	HuggingFaceTB/SmolVLM-256M-Instruct	Freeze before live use

The runner retains actual request bytes, supplied parameters, image dimensions, tool permissions, timeouts and raw responses. Unsupported provider settings are errors or recorded differences, never silently treated as identical experimental conditions. Historical results from earlier source releases are not transferred into these rows.

Condition	N	Exact success
standard	617	
text-without-image	140	
graph-edges-withheld	146	
full-scene	140	
background-mask	140	
operand-only	140	
choice-order	469	
edge-order	146	
id-permutation	576	
wrong-image	140	
camera-perturbation	140	
color-nuisance	67	
multi-view	140	
graph-intervention	146	

Empty success cells in this condition inventory are unavailable measurements. Their denominators are selected by the unchanged plan. The learned-model result index has 84 planned model/condition records and no completed real record in the inspected workspace.

3 Paired Evidence and Statistical Protocol

Table 7: Planned visual pair inventory. These aggregate counts describe coverage; Color and Part-type must be analyzed separately. The generated columns summarize paired correctness and B accuracy. Access contrasts also require paired accuracy differences and QA status.

Comparison	N	A, B correct (%)	B correct (%)
visual-mismatch	140	–	–
background-only	140	–	–
label-binding	140	–	–
camera	140	–	–
color-nuisance	67	–	–
view-access	140	–	–

Table 8: Planned graph pair inventory. Neighbors and deletion are separate analyses even when a comparison covers both.

Comparison	N	A, B correct (%)	B correct (%)
edge-causal-control	146	–	–
edge-equivalence	146	–	–

Binding. Each pair retains pair, parent and source IDs; source and dataset hashes; model revision, adapter and prompt hashes; condition A/B; both exact requests; canonical predictions; both golds and provenance; truth policy; and explicit failure reasons. A result pair cannot be formed merely because two files share a task ID. A missing response becomes a failed observation; it never shrinks the denominator.

Fact change. Wrong-image binds gold B to the stored `mismatch.alternateChoiceId` and verifies its relation to the B request. Graph-intervention recomputes gold from each supplied graph. The original source-gold wrong-image score is preserved in a separate field. The primary new-gold rate and both-correct rate use every declared pair. Conditional adaptation has an explicitly displayed A-correct denominator and is null if that denominator is zero. The B response partition is new gold / old gold / another legal answer / invalid or missing, with counts summing to the fixed N.

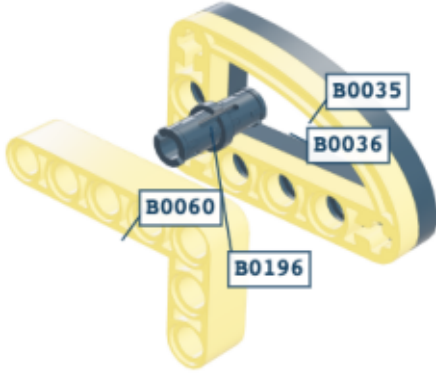
Invariance. ID-renamed answers are mapped to canonical identities. Choice letters are only comparable after mapping to their option meanings. Exact sets ignore order while retaining duplicate-invalid status; action success uses legal execution. The five transitions are both correct, A-only correct, B-only correct, both wrong with the same semantic answer, and both wrong otherwise. Missing or invalid answers are explicitly identified and cannot count as valid semantic agreement.

Withdrawal and access. Removing images or edges can make a task underdetermined. Scores retain original gold only as dependency measurements. Full/Mask measures a total access effect, and fixed two-view access is not active exploration. Semantic invariance under camera or color changes does not prove visual decidability.

Weighting and uncertainty. First average variants within each original parent, then parents within source, then sources equally. Micro and parent-macro rates are auxiliary. Report items, parents and sources for every family and QA stratum. Resample source clusters with replacement 10,000 times (seed 20260919), using percentile 95% intervals. Calculate model-minus-baseline differences on the same records before resampling. No equivalence claim follows from an interval including zero. Subgroups chosen after inspecting model performance must be marked exploratory.

4 Real-source Worked Pair

(a) A: operand-only; gold C



(b) B: wrong-image; gold A

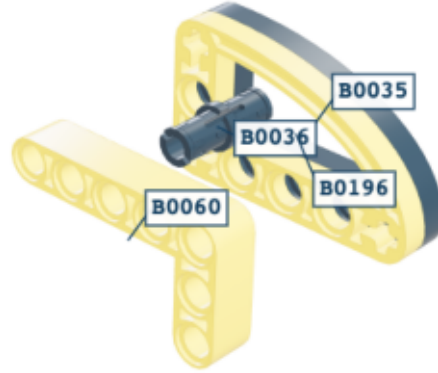


Figure 1: Actual operand-only and wrong-image render details for `ld2-omr-42004-shape-match-1`. B0035 remains the target. The candidate-label swap changes the image-conditioned answer from C to A. The paper enlarges the pixels; these details are not additional inputs or human answerability approvals.

The Mini Backhoe Loader has 246 placed parts. B0035 is `brick_000035`, black part 32250, at source line 206 in `42004 - front right side.ldr`. B0036 is yellow part 32250 at line 207. B0196 is a black 2780 pin from another submodel. The options are A:B0196, B:B0116, C:B0036, D:B0060. The wrong-image render exchanges labels B0036 and B0196 while the question and options remain fixed. Thus the original gold C and new-image gold A describe different observation-conditioned bindings, not contradictory claims about source CAD truth.

For this pair, $C \rightarrow A$ yields both-correct, $C \rightarrow C$ is old-answer persistence, $C \rightarrow B$ is another legal error, and a missing B is a failure. These are hypothetical response examples. No model predictions have been observed. The same identity links recognized axle edges to B0034/B0037/B0038/B0051 and the independent `restore-B0035` display action. Those are separate contracts.

Reproduction hashes. Source SHA-256:

`a32b924f302a1c028c1131f78e438ad2d78e70a041999690a1e4285e851f7dbf`

Operand image:

`cc2826da13f55fd9234010d860f3894f9f7fffb692502d9018f3368fcb1551852`

Wrong image:

`cd27939fd7988f8ab534a5115347cca2b6b0e0afa06195141f49c613605f59c1`

Materialized offline B request:

`16bbda47009ab11e28c7fe37d18bcf464a6f606ab6b9fe6c3d765f45318c30fc`

5 Graph Audit and Matched Challenge

Table 9: Algorithmic baselines on all natural deletion items and the original/intervened pairs, with identical denominators.

Algorithmic baseline	Micro (%)	Source macro (%)	Both (%)
Node count only	86.30	83.75	0.00
Always four	60.27	61.11	0.00
BFS / union-find	100.00	100.00	100.00

Node-only source macro is 83.75%, with 95% source-cluster interval [72.92,93.12]. Always-four source macro is 61.11%, interval [46.32,75.07]. BFS/union-find source macro is 100%, interval [100,100]. Both-correct intervals are [0,0] for the two priors and [100,100] for the graph algorithm. Degenerate bootstrap intervals reflect identical observed cluster outcomes, not certainty about arbitrary unseen graph distributions.

Table 10: Per-source census of node-only results. No source or question is removed because of its score.

Source	N	Node-only correct
10001	4	4
10014	2	2
10036	2	1
10128	2	1
10156	2	2
10159	3	3
10213	5	5
10220	5	5
31009	2	2
31027	2	2
31028	2	0
5867	2	2
omr-10242	4	3
omr-10265	5	5
omr-10269	4	3
omr-21309	5	4
omr-31031	2	1
omr-31088	2	1
omr-42004	2	2
omr-42020	2	2
omr-42054	5	4
omr-42061	2	2
omr-42066	5	5
omr-42102	2	2

Table 11: Every natural node-only failure. These ten items across nine sources are the complete positive-deficit stratum.

Task	Nodes after	Edges after	Gold
ld2-31028-graph-removal-1	5	1	4
ld2-31028-graph-removal-2	5	2	3
ld2-10036-graph-removal-2	5	1	4
ld2-omr-31031-graph-removal-2	4	1	3
ld2-omr-31088-graph-removal-1	5	1	4
ld2-10128-graph-removal-2	4	2	2
ld2-omr-10269-graph-removal-2	6	1	5
ld2-omr-10242-graph-removal-1	5	1	4
ld2-omr-21309-graph-removal-1	4	1	3
ld2-omr-42054-graph-removal-5	5	2	3

Challenge rules. The extension version is `ldraw2-matched-graphs-v1`, seed 20260919. Each parent contributes three hypothetical observations; the maximum per source is 15. A target-to-survivors star is combined with either a six-cycle/two-triangle pair (at least six survivors) or a four-node path/three-node triangle pair (four or five survivors). Other survivors are isolated after target deletion. The third observation permutes survivor labels in the anchor graph. Seeded parent ordering alternates the direction of the answer change.

The 219 observations produce 146 pairs: 37 increases, 36 decreases and 73 unchanged answers. All match node set, unique edge count and target degree. 12 changed pairs also match survivor degree sequences; 61 do not, and the manifest records that gap. Edge counts need not match the original natural graph. Natural, single-edge intervention and hypothetical challenge results remain separate. Opaque request IDs and independent requests do not expose pair membership or transformation direction to the model.

Independent checks. BFS and union-find agree on every original, original-intervention and challenge graph. Python and the frozen TypeScript intervention yield identical original-intervention edge lists. A synthetic six-cycle versus two-triangle fixture verifies that node/edge counts and every survivor degree can agree while the component count differs. Duplicate, reversed and isolated-edge cases are included. Synthetic fixtures are software tests, never counted as real sources.

6 Related-work Evidence at Protocol Level

All comparisons use pinned full papers, read including appendices: BrickNet, CVPR 2026 proceedings, ten PDF pages; LEGO-Puzzles arXiv:2503.19990v3, June 20, 2025, eleven pages; PhyBlock arXiv:2506.08708v2, November 21, 2025, forty pages. Page numbers below refer to PDF pages. Official code snapshots are separately identified and are not substituted for the papers’ experimental recipes.

6.1 BrickNet

BrickNet [Kulits and Schmid(2026)] evaluates unconditional and caption-conditioned object/build-sequence generation and held-out sequence perplexity (pp. 5–7, Sec. 5). Conditional generation receives a caption; unconditional generation receives a start token. Graph-backed part/connector programs reconstruct poses (pp. 3–4, Sec. 3.4). Eight views are used for captioning (p. 4), and render–text agreement evaluates generated objects (p. 7).

Its placed-part graph vertices and letter-named spanning-tree programs distinguish repeated instances (pp. 3–4, Fig. 4). Connector annotations combine procedural and manual authoring and are reviewed; collision geometry uses inset meshes with caveats about non-watertight surfaces and deformation (pp. 3–4). The paper uses human-authored online LDraw sources, reporting 320,808 samples and 9,743 variants, with 512 held-out single-object evaluation files (pp. 2,4–5).

Graph/pose representation, sampling, scale and training ablations are reported (pp. 5–7). Same-target answer-changing and answer-preserving VQA with the two semantic adaptation measures is not reported. This does not imply the assets could not support that experiment. The code snapshot at commit `b09a689c4929bd27f59c9bb9446237967aa332c9` (August 19, 2026) preserves instance/connector identity through `core.Graph`, `core.Edge` and `tree.id2i`. `score.score_text` returns action count, first invalid action and collision indices; `score.check_tree` decodes placements. The paper regenerates unparseable caption-conditioned outputs before perceptual scoring (p. 7), unlike our fixed-denominator failure rule. Later `DATA.md` training splits and packing lengths differ from the paper; those recipes are not merged in our comparison.

6.2 LEGO-Puzzles

LEGO-Puzzles [Tang et al.(2025)Tang, Gao, Zeng, Duan, Sun, Xing, Liu, Lyu, and Chen] v3 has 1,100 questions from 407 building instructions across eleven families (pp. 3–4). Task-dependent inputs include current/target states, next pieces and candidate states (Fig. 1, p. 2; Sec. 3.2, p. 4). Height/Adjacency use arrows, while states and choices use letters. Unified source naming is described (p. 4), but a stable scene-instance identifier shared across image, record, graph and action contracts is not reported in this version. Several tasks integrate multiple image sources; describing them all as independent single-image lookups would be incorrect.

Studio fixes cameras across steps, and type, number, color, position and viewpoint can be edited during construction (p. 4). Duplicate/image checks and three-annotator verification resolve or remove ambiguous items (p. 4). VQA scoring combines rule-based extraction with a ChatGPT fallback; five experts rate image generation (pp. 5,7). Next-k-Step uses $k = 1, \dots, 5$ and compares CoT conditions (pp. 7–8); the version has up to seven intermediate stages (pp. 3–4). We do not import counts from a later OpenReview revision.

Official code at commit `1413d2c236879573cc0afff646bae5e3d04023bd` (June 20, 2025) uses `LEGO.split_LEGO` to interleave image tokens and `build_prompt_sort` for ordered letters. These builders supply images and text, not curation CAD records. `extract_answer_from_item` calls `can_infer_lego`; an enabled judge can retry three times and fall back to a random option after exhausted matching. This is a code-snapshot observation, not a rerun score. The same-target new-gold and both-correct estimand is not reported in v3, although the assets are reusable.

6.3 PhyBlock

PhyBlock [Ma et al.(2025)Ma, Wen, Lin, Xu, Liang, Lin, Ma, Wang, Wei, Lin, Han, Cao, Chen, Laptev, and Liang] v2 separates 400 assembly tasks from 2,200 physical VQA questions across sixteen categories (pp. 5–6). Planning uses target and candidate-block images; interactive planning adds observations and history (pp. 7–8,25–27). The output selects reusable candidate IDs in order without rotation; exact pose prediction is deferred (pp. 39–40). The evaluator uses AOV dependencies and target poses.

Scene records have unique `order`, `type`, `color`, `dependencies`, `position` and `orientation` (pp. 17–18). Candidate IDs are different: the prompt explicitly permits the same block index multiple times (pp. 25–26). Planning jointly involves candidate matching and legal ordering (pp. 22–23), while VQA separately probes attributes, relations, scenes and dynamics (pp. 5–6). Eight geometries and five colors are used with Genesis, manual scene annotation and augmentation (pp. 6,15–17).

AOV evaluation matches unused legal reference blocks and counts TP/FP/FN (Algorithm 1, pp. 22–23). Static VQA uses human-verified LLM generation; dynamic VQA uses simulated perturbations, frames and human checks (p. 19). The paper

reports topology/pose scoring, full/stepwise planning, views/backgrounds and physical perturbations (pp. 8,19,22–28), with no real-robot evaluation (p. 40). Our specific paired visual estimand is not reported.

The official commit `da81ed9cad8184509aadb6f01af83d53e5c9d6fa` (August 20, 2025) predates v2. Its sample `000.json` confirms `order/depend/euler` scene fields. `are_blocks_equal` compares `type/color/euler` in protocol A and `type/color` in B. `validate_block_placement` assigns the earliest unused legal reference instance. `extract_blocks_sequence` filters invalid candidate indices. `BlockEnv.move_cube` selects a candidate using a target position/euler and `step` renders feedback. This is not continuous pose/control synthesis. The top-level evaluator does not separately emit all four paper error types.

What the additional assets establish. IDs and templates can support attribute lookup on these prior assets. Our additional unit is a bound pair: canonical target, exact visible evidence, both golds, truth policy, raw predictions and conditional answerability decisions. It allows testing correct adaptation instead of mere score change. The natural graph audit demonstrates why this distinction matters for the present distribution; the analogous visual finding remains a research objective. No exclusivity, firstness or cross-benchmark percentage comparison is asserted.

7 Quality, Failure Handling and Software Validation

The required queue contains 140 visual items and 111 other flagged items. The physical queue contains 573 intersection candidates without recognized mating interfaces. Intended mating, approximate mesh overlap, source defects and unresolved cases require separate decisions. All are pending human adjudication; an automated image or graph check does not resolve them.

Exact-wire visual review records task, condition, request hash, image hashes/dimensions, reviewer, date, native-size/blinding attestations and reason. A browser review export shows actual wire pixels without references. Pending/failed QA is reported and cannot silently delete an unfavorable model result. Changing the input or gold requires a new version and affected runs; old scores cannot be ported without verification.

The adaptation test suite covers perfect adaptation with zero accuracy difference, persistent old answers, other legal answers, invalid outputs, a zero conditional denominator, mismatched models/data, duplicate pairs and missing B observations. Raw-receipt fixtures exercise all eleven families, including executable actions. Tampered request bindings, numeric metrics, counts, states and hashes are rejected. Synthetic receipts are explicitly labeled and rejected by production ingestion. These passing tests demonstrate analysis behavior, not model performance or human agreement.

Missing, timed-out, refused, invalid-JSON and invalid-format responses retain distinct reasons and count as failures in the primary denominator. Format validity and semantic accuracy conditional on valid format are auxiliary. No hidden retries or answer repair inflate the main rate. Unknown cost remains null.

8 Reproduction, Preservation and Figure Provenance

The pre-revision archive preserves 2,616 actual file entries, including previously untracked content. Its baseline lock checks all four guide fingerprints and the earlier 2,131-file v1 freeze. The new namespace contains analysis code, result indices, paired gold bindings and the hypothetical extension; no original source, v1/v2 request, response, gold, score or manuscript is rewritten.

The publication generator reads hash-checked frozen inputs, produces numeric macros and imports the state-driven tables. It records every source render and paper crop. The official CVPR 2026 style is copied unchanged and used in anonymous review mode with a nonregistered draft ID. Source archives include TeX, bibliography, figure PDFs, complete dossiers and hashes. Local build/verification instructions are supplied with the release; machine-specific paths are not part of this PDF.

All figure images are derivatives of real source geometry renders. Paper-only crop rectangles and zooms never change model inputs. The graph drawing is explicitly schematic; state arrows show visibility effects, not physical motion. Six-step images use one shared rectangle from unchanged saved frames. Full/Mask share the exact camera, projection and crop window; Crop reframes operands. No additional artwork substitutes for a source model.

9 Detailed Family Contracts

9.1 Color

73 items; visual; atomic; single-choice.

Public input. `{}`; `visualInput.numberedView` supplies a frozen operand-isolation PNG; `isolationAllowed=true`

Operation and scoring. Bind the target B-number in the image; identify its source color name. Exact anonymous `choiceId` of the semantic answer.

Interpretation. Source color equality is an oracle, not proof of perceptual distinguishability.

Control. No known deterministic S1–S4 request shortcut; semantic priors reported separately.

9.2 Part type

67 items; visual; atomic; single-choice.

Public input. `{}`; `visualInput.numberedView` supplies a frozen operand-isolation PNG; `isolationAllowed=true`

Operation and scoring. Bind target and candidates in the image; match the source part type while ignoring color and pose. Exact anonymous choiceId of the semantic answer.

Interpretation. Near-identical source types require human adjudication; no unseen geometry is inferred.

Control. No known deterministic S1–S4 request shortcut; semantic priors reported separately.

9.3 Distance

73 items; source-data; atomic; single-choice.

Public input. centers, coordinateFrame

Operation and scoring. Read supplied bounding-box centers, compute Euclidean distance, select the unique minimum. Exact anonymous choiceId of the semantic answer.

Interpretation. Coordinates are explicit, not visually estimated. Candidates are hash sampled, not nearest spatial neighbors.

Control. No known deterministic S1–S4 request shortcut; semantic priors reported separately.

9.4 Interface

87 items; connector-graph; metacognitive; single-choice.

Public input. connectorRecord.{family,aConnector,bConnector}; scope

Operation and scoring. Read connectorRecord.family and select the identical label. Exact anonymous choiceId of the semantic answer.

Interpretation. An explicit record-reading control, not contact perception or load-capacity reasoning.

Control. No known deterministic S1–S4 request shortcut; semantic priors reported separately.

9.5 Neighbors

73 items; connector-graph; metacognitive; multiple-choice.

Public input. edges, graphMeaning

Operation and scoring. Collect candidate labels joined to the target by any supplied undirected edge; deduplicate. Exact set of choiceIds; order ignored; duplicate IDs invalid.

Interpretation. One-hop set extraction within the recognized graph, not physical connectivity discovery.

Control. No known deterministic S1–S4 request shortcut; semantic priors reported separately.

9.6 Graph deletion

73 items; connector-graph; graph-internal; integer.

Public input. edges, graphMeaning, nodes

Operation and scoring. Delete the named vertex and incident edges; count connected components including remaining isolated vertices. A safe JSON integer value exactly equal to the independent component count.

Interpretation. An operation within a supplied local graph; no extraction trajectory or cross-contract chain.

Control. No known deterministic S1–S4 request shortcut; semantic priors reported separately.

9.7 Coverage

17 items; source-data; metacognitive; single-choice.

Public input. coverage[].{label,supported}

Operation and scoring. Select the sole row whose supplied supported flag is false. Exact anonymous choiceId of the semantic answer.

Interpretation. A definition-coverage control; unsupported does not imply floating or disconnected.

Control. No known deterministic S1–S4 request shortcut; semantic priors reported separately.

9.8 Evidence limit

24 items; source-data; metacognitive; single-choice.

Public input. available

Operation and scoring. Select Not established by this evidence for the stated gravitational-stability claim. Exact anonymous choiceId of the semantic answer.

Interpretation. The semantic answer is constant; accuracy alone is not reasoning evidence.

Control. No known deterministic S1–S4 request shortcut; semantic priors reported separately. Constant semantic answer.

9.9 Step lookup

55 items; source-data; procedural; single-choice.

Public input. steps[].{index,numbers}; provenance

Operation and scoring. Find the target label in the supplied author-step table; select the smallest matching expanded index. Exact anonymous choiceId of the semantic answer.

Interpretation. Table lookup; source order is not evidence of collision-free assembly.

Control. No known deterministic S1–S4 request shortcut; semantic priors reported separately.

9.10 Step sequence

17 items; scene-edit; procedural; actions.

Public input. initialFacts; goalFacts; absentFacts; budget; actions[].{id,label,requires,forbids?,adds,deletes,cost}; editMode

Operation and scoring. Follow the public predecessor facts and reveal the supplied author window in order. String actionIds; legal full replay under prerequisites, forbids and budget; all goal facts and no prohibited facts.

Interpretation. A small explicit fact chain, not mechanical planning; the current six-step response template is constant.

Control. No known deterministic S1–S4 request shortcut; semantic priors reported separately. Identical six-step action program in this release.

9.11 Restoration

58 items; scene-edit; procedural; actions.

Public input. initialFacts; goalFacts; absentFacts; budget; actions[].{id,label,requires,forbids?,adds,deletes,cost}; editMode

Operation and scoring. Choose the restore action whose missing fact holds; execute it within budget and satisfy all goals. String actionIds; legal full replay under prerequisites, forbids and budget; all goal facts and no prohibited facts.

Interpretation. One visibility edit at an original source pose, not insertion-feasibility planning.

Control. No known deterministic S1–S4 request shortcut; semantic priors reported separately.

References

- [Kulits and Schmid(2026)] Peter Kulits and Cordelia Schmid. BrickNet: Graph-backed generative brick assembly. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 39252–39261, 2026.
- [Ma et al.(2025)Ma, Wen, Lin, Xu, Liang, Lin, Ma, Wang, Wei, Lin, Han, Cao, Chen, Laptev, and Liang] Liang Ma, Jiajun Wen, Min Lin, Rongtao Xu, Xiwen Liang, Bingqian Lin, Jun Ma, Yongxin Wang, Ziming Wei, Haokun Lin, Mingfei Han, Meng Cao, Bokui Chen, Ivan Laptev, and Xiaodan Liang. PhyBlock: A progressive benchmark for physical understanding and planning via 3d block assembly. *arXiv:2506.08708v2*, 2025. Version dated November 21, 2025; NeurIPS Datasets and Benchmarks.
- [Tang et al.(2025)Tang, Gao, Zeng, Duan, Sun, Xing, Liu, Lyu, and Chen] Kexian Tang, Junyao Gao, Yanhong Zeng, Haodong Duan, Yanan Sun, Zhening Xing, Wenran Liu, Kaifeng Lyu, and Kai Chen. LEGO-Puzzles: How good are MLLMs at multi-step spatial reasoning? *arXiv:2503.19990v3*, 2025. Version dated June 20, 2025.

10 Complete Original Source Dossiers

The following dossiers are copied verbatim from the protected v2 question index. Their source/output semantics and pending review status remain unchanged. Reporting roles and paired analysis are defined in the preceding sections. Dossier source authors are third-party asset attributions, not manuscript author identities.

11 31028: Sea Plane

53 source instances; 15 tasks; D1 size band. 27 type strings; 10 color codes; 0 unsupported instances; 0 intersection candidates. 0 expanded author steps. Independent reparse: passed.



Attribution. Merlijn Wissink [legolijntje]. CC BY 2.0.
<https://library.ldraw.org/omr/sets/259>

Source SHA256. d2619a85e973430a1b256d607fe8e44a865beb05ef63ac8028a5109c228ee97c

Connector records. stud: 108; axle: 5; hinge: 0; fixed: 0; ball: 0

Public input lock. f5df0685f7d89d74435eef34a311b18fc9d14ce4bc39a3ab7bbf43c78b3eb9ba

Status. Source preservation does not certify stability. Reference outputs below are internal review material. Full model inputs are the versioned public JSON and numbered PNGs.

1. **ld2-31028-color-1.** Identify the original color of B0018; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B0018 = brick_000018

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-31028-color-1.png; structured input = {}.

Options: A: Black; B: Orange; C: Light Bluish Grey; D: Trans Brown

Reference output: {"choiceId": "B"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

2. **ld2-31028-shape-match-1.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0018?
shape-match; visual; atomic; single-choice.

Operands: B0008 = brick_000008, B0018 = brick_000018, B0031 = brick_000031, B0042 = brick_000042, B0004 = brick_000004

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-31028-shape-match-1.png; structured input = {}.

Options: A: B0008; B: B0031; C: B0004; D: B0042

Reference output: {"choiceId": "B"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

3. **ld2-31028-distance-1.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0018?

distance; source-data; atomic; single-choice.

Operands: B0024 = brick_000024, B0018 = brick_000018, B0028 = brick_000028, B0043 = brick_000043

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0018": [0, 8.8, -36], "B0028": [0, 15.2, -24], "B0043": [-17.2, -12, 0], "B0024": [-20, 15.2, -56]}}

Options: A: B0043; B: B0024; C: B0028

Reference output: {"choiceId": "C"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

4. **ld2-31028-color-2.** Identify the original color of B0035; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B0035 = brick_000035

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-31028-color-2.png; structured input = {}.

Options: A: Trans Orange; B: Black; C: Trans Brown; D: White

Reference output: {"choiceId": "C"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

5. **ld2-31028-shape-match-2.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0035?

shape-match; visual; atomic; single-choice.

Operands: B0034 = brick_000034, B0038 = brick_000038, B0045 = brick_000045, B0035 = brick_000035, B0012 = brick_000012

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-31028-shape-match-2.png; structured input = {}.

Options: A: B0034; B: B0045; C: B0038; D: B0012

Reference output: {"choiceId": "A"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

6. **ld2-31028-distance-2.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0035?

distance; source-data; atomic; single-choice.

Operands: B0035 = brick_000035, B0027 = brick_000027, B0006 = brick_000006, B0014 = brick_000014

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0035": [4, 15.92, 0], "B0014": [-12, 0.8, 8], "B0006": [0, -0.8, 0], "B0027": [-24, 12, -23.8]}}

Options: A: B0027; B: B0006; C: B0014

Reference output: {"choiceId": "B"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

7. **Id2-31028-interface-1.** Classify the supplied connector record for B0053 and B0006. This does not certify load capacity.
 interface; connector-graph; metacognitive; single-choice.
Operands: B0053 = brick_000053, B0006 = brick_000006
Public evidence summary: {"connectorRecord": {"family": "axle", "aConnector": 0, "bConnector": 1}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
Options: A: fixed; B: hinge; C: stud; D: axle; E: ball
Reference output: {"choiceId": "D"}
Derivation: public connectorRecord.family equality
Equivalence: exact declared response
Review: machine checks passed; no human certification.
8. **Id2-31028-interface-2.** Classify the supplied connector record for B0042 and B0044. This does not certify load capacity.
 interface; connector-graph; metacognitive; single-choice.
Operands: B0044 = brick_000044, B0042 = brick_000042
Public evidence summary: {"connectorRecord": {"family": "stud", "aConnector": 11, "bConnector": 10}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
Options: A: fixed; B: ball; C: stud; D: hinge; E: axle
Reference output: {"choiceId": "C"}
Derivation: public connectorRecord.family equality
Equivalence: exact declared response
Review: machine checks passed; no human certification.
9. **Id2-31028-neighbors-1.** Select every candidate directly adjacent to B0027 in the supplied connector graph.
 neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0038 = brick_000038, B0018 = brick_000018, B0031 = brick_000031, B0027 = brick_000027, B0047 = brick_000047, B0028 = brick_000028
Public evidence summary: 4 unique undirected pairs
Options: A: B0028; B: B0031; C: B0038; D: B0018; E: B0047
Reference output: {"choiceIds": ["A", "B", "D"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: machine checks passed; no human certification.
10. **Id2-31028-graph-removal-1.** Delete B0027 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
 graph-removal; connector-graph; graph-internal; integer.
Operands: B0031 = brick_000031, B0027 = brick_000027, B0028 = brick_000028, B0047 = brick_000047, B0018 = brick_000018, B0038 = brick_000038
Public evidence summary: 4 unique undirected pairs; 6 nodes
Reference output: {"value": 4}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: machine checks passed; no human certification.
11. **Id2-31028-neighbors-2.** Select every candidate directly adjacent to B0047 in the supplied connector graph.
 neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0044 = brick_000044, B0049 = brick_000049, B0050 = brick_000050, B0001 = brick_000001, B0048 = brick_000048, B0047 = brick_000047
Public evidence summary: 5 unique undirected pairs

Options: A: B0050; B: B0044; C: B0049; D: B0001; E: B0048

Reference output: {"choiceIds": ["A", "C", "E"]}

Derivation: public incident edge endpoint set

Equivalence: unordered exact set; duplicate IDs invalid

Review: machine checks passed; no human certification.

12. **ld2-31028-graph-removal-2.** Delete B0047 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.

graph-removal; connector-graph; graph-internal; integer.

Operands: B0048 = brick_000048, B0044 = brick_000044, B0050 = brick_000050, B0049 = brick_000049, B0047 = brick_000047, B0001 = brick_000001

Public evidence summary: 5 unique undirected pairs; 6 nodes

Reference output: {"value": 3}

Derivation: union-find component count after public target deletion

Equivalence: exact declared response

Review: machine checks passed; no human certification.

13. **ld2-31028-evidence-limit-1.** Only source geometry, connector matching and mesh intersection audits are available. Without mass, friction, material or dynamic tests, is gravitational stability established?

evidence-limit; source-data; metacognitive; single-choice.

Operands: none

Public evidence summary: {"available": ["Original CAD", "Connector inference", "Inset-mesh intersection candidates"]}

Options: A: Not established by this evidence; B: Established

Reference output: {"choiceId": "A"}

Derivation: fixed epistemic control label

Equivalence: exact declared response

Review: machine checks passed; no human certification.

14. **ld2-31028-restore-instance-1.** The display copy hides B0018. Restore only that numbered instance, preserving all other instances and source poses.

restore-instance; scene-edit; procedural; actions.

Operands: B0018 = brick_000018, B0001 = brick_000001

Public evidence summary: 2 public actions; budget 1; {"initialFacts": ["missing-brick_000018", "present-brick_000001"], "goalFacts": ["present-brick_000018", "present-brick_000001"], "absentFacts": ["missing-brick_000018"]}

Reference output: {"actionIds": ["restore-B0018"]}

Derivation: public precondition solver

Equivalence: any legal sequence reaching all goals within budget; display-only semantics

Review: machine checks passed; no human certification.

15. **ld2-31028-restore-instance-2.** The display copy hides B0035. Restore only that numbered instance, preserving all other instances and source poses.

restore-instance; scene-edit; procedural; actions.

Operands: B0001 = brick_000001, B0035 = brick_000035

Public evidence summary: 2 public actions; budget 1; {"initialFacts": ["missing-brick_000035", "present-brick_000001"], "goalFacts": ["present-brick_000035", "present-brick_000001"], "absentFacts": ["missing-brick_000035"]}

Reference output: {"actionIds": ["restore-B0035"]}

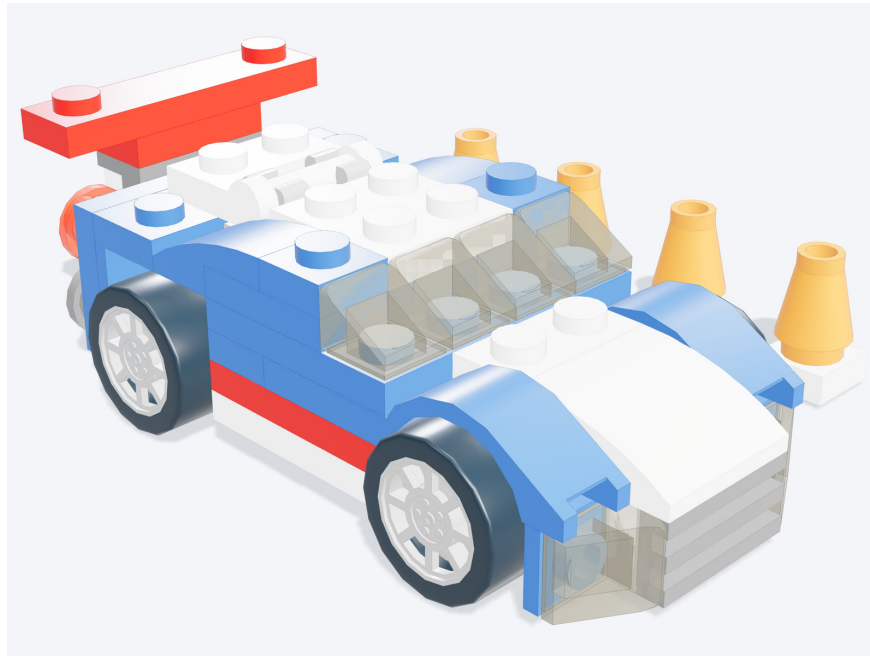
Derivation: public precondition solver

Equivalence: any legal sequence reaching all goals within budget; display-only semantics

Review: machine checks passed; no human certification.

12 31027: Blue Racer

67 source instances; 17 tasks; D1 size band. 24 type strings; 10 color codes; 0 unsupported instances; 0 intersection candidates. 0 expanded author steps. Independent reparse: passed.



Attribution. Merlijn Wissink [legolijntje]. CC BY 2.0.
<https://library.ldraw.org/omr/sets/223>

Source SHA256. 1f1975602ce9a63f2954c8da97f4fc13a473f709afa4a25b1022add33b55cfd1

Connector records. stud: 112; axle: 6; hinge: 4; fixed: 16; ball: 0

Public input lock. 8a8536853ab343c715c653f7f072a11d937291636d5fa3e2be647c8405c190bc

Status. Source preservation does not certify stability. Reference outputs below are internal review material. Full model inputs are the versioned public JSON and numbered PNGs.

1. **ld2-31027-color-1.** Identify the original color of B0045; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B0045 = brick_000045

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-31027-color-1.png; structured input = {}.

Options: A: Trans Brown; B: Trans Red; C: Orange; D: Light Bluish Grey

Reference output: {"choiceId": "A"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

2. **ld2-31027-shape-match-1.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0045?
shape-match; visual; atomic; single-choice.

Operands: B0045 = brick_000045, B0054 = brick_000054, B0019 = brick_000019, B0036 = brick_000036, B0062 = brick_000062

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-31027-shape-match-1.png; structured input = {}.

Options: A: B0062; B: B0019; C: B0036; D: B0054

Reference output: {"choiceId": "B"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

3. **ld2-31027-distance-1.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0045?

distance; source-data; atomic; single-choice.

Operands: B0036 = brick_000036, B0045 = brick_000045, B0051 = brick_000051, B0040 = brick_000040

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0045": [12, 9.52, 12], "B0040": [12, 9.600012, -8], "B0051": [12, -2.4, -31.2], "B0036": [8, 8.8, 0]}}

Options: A: B0051; B: B0040; C: B0036

Reference output: {"choiceId": "C"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

4. **ld2-31027-color-2.** Identify the original color of B0050; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B0050 = brick_000050

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-31027-color-2.png; structured input = {}.

Options: A: Black; B: Red; C: White; D: Light Bluish Grey

Reference output: {"choiceId": "C"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

5. **ld2-31027-shape-match-2.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0050?

shape-match; visual; atomic; single-choice.

Operands: B0002 = brick_000002, B0034 = brick_000034, B0050 = brick_000050, B0014 = brick_000014, B0043 = brick_000043

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-31027-shape-match-2.png; structured input = {}.

Options: A: B0014; B: B0002; C: B0043; D: B0034

Reference output: {"choiceId": "D"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

6. **ld2-31027-distance-2.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0050?

distance; source-data; atomic; single-choice.

Operands: B0058 = brick_000058, B0046 = brick_000046, B0050 = brick_000050, B0019 = brick_000019

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0050": [0, 1.6, -32.000012], "B0019": [-12, -4, 36.72], "B0046": [-12, 12, 4], "B0058": [-8, 2.4, -21.6]}}

Options: A: B0046; B: B0058; C: B0019

Reference output: {"choiceId": "B"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

7. **Id2-31027-interface-1.** Classify the supplied connector record for B0037 and B0059. This does not certify load capacity.
interface; connector-graph; metacognitive; single-choice.
Operands: B0059 = brick_000059, B0037 = brick_000037
Public evidence summary: {"connectorRecord": {"family": "axle", "aConnector": 1, "bConnector": 0}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
Options: A: axle; B: fixed; C: stud; D: hinge; E: ball
Reference output: {"choiceId": "A"}
Derivation: public connectorRecord.family equality
Equivalence: exact declared response
Review: machine checks passed; no human certification.
8. **Id2-31027-interface-2.** Classify the supplied connector record for B0026 and B0027. This does not certify load capacity.
interface; connector-graph; metacognitive; single-choice.
Operands: B0027 = brick_000027, B0026 = brick_000026
Public evidence summary: {"connectorRecord": {"family": "fixed", "aConnector": 1, "bConnector": 1}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
Options: A: fixed; B: ball; C: stud; D: hinge; E: axle
Reference output: {"choiceId": "A"}
Derivation: public connectorRecord.family equality
Equivalence: exact declared response
Review: machine checks passed; no human certification.
9. **Id2-31027-interface-3.** Classify the supplied connector record for B0002 and B0024. This does not certify load capacity.
interface; connector-graph; metacognitive; single-choice.
Operands: B0002 = brick_000002, B0024 = brick_000024
Public evidence summary: {"connectorRecord": {"family": "hinge", "aConnector": 2, "bConnector": 3}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
Options: A: fixed; B: ball; C: stud; D: axle; E: hinge
Reference output: {"choiceId": "E"}
Derivation: public connectorRecord.family equality
Equivalence: exact declared response
Review: machine checks passed; no human certification.
10. **Id2-31027-interface-4.** Classify the supplied connector record for B0060 and B0061. This does not certify load capacity.
interface; connector-graph; metacognitive; single-choice.
Operands: B0060 = brick_000060, B0061 = brick_000061
Public evidence summary: {"connectorRecord": {"family": "stud", "aConnector": 1, "bConnector": 3}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
Options: A: fixed; B: hinge; C: ball; D: stud; E: axle
Reference output: {"choiceId": "D"}
Derivation: public connectorRecord.family equality
Equivalence: exact declared response
Review: machine checks passed; no human certification.

11. **Id2-31027-neighbors-1.** Select every candidate directly adjacent to B0012 in the supplied connector graph.
neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0001 = brick_000001, B0012 = brick_000012, B0031 = brick_000031, B0059 = brick_000059, B0055 = brick_000055
Public evidence summary: 2 unique undirected pairs
Options: A: B0059; B: B0001; C: B0055; D: B0031
Reference output: {"choiceIds": ["B", "D"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: machine checks passed; no human certification.
12. **Id2-31027-graph-removal-1.** Delete B0012 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
graph-removal; connector-graph; graph-internal; integer.
Operands: B0012 = brick_000012, B0059 = brick_000059, B0055 = brick_000055, B0031 = brick_000031, B0001 = brick_000001
Public evidence summary: 2 unique undirected pairs; 5 nodes
Reference output: {"value": 4}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: machine checks passed; no human certification.
13. **Id2-31027-neighbors-2.** Select every candidate directly adjacent to B0024 in the supplied connector graph.
neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0053 = brick_000053, B0024 = brick_000024, B0025 = brick_000025, B0002 = brick_000002, B0058 = brick_000058
Public evidence summary: 2 unique undirected pairs
Options: A: B0053; B: B0025; C: B0002; D: B0058
Reference output: {"choiceIds": ["B", "C"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: machine checks passed; no human certification.
14. **Id2-31027-graph-removal-2.** Delete B0024 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
graph-removal; connector-graph; graph-internal; integer.
Operands: B0024 = brick_000024, B0058 = brick_000058, B0053 = brick_000053, B0002 = brick_000002, B0025 = brick_000025
Public evidence summary: 2 unique undirected pairs; 5 nodes
Reference output: {"value": 4}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: machine checks passed; no human certification.
15. **Id2-31027-evidence-limit-1.** Only source geometry, connector matching and mesh intersection audits are available. Without mass, friction, material or dynamic tests, is gravitational stability established?
evidence-limit; source-data; metacognitive; single-choice.
Operands: none
Public evidence summary: {"available": ["Original CAD", "Connector inference", "Inset-mesh intersection candidates"]}
Options: A: Established; B: Not established by this evidence
Reference output: {"choiceId": "B"}
Derivation: fixed epistemic control label
Equivalence: exact declared response
Review: machine checks passed; no human certification.

16. **ld2-31027-restore-instance-1.** The display copy hides B0045. Restore only that numbered instance, preserving all other instances and source poses.
- restore-instance; scene-edit; procedural; actions.
- Operands:* B0001 = brick_000001, B0045 = brick_000045
- Public evidence summary:* 2 public actions; budget 1; {"initialFacts": ["missing-brick_000045", "present-brick_000001"], "goalFacts": ["present-brick_000045", "present-brick_000001"], "absentFacts": ["missing-brick_000045"]}
- Reference output:* {"actionIds": ["restore-B0045"]}
- Derivation:* public precondition solver
- Equivalence:* any legal sequence reaching all goals within budget; display-only semantics
- Review:* machine checks passed; no human certification.
17. **ld2-31027-restore-instance-2.** The display copy hides B0050. Restore only that numbered instance, preserving all other instances and source poses.
- restore-instance; scene-edit; procedural; actions.
- Operands:* B0050 = brick_000050, B0001 = brick_000001
- Public evidence summary:* 2 public actions; budget 1; {"initialFacts": ["missing-brick_000050", "present-brick_000001"], "goalFacts": ["present-brick_000050", "present-brick_000001"], "absentFacts": ["missing-brick_000050"]}
- Reference output:* {"actionIds": ["restore-B0050"]}
- Derivation:* public precondition solver
- Equivalence:* any legal sequence reaching all goals within budget; display-only semantics
- Review:* machine checks passed; no human certification.

13 10156: LEGO Truck

111 source instances; 20 tasks; D1 size band. 51 type strings; 10 color codes; 1 unsupported instances; 2 intersection candidates. 19 expanded author steps. Independent reparse: passed.



Attribution. Robert Paciorek [bercik]. CC BY 2.0.

<https://library.ldraw.org/omr/sets/659>

Source SHA256. a58a784e6f6ce9d39999db1a4505941b5db5ebd1798af979bae891bc520a1a45

Connector records. stud: 213; axle: 13; hinge: 13; fixed: 26; ball: 0

Public input lock. 12d3d37b291161ed18adacf5f133e99539c928536ed76fb5ef1f585f54f5ea01

Status. Source preservation does not certify stability. Reference outputs below are internal review material. Full model inputs are the versioned public JSON and numbered PNGs.

1. **ld2-10156-color-1.** Identify the original color of B0069; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B0069 = brick_000069

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10156-color-1.png; structured input = {}.

Options: A: Chrome Silver; B: White; C: Yellow; D: Red

Reference output: {"choiceId": "C"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

2. **ld2-10156-shape-match-1.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0069?

shape-match; visual; atomic; single-choice.

Operands: B0054 = brick_000054, B0107 = brick_000107, B0069 = brick_000069, B0016 = brick_000016, B0088 = brick_000088

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10156-shape-match-1.png; structured input = {}.

Options: A: B0054; B: B0088; C: B0016; D: B0107

Reference output: {"choiceId": "A"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

3. **ld2-10156-distance-1.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0069?

distance; source-data; atomic; single-choice.

Operands: B0093 = brick_000093, B0069 = brick_000069, B0045 = brick_000045, B0067 = brick_000067

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0069": [68.105528, 47.2, -7.879336], "B0045": [38.606308, 15.2, -36.974207], "B0067": [-3.769975, 28.8, 11.199482], "B0093": [60.18416, 72.8, -26.399917]}}

Options: A: B0045; B: B0067; C: B0093

Reference output: {"choiceId": "C"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

4. **ld2-10156-color-2.** Identify the original color of B0019; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B0019 = brick_000019

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10156-color-2.png; structured input = {}.

Options: A: Chrome Silver; B: Light Grey; C: Black; D: Trans Neon Orange

Reference output: {"choiceId": "B"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

5. **ld2-10156-distance-2.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0019?

distance; source-data; atomic; single-choice.

Operands: B0031 = brick_000031, B0019 = brick_000019, B0086 = brick_000086, B0101 = brick_000101

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0019": [-7.119767, 18.4, 12.499461], "B0086": [65.426448, 48, -52.520376], "B0101": [97.003521, 10.8, -69.827264], "B0031": [38.3926, 15.2, 4.949118]}}

Options: A: B0101; B: B0086; C: B0031

Reference output: {"choiceId": "C"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

6. **ld2-10156-interface-1.** Classify the supplied connector record for B0099 and B0018. This does not certify load capacity.

interface; connector-graph; metacognitive; single-choice.

Operands: B0099 = brick_000099, B0018 = brick_000018

Public evidence summary: {"connectorRecord": {"family": "axle", "aConnector": 0, "bConnector": 1}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}

Options: A: stud; B: axle; C: hinge; D: ball; E: fixed

Reference output: {"choiceId": "B"}

Derivation: public connectorRecord.family equality

Equivalence: exact declared response

Review: machine checks passed; no human certification.

7. **Id2-10156-interface-2.** Classify the supplied connector record for B0107 and B0108. This does not certify load capacity.
interface; connector-graph; metacognitive; single-choice.
Operands: B0107 = brick_000107, B0108 = brick_000108
Public evidence summary: {"connectorRecord": {"family": "fixed", "aConnector": 2, "bConnector": 1}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
Options: A: ball; B: fixed; C: hinge; D: axle; E: stud
Reference output: {"choiceId": "B"}
Derivation: public connectorRecord.family equality
Equivalence: exact declared response
Review: machine checks passed; no human certification.
8. **Id2-10156-interface-3.** Classify the supplied connector record for B0004 and B0010. This does not certify load capacity.
interface; connector-graph; metacognitive; single-choice.
Operands: B0004 = brick_000004, B0010 = brick_000010
Public evidence summary: {"connectorRecord": {"family": "hinge", "aConnector": 2, "bConnector": 1}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
Options: A: stud; B: hinge; C: ball; D: axle; E: fixed
Reference output: {"choiceId": "B"}
Derivation: public connectorRecord.family equality
Equivalence: exact declared response
Review: machine checks passed; no human certification.
9. **Id2-10156-interface-4.** Classify the supplied connector record for B0080 and B0088. This does not certify load capacity.
interface; connector-graph; metacognitive; single-choice.
Operands: B0088 = brick_000088, B0080 = brick_000080
Public evidence summary: {"connectorRecord": {"family": "stud", "aConnector": 3, "bConnector": 1}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
Options: A: ball; B: fixed; C: axle; D: stud; E: hinge
Reference output: {"choiceId": "D"}
Derivation: public connectorRecord.family equality
Equivalence: exact declared response
Review: machine checks passed; no human certification.
10. **Id2-10156-neighbors-1.** Select every candidate directly adjacent to B0105 in the supplied connector graph.
neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0016 = brick_000016, B0006 = brick_000006, B0098 = brick_000098, B0105 = brick_000105, B0106 = brick_000106
Public evidence summary: 2 unique undirected pairs
Options: A: B0006; B: B0098; C: B0016; D: B0106
Reference output: {"choiceIds": ["C", "D"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: machine checks passed; no human certification.
11. **Id2-10156-graph-removal-1.** Delete B0105 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
graph-removal; connector-graph; graph-internal; integer.
Operands: B0106 = brick_000106, B0105 = brick_000105, B0098 = brick_000098, B0016 = brick_000016, B0006 = brick_000006

Public evidence summary: 2 unique undirected pairs; 5 nodes

Reference output: {"value": 4}

Derivation: union-find component count after public target deletion

Equivalence: exact declared response

Review: machine checks passed; no human certification.

12. **ld2-10156-neighbors-2.** Select every candidate directly adjacent to B0070 in the supplied connector graph.

neighbors; connector-graph; metacognitive; multiple-choice.

Operands: B0070 = brick_000070, B0082 = brick_000082, B0063 = brick_000063, B0035 = brick_000035, B0076 = brick_000076

Public evidence summary: 2 unique undirected pairs

Options: A: B0076; B: B0035; C: B0082; D: B0063

Reference output: {"choiceIds": ["C", "D"]}

Derivation: public incident edge endpoint set

Equivalence: unordered exact set; duplicate IDs invalid

Review: pending required human review.

13. **ld2-10156-graph-removal-2.** Delete B0070 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.

graph-removal; connector-graph; graph-internal; integer.

Operands: B0070 = brick_000070, B0063 = brick_000063, B0082 = brick_000082, B0035 = brick_000035, B0076 = brick_000076

Public evidence summary: 2 unique undirected pairs; 5 nodes

Reference output: {"value": 4}

Derivation: union-find component count after public target deletion

Equivalence: exact declared response

Review: pending required human review.

14. **ld2-10156-coverage-1.** Which numbered instance lacks a connector definition in the coverage record, so this audit cannot establish that it is floating?

coverage; source-data; metacognitive; single-choice.

Operands: B0081 = brick_000081, B0069 = brick_000069, B0111 = brick_000111, B0019 = brick_000019

Public evidence summary: {"coverage": [{"label": "B0081", "supported": true}, {"label": "B0019", "supported": true}, {"label": "B0111", "supported": false}, {"label": "B0069", "supported": true}]}

Options: A: B0019; B: B0111; C: B0069; D: B0081

Reference output: {"choiceId": "B"}

Derivation: public supported=false record

Equivalence: exact declared response

Review: pending required human review.

15. **ld2-10156-evidence-limit-1.** Only source geometry, connector matching and mesh intersection audits are available. Without mass, friction, material or dynamic tests, is gravitational stability established?

evidence-limit; source-data; metacognitive; single-choice.

Operands: none

Public evidence summary: {"available": ["Original CAD", "Connector inference", "Inset-mesh intersection candidates"]}

Options: A: Not established by this evidence; B: Established

Reference output: {"choiceId": "A"}

Derivation: fixed epistemic control label

Equivalence: exact declared response

Review: machine checks passed; no human certification.

16. **ld2-10156-source-step-1.** At which expanded author step does B0075 first appear in the supplied source-step table?
Source order is not a collision-free assembly proof.
source-step; source-data; procedural; single-choice.
Operands: B0001 = brick_000001, B0075 = brick_000075
Public evidence summary: 2 supplied author-step records; exact table in public JSON
Options: A: 14; B: 16; C: 11; D: 9
Reference output: {"choiceId": "A"}
Derivation: minimum public step index containing target
Equivalence: exact declared response
Review: machine checks passed; no human certification.
17. **ld2-10156-source-step-2.** At which expanded author step does B0050 first appear in the supplied source-step table?
Source order is not a collision-free assembly proof.
source-step; source-data; procedural; single-choice.
Operands: B0001 = brick_000001, B0050 = brick_000050
Public evidence summary: 2 supplied author-step records; exact table in public JSON
Options: A: 12; B: 14; C: 19; D: 10
Reference output: {"choiceId": "D"}
Derivation: minimum public step index containing target
Equivalence: exact declared response
Review: machine checks passed; no human certification.
18. **ld2-10156-source-sequence-1.** Reveal the supplied source-step window in author order. Actions change visibility only, without simulating insertion trajectories.
source-sequence; scene-edit; procedural; actions.
Operands: B0004 = brick_000004, B0010 = brick_000010, B0011 = brick_000011, B0020 = brick_000020, B0017 = brick_000017, B0019 = brick_000019, B0003 = brick_000003, B0002 = brick_000002, B0025 = brick_000025, B0021 = brick_000021, B0006 = brick_000006, B0023 = brick_000023, B0024 = brick_000024, B0009 = brick_000009, B0013 = brick_000013, B0026 = brick_000026, B0007 = brick_000007, B0028 = brick_000028, B0001 = brick_000001, B0027 = brick_000027, B0016 = brick_000016, B0008 = brick_000008, B0022 = brick_000022, B0014 = brick_000014, B0029 = brick_000029, B0012 = brick_000012, B0005 = brick_000005, B0018 = brick_000018, B0015 = brick_000015
Public evidence summary: 6 public actions; budget 6; {"initialFacts": ["start"], "goalFacts": ["done-6"], "absentFacts": []}
Reference output: {"actionIds": ["step-1", "step-2", "step-3", "step-4", "step-5", "step-6"]}
Derivation: public precondition solver
Equivalence: any legal sequence reaching all goals within budget; display-only semantics
Review: machine checks passed; no human certification.
19. **ld2-10156-restore-instance-1.** The display copy hides B0069. Restore only that numbered instance, preserving all other instances and source poses.
restore-instance; scene-edit; procedural; actions.
Operands: B0001 = brick_000001, B0069 = brick_000069
Public evidence summary: 2 public actions; budget 1; {"initialFacts": ["missing-brick_000069", "present-brick_000001"], "goalFacts": ["present-brick_000069", "present-brick_000001"], "absentFacts": ["missing-brick_000069"]}
Reference output: {"actionIds": ["restore-B0069"]}
Derivation: public precondition solver
Equivalence: any legal sequence reaching all goals within budget; display-only semantics
Review: machine checks passed; no human certification.

20. **ld2-10156-restore-instance-2.** The display copy hides B0019. Restore only that numbered instance, preserving all other instances and source poses.

restore-instance; scene-edit; procedural; actions.

Operands: B0019 = brick_000019, B0001 = brick_000001

Public evidence summary: 2 public actions; budget 1; {"initialFacts": ["missing-brick_000019", "present-brick_000001"], "goalFacts": ["present-brick_000019", "present-brick_000001"], "absentFacts": ["missing-brick_000019"]}

Reference output: {"actionIds": ["restore-B0019"]}

Derivation: public precondition solver

Equivalence: any legal sequence reaching all goals within budget; display-only semantics

Review: machine checks passed; no human certification.

14 42102: Mini CLAAS XERION

129 source instances; 17 tasks; D1 size band. 40 type strings; 11 color codes; 6 unsupported instances; 0 intersection candidates. 56 expanded author steps. Independent reparse: passed.



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<https://library.ldraw.org/library/omr/42102-1.mpd>

Source SHA256. 096deaf98e20c3e65b547c91e972005d853b1b1c3cf7a619910682e6fe2d2deb

Connector records. stud: 0; axle: 152; hinge: 0; fixed: 0; ball: 0

Public input lock. 233161ca642864d13de8219f04b975eb3db1c64f75f2279f803bfd4c40df1f98

Status. Source preservation does not certify stability. Reference outputs below are internal review material. Full model inputs are the versioned public JSON and numbered PNGs.

1. **ld2-omr-42102-color-1.** Identify the original color of B0094; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B0094 = brick_000094

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-42102-color-1.png; structured input = {}.

Options: A: Black; B: Red; C: Light Bluish Grey; D: Lime

Reference output: {"choiceId": "D"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

2. **ld2-omr-42102-shape-match-1.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0094?

shape-match; visual; atomic; single-choice.

Operands: B0061 = brick_000061, B0003 = brick_000003, B0020 = brick_000020, B0094 = brick_000094, B0065 = brick_000065

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-42102-shape-match-1.png; structured input = {}.

Options: A: B0003; B: B0020; C: B0061; D: B0065

Reference output: {"choiceId": "C"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

3. **ld2-omr-42102-distance-1.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0094?

distance; source-data; atomic; single-choice.

Operands: B0094 = brick_000094, B0019 = brick_000019, B0053 = brick_000053, B0045 = brick_000045

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0094": [8, 20.569662, -71.488617], "B0045": [8, 28.592, 18.5592], "B0019": [12, 32, -56], "B0053": [8, 32.496, -13.2168]}}

Options: A: B0045; B: B0053; C: B0019

Reference output: {"choiceId": "C"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

4. **ld2-omr-42102-color-2.** Identify the original color of B0088; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B0088 = brick_000088

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-42102-color-2.png; structured input = {}.

Options: A: Black; B: Lime; C: Tan; D: Dark Bluish Grey

Reference output: {"choiceId": "A"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

5. **ld2-omr-42102-distance-2.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0088?

distance; source-data; atomic; single-choice.

Operands: B0088 = brick_000088, B0113 = brick_000113, B0124 = brick_000124, B0059 = brick_000059

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0088": [8, 1.4752, -75.9456], "B0113": [6, -0.0048, -56.0056], "B0059": [28, 16, -40], "B0124": [31.4, 0, -8]}}

Options: A: B0113; B: B0124; C: B0059

Reference output: {"choiceId": "A"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

6. **ld2-omr-42102-interface-1.** Classify the supplied connector record for B0037 and B0040. This does not certify load capacity.

interface; connector-graph; metacognitive; single-choice.

Operands: B0037 = brick_000037, B0040 = brick_000040

Public evidence summary: {"connectorRecord": {"family": "axle", "aConnector": 2, "bConnector": 1}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}

Options: A: hinge; B: axle; C: stud; D: fixed; E: ball

Reference output: {"choiceId": "B"}

Derivation: public connectorRecord.family equality

Equivalence: exact declared response

Review: machine checks passed; no human certification.

7. **ld2-omr-42102-neighbors-1.** Select every candidate directly adjacent to B0118 in the supplied connector graph.
 neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0091 = brick_000091, B0118 = brick_000118, B0055 = brick_000055, B0013 = brick_000013, B0119 = brick_000119
Public evidence summary: 2 unique undirected pairs
Options: A: B0055; B: B0119; C: B0091; D: B0013
Reference output: {"choiceIds": ["B", "D"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: pending required human review.
8. **ld2-omr-42102-graph-removal-1.** Delete B0118 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
 graph-removal; connector-graph; graph-internal; integer.
Operands: B0119 = brick_000119, B0013 = brick_000013, B0091 = brick_000091, B0118 = brick_000118, B0055 = brick_000055
Public evidence summary: 2 unique undirected pairs; 5 nodes
Reference output: {"value": 4}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: pending required human review.
9. **ld2-omr-42102-neighbors-2.** Select every candidate directly adjacent to B0075 in the supplied connector graph.
 neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0129 = brick_000129, B0120 = brick_000120, B0075 = brick_000075, B0074 = brick_000074, B0071 = brick_000071
Public evidence summary: 2 unique undirected pairs
Options: A: B0120; B: B0071; C: B0074; D: B0129
Reference output: {"choiceIds": ["B", "C"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: pending required human review.
10. **ld2-omr-42102-graph-removal-2.** Delete B0075 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
 graph-removal; connector-graph; graph-internal; integer.
Operands: B0120 = brick_000120, B0071 = brick_000071, B0075 = brick_000075, B0129 = brick_000129, B0074 = brick_000074
Public evidence summary: 2 unique undirected pairs; 5 nodes
Reference output: {"value": 4}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: pending required human review.
11. **ld2-omr-42102-coverage-1.** Which numbered instance lacks a connector definition in the coverage record, so this audit cannot establish that it is floating?
 coverage; source-data; metacognitive; single-choice.
Operands: B0114 = brick_000114, B0094 = brick_000094, B0009 = brick_000009, B0088 = brick_000088
Public evidence summary: {"coverage": [{"label": "B0088", "supported": true}, {"label": "B0114", "supported": true}, {"label": "B0009", "supported": false}, {"label": "B0094", "supported": true}]}
Options: A: B0009; B: B0094; C: B0114; D: B0088
Reference output: {"choiceId": "A"}
Derivation: public supported=false record
Equivalence: exact declared response
Review: pending required human review.

12. **ld2-omr-42102-evidence-limit-1.** Only source geometry, connector matching and mesh intersection audits are available. Without mass, friction, material or dynamic tests, is gravitational stability established?
evidence-limit; source-data; metacognitive; single-choice.
Operands: none
Public evidence summary: {"available": ["Original CAD", "Connector inference", "Inset-mesh intersection candidates"]}
Options: A: Established; B: Not established by this evidence
Reference output: {"choiceId": "B"}
Derivation: fixed epistemic control label
Equivalence: exact declared response
Review: machine checks passed; no human certification.
13. **ld2-omr-42102-source-step-1.** At which expanded author step does B0094 first appear in the supplied source-step table? Source order is not a collision-free assembly proof.
source-step; source-data; procedural; single-choice.
Operands: B0001 = brick_000001, B0094 = brick_000094
Public evidence summary: 2 supplied author-step records; exact table in public JSON
Options: A: 22; B: 43; C: 52; D: 55
Reference output: {"choiceId": "B"}
Derivation: minimum public step index containing target
Equivalence: exact declared response
Review: machine checks passed; no human certification.
14. **ld2-omr-42102-source-step-2.** At which expanded author step does B0078 first appear in the supplied source-step table? Source order is not a collision-free assembly proof.
source-step; source-data; procedural; single-choice.
Operands: B0078 = brick_000078, B0001 = brick_000001
Public evidence summary: 2 supplied author-step records; exact table in public JSON
Options: A: 35; B: 11; C: 16; D: 17
Reference output: {"choiceId": "A"}
Derivation: minimum public step index containing target
Equivalence: exact declared response
Review: machine checks passed; no human certification.
15. **ld2-omr-42102-source-sequence-1.** Reveal the supplied source-step window in author order. Actions change visibility only, without simulating insertion trajectories.
source-sequence; scene-edit; procedural; actions.
Operands: B0001 = brick_000001, B0010 = brick_000010, B0014 = brick_000014, B0012 = brick_000012, B0011 = brick_000011, B0003 = brick_000003, B0002 = brick_000002, B0004 = brick_000004, B0013 = brick_000013, B0005 = brick_000005, B0009 = brick_000009, B0008 = brick_000008, B0007 = brick_000007, B0006 = brick_000006
Public evidence summary: 6 public actions; budget 6; {"initialFacts": ["start"], "goalFacts": ["done-6"], "absentFacts": []}
Reference output: {"actionIds": ["step-1", "step-2", "step-3", "step-4", "step-5", "step-6"]}
Derivation: public precondition solver
Equivalence: any legal sequence reaching all goals within budget; display-only semantics
Review: pending required human review.
16. **ld2-omr-42102-restore-instance-1.** The display copy hides B0094. Restore only that numbered instance, preserving all other instances and source poses.
restore-instance; scene-edit; procedural; actions.
Operands: B0001 = brick_000001, B0094 = brick_000094
Public evidence summary: 2 public actions; budget 1; {"initialFacts": ["missing-brick_000094", "present-brick_000001"], "goalFacts": ["present-brick_000094", "present-brick_000001"], "absentFacts": ["missing-brick_000094"]}

Reference output: {"actionIds": ["restore-B0094"]}

Derivation: public precondition solver

Equivalence: any legal sequence reaching all goals within budget; display-only semantics

Review: machine checks passed; no human certification.

17. **ld2-omr-42102-restore-instance-2.** The display copy hides B0088. Restore only that numbered instance, preserving all other instances and source poses.

restore-instance; scene-edit; procedural; actions.

Operands: B0088 = brick_000088, B0001 = brick_000001

Public evidence summary: 2 public actions; budget 1; {"initialFacts": ["missing-brick_000088", "present-brick_000001"], "goalFacts": ["present-brick_000088", "present-brick_000001"], "absentFacts": ["missing-brick_000088"]}

Reference output: {"actionIds": ["restore-B0088"]}

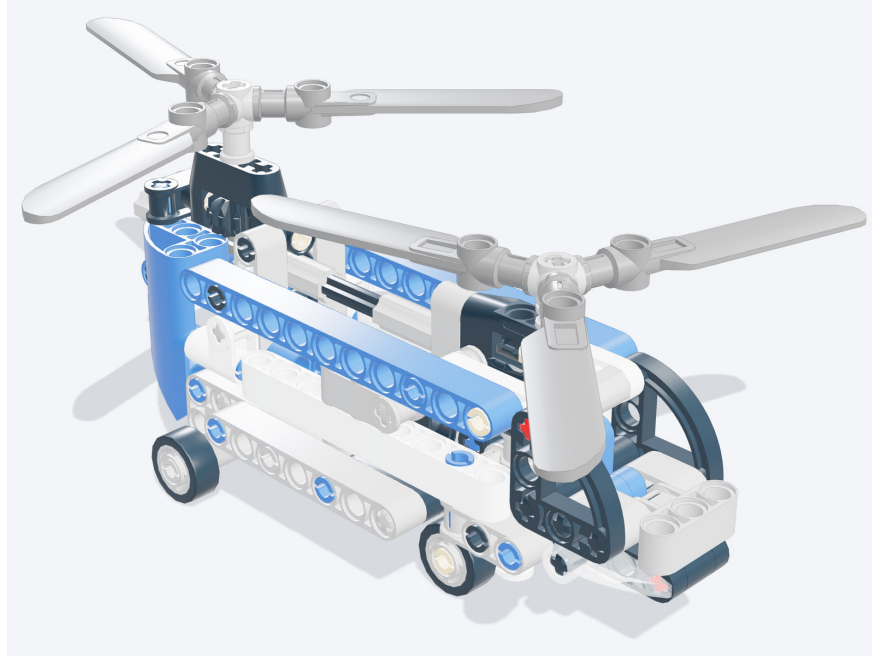
Derivation: public precondition solver

Equivalence: any legal sequence reaching all goals within budget; display-only semantics

Review: machine checks passed; no human certification.

15 42020: Twin-Rotor Helicopter

141 source instances; 14 tasks; D1 size band. 45 type strings; 8 color codes; 0 unsupported instances; 0 intersection candidates. 0 expanded author steps. Independent reparse: passed.



Attribution. Faramond Florent [Makou]. CC BY 2.0.

<https://library.ldraw.org/library/omr/42020-1.mpd>

Source SHA256. 03ce9e9f6e5e8fe9b35ec612407be6ac51996d369a341775b5ae80b45d7017a2

Connector records. stud: 0; axle: 181; hinge: 0; fixed: 16; ball: 0

Public input lock. 8d837ec764a1e894d4e133636396377ebda7d89e796f297d18b746bb8a6ee733

Status. Source preservation does not certify stability. Reference outputs below are internal review material. Full model inputs are the versioned public JSON and numbered PNGs.

1. **ld2-omr-42020-color-1.** Identify the original color of B0064; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B0064 = brick_000064

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-42020-color-1.png; structured input = {}.

Options: A: Dark Bluish Grey; B: Light Bluish Grey; C: Red; D: Black

Reference output: {"choiceId": "B"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

2. **ld2-omr-42020-shape-match-1.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0064?

shape-match; visual; atomic; single-choice.

Operands: B0120 = brick_000120, B0117 = brick_000117, B0064 = brick_000064, B0109 = brick_000109, B0063 = brick_000063

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-42020-shape-match-1.png; structured input = {}.

Options: A: B0109; B: B0120; C: B0117; D: B0063

Reference output: {"choiceId": "D"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

3. **ld2-omr-42020-distance-1.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0064?

distance; source-data; atomic; single-choice.

Operands: B0107 = brick_000107, B0053 = brick_000053, B0064 = brick_000064, B0051 = brick_000051

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0064": [16, -40, -20], "B0053": [32, -48, -8], "B0107": [22, -16, 88], "B0051": [-3.8, -23.2, -23.800047]}}

Options: A: B0107; B: B0053; C: B0051

Reference output: {"choiceId": "B"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

4. **ld2-omr-42020-color-2.** Identify the original color of B0030; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B0030 = brick_000030

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-42020-color-2.png; structured input = {}.

Options: A: Red; B: Light Bluish Grey; C: Dark Bluish Grey; D: White

Reference output: {"choiceId": "B"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

5. **ld2-omr-42020-distance-2.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0030?

distance; source-data; atomic; single-choice.

Operands: B0052 = brick_000052, B0017 = brick_000017, B0132 = brick_000132, B0030 = brick_000030

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0030": [8, -32, 24], "B0132": [0, -24, 92], "B0052": [8, -16, -32], "B0017": [24, -32, 8]}}

Options: A: B0132; B: B0052; C: B0017

Reference output: {"choiceId": "C"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

6. **ld2-omr-42020-interface-1.** Classify the supplied connector record for B0004 and B0005. This does not certify load capacity.

interface; connector-graph; metacognitive; single-choice.

Operands: B0005 = brick_000005, B0004 = brick_000004

Public evidence summary: {"connectorRecord": {"family": "axle", "aConnector": 0, "bConnector": 0}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}

Options: A: ball; B: fixed; C: hinge; D: axle; E: stud

Reference output: {"choiceId": "D"}

Derivation: public connectorRecord.family equality

Equivalence: exact declared response

Review: machine checks passed; no human certification.

7. **ld2-omr-42020-interface-2.** Classify the supplied connector record for B0053 and B0059. This does not certify load capacity.
 interface; connector-graph; metacognitive; single-choice.
Operands: B0059 = brick_000059, B0053 = brick_000053
Public evidence summary: {"connectorRecord": {"family": "fixed", "aConnector": 1, "bConnector": 0}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
Options: A: ball; B: fixed; C: stud; D: axle; E: hinge
Reference output: {"choiceId": "B"}
Derivation: public connectorRecord.family equality
Equivalence: exact declared response
Review: machine checks passed; no human certification.
8. **ld2-omr-42020-neighbors-1.** Select every candidate directly adjacent to B0007 in the supplied connector graph.
 neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0035 = brick_000035, B0007 = brick_000007, B0012 = brick_000012, B0086 = brick_000086, B0023 = brick_000023
Public evidence summary: 2 unique undirected pairs
Options: A: B0012; B: B0023; C: B0035; D: B0086
Reference output: {"choiceIds": ["A", "C"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: machine checks passed; no human certification.
9. **ld2-omr-42020-graph-removal-1.** Delete B0007 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
 graph-removal; connector-graph; graph-internal; integer.
Operands: B0012 = brick_000012, B0086 = brick_000086, B0035 = brick_000035, B0023 = brick_000023, B0007 = brick_000007
Public evidence summary: 2 unique undirected pairs; 5 nodes
Reference output: {"value": 4}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: machine checks passed; no human certification.
10. **ld2-omr-42020-neighbors-2.** Select every candidate directly adjacent to B0079 in the supplied connector graph.
 neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0113 = brick_000113, B0079 = brick_000079, B0059 = brick_000059, B0075 = brick_000075, B0081 = brick_000081
Public evidence summary: 2 unique undirected pairs
Options: A: B0081; B: B0059; C: B0075; D: B0113
Reference output: {"choiceIds": ["A", "C"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: machine checks passed; no human certification.
11. **ld2-omr-42020-graph-removal-2.** Delete B0079 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
 graph-removal; connector-graph; graph-internal; integer.
Operands: B0081 = brick_000081, B0113 = brick_000113, B0059 = brick_000059, B0079 = brick_000079, B0075 = brick_000075
Public evidence summary: 2 unique undirected pairs; 5 nodes
Reference output: {"value": 4}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: machine checks passed; no human certification.

12. **ld2-omr-42020-evidence-limit-1.** Only source geometry, connector matching and mesh intersection audits are available. Without mass, friction, material or dynamic tests, is gravitational stability established?
evidence-limit; source-data; metacognitive; single-choice.
Operands: none
Public evidence summary: {"available": ["Original CAD", "Connector inference", "Inset-mesh intersection candidates"]}
Options: A: Not established by this evidence; B: Established
Reference output: {"choiceId": "A"}
Derivation: fixed epistemic control label
Equivalence: exact declared response
Review: machine checks passed; no human certification.
13. **ld2-omr-42020-restore-instance-1.** The display copy hides B0064. Restore only that numbered instance, preserving all other instances and source poses.
restore-instance; scene-edit; procedural; actions.
Operands: B0001 = brick_000001, B0064 = brick_000064
Public evidence summary: 2 public actions; budget 1; {"initialFacts": ["missing-brick_000064", "present-brick_000001"], "goalFacts": ["present-brick_000064", "present-brick_000001"], "absentFacts": ["missing-brick_000064"]}
Reference output: {"actionIds": ["restore-B0064"]}
Derivation: public precondition solver
Equivalence: any legal sequence reaching all goals within budget; display-only semantics
Review: machine checks passed; no human certification.
14. **ld2-omr-42020-restore-instance-2.** The display copy hides B0030. Restore only that numbered instance, preserving all other instances and source poses.
restore-instance; scene-edit; procedural; actions.
Operands: B0030 = brick_000030, B0001 = brick_000001
Public evidence summary: 2 public actions; budget 1; {"initialFacts": ["missing-brick_000030", "present-brick_000001"], "goalFacts": ["present-brick_000030", "present-brick_000001"], "absentFacts": ["missing-brick_000030"]}
Reference output: {"actionIds": ["restore-B0030"]}
Derivation: public precondition solver
Equivalence: any legal sequence reaching all goals within budget; display-only semantics
Review: machine checks passed; no human certification.

16 10036: Pizza To Go

166 source instances; 20 tasks; D2 size band. 71 type strings; 12 color codes; 0 unsupported instances; 3 intersection candidates. 29 expanded author steps. Independent reparse: passed.



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<https://library.ldraw.org/omr/sets/658>

Source SHA256. bb9eafb3fc283a4b3b28eef58d6f9208528ee3d9b3debe4a7db8bbadc85f63f8

Connector records. stud: 361; axle: 11; hinge: 24; fixed: 22; ball: 0

Public input lock. d897373fa4507a329b51bf19d067a8237781e3c2504acce2db5b2903523db9a3

Status. Source preservation does not certify stability. Reference outputs below are internal review material. Full model inputs are the versioned public JSON and numbered PNGs.

1. **ld2-10036-color-1.** Identify the original color of B0063; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B0063 = brick_000063

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10036-color-1.png; structured input = {}.

Options: A: Trans Light Blue; B: White; C: Brown; D: Dark Grey

Reference output: {"choiceId": "B"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

2. **ld2-10036-shape-match-1.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0063?

shape-match; visual; atomic; single-choice.

Operands: B0120 = brick_000120, B0063 = brick_000063, B0103 = brick_000103, B0093 = brick_000093, B0029 = brick_000029

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10036-shape-match-1.png; structured input = {}.

Options: A: B0029; B: B0093; C: B0103; D: B0120

Reference output: {"choiceId": "B"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

3. **ld2-10036-distance-1.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0063?

distance; source-data; atomic; single-choice.

Operands: B0142 = brick_000142, B0018 = brick_000018, B0063 = brick_000063, B0149 = brick_000149

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0063": [32, 66.4, -20], "B0149": [45.643616, 13.999961, 60.094864], "B0142": [-54.752249, 18.188798, 57.747503], "B0018": [56, 24.8, -32]}}

Options: A: B0142; B: B0149; C: B0018

Reference output: {"choiceId": "C"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: pending required human review.

4. **ld2-10036-color-2.** Identify the original color of B0162; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B0162 = brick_000162

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10036-color-2.png; structured input = {}.

Options: A: Yellow; B: Red; C: Black; D: Dark Grey

Reference output: {"choiceId": "A"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

5. **ld2-10036-shape-match-2.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0162?

shape-match; visual; atomic; single-choice.

Operands: B0164 = brick_000164, B0054 = brick_000054, B0067 = brick_000067, B0140 = brick_000140, B0162 = brick_000162

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10036-shape-match-2.png; structured input = {}.

Options: A: B0054; B: B0140; C: B0067; D: B0164

Reference output: {"choiceId": "B"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

6. **ld2-10036-distance-2.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0162?

distance; source-data; atomic; single-choice.

Operands: B0162 = brick_000162, B0054 = brick_000054, B0151 = brick_000151, B0091 = brick_000091

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0162": [-11.3066, 22.571561, 56.5872], "B0091": [83.576354, -2.4, 76.832951], "B0054": [-28, 56.8, -32], "B0151": [40.744899, 22.572178, 54.210612]}}

Options: A: B0151; B: B0091; C: B0054

Reference output: {"choiceId": "A"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

7. **Id2-10036-interface-1.** Classify the supplied connector record for B0147 and B0148. This does not certify load capacity.
interface; connector-graph; metacognitive; single-choice.
Operands: B0148 = brick_000148, B0147 = brick_000147
Public evidence summary: {"connectorRecord": {"family": "axle", "aConnector": 2, "bConnector": 0}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
Options: A: hinge; B: fixed; C: ball; D: axle; E: stud
Reference output: {"choiceId": "D"}
Derivation: public connectorRecord.family equality
Equivalence: exact declared response
Review: machine checks passed; no human certification.
8. **Id2-10036-interface-2.** Classify the supplied connector record for B0128 and B0127. This does not certify load capacity.
interface; connector-graph; metacognitive; single-choice.
Operands: B0127 = brick_000127, B0128 = brick_000128
Public evidence summary: {"connectorRecord": {"family": "fixed", "aConnector": 2, "bConnector": 0}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
Options: A: ball; B: axle; C: hinge; D: stud; E: fixed
Reference output: {"choiceId": "E"}
Derivation: public connectorRecord.family equality
Equivalence: exact declared response
Review: machine checks passed; no human certification.
9. **Id2-10036-interface-3.** Classify the supplied connector record for B0140 and B0137. This does not certify load capacity.
interface; connector-graph; metacognitive; single-choice.
Operands: B0140 = brick_000140, B0137 = brick_000137
Public evidence summary: {"connectorRecord": {"family": "hinge", "aConnector": 0, "bConnector": 5}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
Options: A: axle; B: stud; C: hinge; D: ball; E: fixed
Reference output: {"choiceId": "C"}
Derivation: public connectorRecord.family equality
Equivalence: exact declared response
Review: machine checks passed; no human certification.
10. **Id2-10036-interface-4.** Classify the supplied connector record for B0090 and B0094. This does not certify load capacity.
interface; connector-graph; metacognitive; single-choice.
Operands: B0094 = brick_000094, B0090 = brick_000090
Public evidence summary: {"connectorRecord": {"family": "stud", "aConnector": 8, "bConnector": 3}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
Options: A: stud; B: axle; C: hinge; D: fixed; E: ball
Reference output: {"choiceId": "A"}
Derivation: public connectorRecord.family equality
Equivalence: exact declared response
Review: machine checks passed; no human certification.

11. **ld2-10036-neighbors-1.** Select every candidate directly adjacent to B0161 in the supplied connector graph.
 neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0023 = brick_000023, B0081 = brick_000081, B0163 = brick_000163, B0161 = brick_000161, B0159 = brick_000159
Public evidence summary: 2 unique undirected pairs
Options: A: B0163; B: B0159; C: B0023; D: B0081
Reference output: {"choiceIds": ["A", "B"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: machine checks passed; no human certification.
12. **ld2-10036-graph-removal-1.** Delete B0161 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
 graph-removal; connector-graph; graph-internal; integer.
Operands: B0023 = brick_000023, B0159 = brick_000159, B0163 = brick_000163, B0081 = brick_000081, B0161 = brick_000161
Public evidence summary: 2 unique undirected pairs; 5 nodes
Reference output: {"value": 4}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: machine checks passed; no human certification.
13. **ld2-10036-neighbors-2.** Select every candidate directly adjacent to B0111 in the supplied connector graph.
 neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0111 = brick_000111, B0101 = brick_000101, B0099 = brick_000099, B0136 = brick_000136, B0162 = brick_000162, B0115 = brick_000115
Public evidence summary: 4 unique undirected pairs
Options: A: B0115; B: B0101; C: B0136; D: B0162; E: B0099
Reference output: {"choiceIds": ["A", "B", "E"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: machine checks passed; no human certification.
14. **ld2-10036-graph-removal-2.** Delete B0111 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
 graph-removal; connector-graph; graph-internal; integer.
Operands: B0115 = brick_000115, B0136 = brick_000136, B0111 = brick_000111, B0101 = brick_000101, B0162 = brick_000162, B0099 = brick_000099
Public evidence summary: 4 unique undirected pairs; 6 nodes
Reference output: {"value": 4}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: machine checks passed; no human certification.
15. **ld2-10036-evidence-limit-1.** Only source geometry, connector matching and mesh intersection audits are available. Without mass, friction, material or dynamic tests, is gravitational stability established?
 evidence-limit; source-data; metacognitive; single-choice.
Operands: none
Public evidence summary: {"available": ["Original CAD", "Connector inference", "Inset-mesh intersection candidates"]}
Options: A: Established; B: Not established by this evidence
Reference output: {"choiceId": "B"}
Derivation: fixed epistemic control label
Equivalence: exact declared response
Review: machine checks passed; no human certification.

16. **ld2-10036-source-step-1.** At which expanded author step does B0016 first appear in the supplied source-step table?
Source order is not a collision-free assembly proof.
source-step; source-data; procedural; single-choice.
Operands: B0016 = brick_000016, B0001 = brick_000001
Public evidence summary: 2 supplied author-step records; exact table in public JSON
Options: A: 9; B: 13; C: 2; D: 4
Reference output: {"choiceId": "D"}
Derivation: minimum public step index containing target
Equivalence: exact declared response
Review: machine checks passed; no human certification.
17. **ld2-10036-source-step-2.** At which expanded author step does B0044 first appear in the supplied source-step table?
Source order is not a collision-free assembly proof.
source-step; source-data; procedural; single-choice.
Operands: B0044 = brick_000044, B0001 = brick_000001
Public evidence summary: 2 supplied author-step records; exact table in public JSON
Options: A: 22; B: 2; C: 26; D: 9
Reference output: {"choiceId": "D"}
Derivation: minimum public step index containing target
Equivalence: exact declared response
Review: machine checks passed; no human certification.
18. **ld2-10036-source-sequence-1.** Reveal the supplied source-step window in author order. Actions change visibility only, without simulating insertion trajectories.
source-sequence; scene-edit; procedural; actions.
Operands: B0029 = brick_000029, B0027 = brick_000027, B0020 = brick_000020, B0021 = brick_000021, B0015 = brick_000015, B0019 = brick_000019, B0017 = brick_000017, B0002 = brick_000002, B0008 = brick_000008, B0003 = brick_000003, B0004 = brick_000004, B0014 = brick_000014, B0012 = brick_000012, B0028 = brick_000028, B0023 = brick_000023, B0024 = brick_000024, B0025 = brick_000025, B0026 = brick_000026, B0001 = brick_000001, B0007 = brick_000007, B0006 = brick_000006, B0009 = brick_000009, B0010 = brick_000010, B0018 = brick_000018, B0016 = brick_000016, B0005 = brick_000005, B0013 = brick_000013, B0011 = brick_000011, B0022 = brick_000022
Public evidence summary: 6 public actions; budget 6; {"initialFacts": ["start"], "goalFacts": ["done-6"], "absentFacts": []}
Reference output: {"actionIds": ["step-1", "step-2", "step-3", "step-4", "step-5", "step-6"]}
Derivation: public precondition solver
Equivalence: any legal sequence reaching all goals within budget; display-only semantics
Review: pending required human review.
19. **ld2-10036-restore-instance-1.** The display copy hides B0063. Restore only that numbered instance, preserving all other instances and source poses.
restore-instance; scene-edit; procedural; actions.
Operands: B0063 = brick_000063, B0001 = brick_000001
Public evidence summary: 2 public actions; budget 1; {"initialFacts": ["missing-brick_000063", "present-brick_000001"], "goalFacts": ["present-brick_000063", "present-brick_000001"], "absentFacts": ["missing-brick_000063"]}
Reference output: {"actionIds": ["restore-B0063"]}
Derivation: public precondition solver
Equivalence: any legal sequence reaching all goals within budget; display-only semantics
Review: machine checks passed; no human certification.

20. **ld2-10036-restore-instance-2.** The display copy hides B0162. Restore only that numbered instance, preserving all other instances and source poses.

restore-instance; scene-edit; procedural; actions.

Operands: B0162 = brick_000162, B0001 = brick_000001

Public evidence summary: 2 public actions; budget 1; {"initialFacts": ["missing-brick_000162", "present-brick_000001"], "goalFacts": ["present-brick_000162", "present-brick_000001"], "absentFacts": ["missing-brick_000162"]}

Reference output: {"actionIds": ["restore-B0162"]}

Derivation: public precondition solver

Equivalence: any legal sequence reaching all goals within budget; display-only semantics

Review: machine checks passed; no human certification.

17 10014: Caboose

170 source instances; 16 tasks; D2 size band. 46 type strings; 8 color codes; 15 unsupported instances; 1 intersection candidates. 0 expanded author steps. Independent reparse: passed.



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<https://library.ldraw.org/omr/sets/1415>

Source SHA256. f3aead9fe55512171565803f187f3f7a952fb70515983c39970cffd89431644d

Connector records. stud: 440; axle: 10; hinge: 0; fixed: 0; ball: 0

Public input lock. c22c7968e5b12e1190e57801d20a8ae842ea5a797bf07e78548274307c6b5b65

Status. Source preservation does not certify stability. Reference outputs below are internal review material. Full model inputs are the versioned public JSON and numbered PNGs.

1. **ld2-10014-color-1.** Identify the original color of B0093; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B0093 = brick_000093

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10014-color-1.png; structured input = {}.

Options: A: Black; B: Brown; C: Dark Grey; D: Light Grey

Reference output: {"choiceId": "C"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

2. **ld2-10014-shape-match-1.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0093?

shape-match; visual; atomic; single-choice.

Operands: B0004 = brick_000004, B0093 = brick_000093, B0002 = brick_000002, B0094 = brick_000094, B0114 = brick_000114

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10014-shape-match-1.png; structured input = {}.

Options: A: B0004; B: B0114; C: B0094; D: B0002

Reference output: {"choiceId": "C"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

3. **ld2-10014-distance-1.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0093?

distance; source-data; atomic; single-choice.

Operands: B0127 = brick_000127, B0093 = brick_000093, B0004 = brick_000004, B0071 = brick_000071

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0093": [-4, 2.4, 4], "B0127": [62, 9.6, 0], "B0071": [-12, 18.4, 8], "B0004": [-20, 34.4, -20]}}

Options: A: B0004; B: B0127; C: B0071

Reference output: {"choiceId": "C"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

4. **ld2-10014-color-2.** Identify the original color of B0139; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B0139 = brick_000139

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10014-color-2.png; structured input = {}.

Options: A: Red; B: Black; C: Yellow; D: Trans Red

Reference output: {"choiceId": "B"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

5. **ld2-10014-shape-match-2.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0139?

shape-match; visual; atomic; single-choice.

Operands: B0078 = brick_000078, B0165 = brick_000165, B0139 = brick_000139, B0134 = brick_000134, B0137 = brick_000137

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10014-shape-match-2.png; structured input = {}.

Options: A: B0165; B: B0137; C: B0134; D: B0078

Reference output: {"choiceId": "B"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

6. **ld2-10014-distance-2.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0139?

distance; source-data; atomic; single-choice.

Operands: B0056 = brick_000056, B0129 = brick_000129, B0139 = brick_000139, B0141 = brick_000141

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0139": [-20, 44, 12], "B0129": [-8.008, -0.8, -27.2], "B0056": [54.8, 24.8, 20], "B0141": [24, 34.4, 20]}}

Options: A: B0141; B: B0056; C: B0129

Reference output: {"choiceId": "A"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

7. **ld2-10014-interface-1.** Classify the supplied connector record for B0021 and B0019. This does not certify load capacity.
 interface; connector-graph; metacognitive; single-choice.
Operands: B0019 = brick_000019, B0021 = brick_000021
Public evidence summary: {"connectorRecord": {"family": "axle", "aConnector": 0, "bConnector": 0}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
Options: A: fixed; B: axle; C: ball; D: stud; E: hinge
Reference output: {"choiceId": "B"}
Derivation: public connectorRecord.family equality
Equivalence: exact declared response
Review: machine checks passed; no human certification.
8. **ld2-10014-interface-2.** Classify the supplied connector record for B0082 and B0067. This does not certify load capacity.
 interface; connector-graph; metacognitive; single-choice.
Operands: B0067 = brick_000067, B0082 = brick_000082
Public evidence summary: {"connectorRecord": {"family": "stud", "aConnector": 5, "bConnector": 9}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
Options: A: axle; B: fixed; C: hinge; D: stud; E: ball
Reference output: {"choiceId": "D"}
Derivation: public connectorRecord.family equality
Equivalence: exact declared response
Review: machine checks passed; no human certification.
9. **ld2-10014-neighbors-1.** Select every candidate directly adjacent to B0032 in the supplied connector graph.
 neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0015 = brick_000015, B0068 = brick_000068, B0021 = brick_000021, B0163 = brick_000163, B0020 = brick_000020, B0032 = brick_000032, B0165 = brick_000165, B0006 = brick_000006
Public evidence summary: 5 unique undirected pairs
Options: A: B0015; B: B0163; C: B0021; D: B0165; E: B0020; F: B0006; G: B0068
Reference output: {"choiceIds": ["A", "B", "C", "D", "E"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: machine checks passed; no human certification.
10. **ld2-10014-graph-removal-1.** Delete B0032 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
 graph-removal; connector-graph; graph-internal; integer.
Operands: B0032 = brick_000032, B0020 = brick_000020, B0163 = brick_000163, B0068 = brick_000068, B0006 = brick_000006, B0165 = brick_000165, B0021 = brick_000021, B0015 = brick_000015
Public evidence summary: 5 unique undirected pairs; 8 nodes
Reference output: {"value": 7}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: machine checks passed; no human certification.
11. **ld2-10014-neighbors-2.** Select every candidate directly adjacent to B0024 in the supplied connector graph.
 neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0025 = brick_000025, B0024 = brick_000024, B0029 = brick_000029, B0126 = brick_000126, B0030 = brick_000030, B0129 = brick_000129, B0026 = brick_000026
Public evidence summary: 4 unique undirected pairs

Options: A: B0025; B: B0129; C: B0029; D: B0030; E: B0026; F: B0126

Reference output: {"choiceIds": ["A", "C", "D", "E"]}

Derivation: public incident edge endpoint set

Equivalence: unordered exact set; duplicate IDs invalid

Review: pending required human review.

12. **ld2-10014-graph-removal-2.** Delete B0024 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.

graph-removal; connector-graph; graph-internal; integer.

Operands: B0024 = brick_000024, B0026 = brick_000026, B0025 = brick_000025, B0129 = brick_000129, B0029 = brick_000029, B0030 = brick_000030, B0126 = brick_000126

Public evidence summary: 4 unique undirected pairs; 7 nodes

Reference output: {"value": 6}

Derivation: union-find component count after public target deletion

Equivalence: exact declared response

Review: pending required human review.

13. **ld2-10014-coverage-1.** Which numbered instance lacks a connector definition in the coverage record, so this audit cannot establish that it is floating?

coverage; source-data; metacognitive; single-choice.

Operands: B0093 = brick_000093, B0139 = brick_000139, B0074 = brick_000074, B0033 = brick_000033

Public evidence summary: {"coverage": [{"label": "B0033", "supported": false}, {"label": "B0074", "supported": true}, {"label": "B0139", "supported": true}, {"label": "B0093", "supported": true}]}

Options: A: B0074; B: B0033; C: B0139; D: B0093

Reference output: {"choiceId": "B"}

Derivation: public supported=false record

Equivalence: exact declared response

Review: pending required human review.

14. **ld2-10014-evidence-limit-1.** Only source geometry, connector matching and mesh intersection audits are available. Without mass, friction, material or dynamic tests, is gravitational stability established?

evidence-limit; source-data; metacognitive; single-choice.

Operands: none

Public evidence summary: {"available": ["Original CAD", "Connector inference", "Inset-mesh intersection candidates"]}

Options: A: Established; B: Not established by this evidence

Reference output: {"choiceId": "B"}

Derivation: fixed epistemic control label

Equivalence: exact declared response

Review: machine checks passed; no human certification.

15. **ld2-10014-restore-instance-1.** The display copy hides B0093. Restore only that numbered instance, preserving all other instances and source poses.

restore-instance; scene-edit; procedural; actions.

Operands: B0093 = brick_000093, B0001 = brick_000001

Public evidence summary: 2 public actions; budget 1; {"initialFacts": ["missing-brick_000093", "present-brick_000001"], "goalFacts": ["present-brick_000093", "present-brick_000001"], "absentFacts": ["missing-brick_000093"]}

Reference output: {"actionIds": ["restore-B0093"]}

Derivation: public precondition solver

Equivalence: any legal sequence reaching all goals within budget; display-only semantics

Review: machine checks passed; no human certification.

16. **ld2-10014-restore-instance-2.** The display copy hides B0139. Restore only that numbered instance, preserving all other instances and source poses.

restore-instance; scene-edit; procedural; actions.

Operands: B0139 = brick_000139, B0001 = brick_000001

Public evidence summary: 2 public actions; budget 1; {"initialFacts": ["missing-brick_000139", "present-brick_000001"], "goalFacts": ["present-brick_000139", "present-brick_000001"], "absentFacts": ["missing-brick_000139"]}

Reference output: {"actionIds": ["restore-B0139"]}

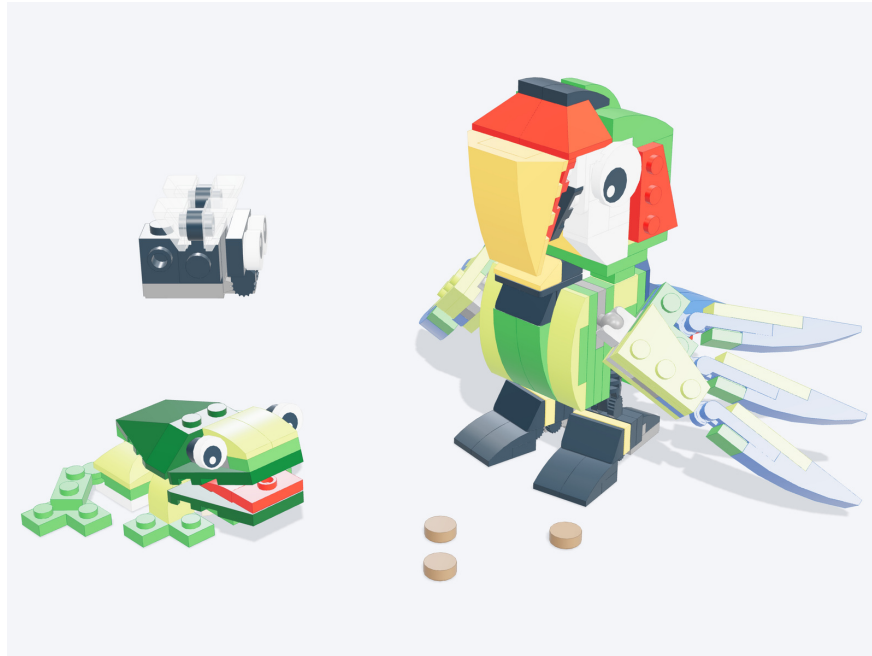
Derivation: public precondition solver

Equivalence: any legal sequence reaching all goals within budget; display-only semantics

Review: machine checks passed; no human certification.

18 31031: Rainforest Animals

215 source instances; 20 tasks; D2 size band. 61 type strings; 15 color codes; 0 unsupported instances; 7 intersection candidates. 130 expanded author steps. Independent reparse: passed.



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<https://library.ldraw.org/library/omr/31031-1.mpd>

Source SHA256. b5b87230d2fe147aa1135baee5d1604a283f1e586a228b27b351253af5565da7

Connector records. stud: 396; axle: 8; hinge: 9; fixed: 0; ball: 3

Public input lock. 2398c12a777d407e5e3d76cccc9e2bd313c0aaa70f039215d7fac29d609b2562

Status. Source preservation does not certify stability. Reference outputs below are internal review material. Full model inputs are the versioned public JSON and numbered PNGs.

1. **ld2-omr-31031-color-1.** Identify the original color of B0182; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B0182 = brick_000182

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-31031-color-1.png; structured input = {}.

Options: A: Trans Clear; B: Green; C: Bright Light Orange; D: Blue

Reference output: {"choiceId": "B"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

2. **ld2-omr-31031-shape-match-1.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0182?

shape-match; visual; atomic; single-choice.

Operands: B0063 = brick_000063, B0177 = brick_000177, B0182 = brick_000182, B0118 = brick_000118, B0071 = brick_000071

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-31031-shape-match-1.png; structured input = {}.

Options: A: B0071; B: B0177; C: B0118; D: B0063

Reference output: {"choiceId": "B"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

3. **ld2-omr-31031-distance-1.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0182?

distance; source-data; atomic; single-choice.

Operands: B0004 = brick_000004, B0096 = brick_000096, B0038 = brick_000038, B0182 = brick_000182

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0182": [11.8912, 42.1888, -13.4232], "B0038": [-80, 22.103829, -19.584571], "B0096": [-6.066, 3.200012, 2.0992], "B0004": [-115.958788, 106.4, 14.991612]}}

Options: A: B0096; B: B0004; C: B0038

Reference output: {"choiceId": "A"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

4. **ld2-omr-31031-color-2.** Identify the original color of B0111; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B0111 = brick_000111

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-31031-color-2.png; structured input = {}.

Options: A: Green; B: Yellow; C: Lime; D: White

Reference output: {"choiceId": "A"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

5. **ld2-omr-31031-shape-match-2.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0111?

shape-match; visual; atomic; single-choice.

Operands: B0097 = brick_000097, B0141 = brick_000141, B0111 = brick_000111, B0005 = brick_000005, B0098 = brick_000098

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-31031-shape-match-2.png; structured input = {}.

Options: A: B0141; B: B0005; C: B0098; D: B0097

Reference output: {"choiceId": "A"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

6. **ld2-omr-31031-distance-2.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0111?

distance; source-data; atomic; single-choice.

Operands: B0094 = brick_000094, B0111 = brick_000111, B0142 = brick_000142, B0149 = brick_000149

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0111": [14.4, 69.6, 12.8], "B0142": [14.4, 92.8, 8.8], "B0094": [1.3218, 2.4, -1.1448], "B0149": [2.4, 85.6, 30.4]}}

Options: A: B0094; B: B0142; C: B0149

Reference output: {"choiceId": "B"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

7. **ld2-omr-31031-interface-1.** Classify the supplied connector record for B0049 and B0110. This does not certify load capacity.
 interface; connector-graph; metacognitive; single-choice.
Operands: B0049 = brick_000049, B0110 = brick_000110
Public evidence summary: {"connectorRecord": {"family": "axle", "aConnector": 1, "bConnector": 0}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
Options: A: hinge; B: fixed; C: axle; D: ball; E: stud
Reference output: {"choiceId": "C"}
Derivation: public connectorRecord.family equality
Equivalence: exact declared response
Review: machine checks passed; no human certification.
8. **ld2-omr-31031-interface-2.** Classify the supplied connector record for B0038 and B0035. This does not certify load capacity.
 interface; connector-graph; metacognitive; single-choice.
Operands: B0038 = brick_000038, B0035 = brick_000035
Public evidence summary: {"connectorRecord": {"family": "ball", "aConnector": 3, "bConnector": 3}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
Options: A: axle; B: fixed; C: hinge; D: stud; E: ball
Reference output: {"choiceId": "E"}
Derivation: public connectorRecord.family equality
Equivalence: exact declared response
Review: machine checks passed; no human certification.
9. **ld2-omr-31031-interface-3.** Classify the supplied connector record for B0015 and B0016. This does not certify load capacity.
 interface; connector-graph; metacognitive; single-choice.
Operands: B0015 = brick_000015, B0016 = brick_000016
Public evidence summary: {"connectorRecord": {"family": "hinge", "aConnector": 1, "bConnector": 2}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
Options: A: stud; B: fixed; C: axle; D: ball; E: hinge
Reference output: {"choiceId": "E"}
Derivation: public connectorRecord.family equality
Equivalence: exact declared response
Review: machine checks passed; no human certification.
10. **ld2-omr-31031-interface-4.** Classify the supplied connector record for B0126 and B0134. This does not certify load capacity.
 interface; connector-graph; metacognitive; single-choice.
Operands: B0126 = brick_000126, B0134 = brick_000134
Public evidence summary: {"connectorRecord": {"family": "stud", "aConnector": 11, "bConnector": 3}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
Options: A: hinge; B: stud; C: ball; D: fixed; E: axle
Reference output: {"choiceId": "B"}
Derivation: public connectorRecord.family equality
Equivalence: exact declared response
Review: machine checks passed; no human certification.

11. **ld2-omr-31031-neighbors-1.** Select every candidate directly adjacent to B0131 in the supplied connector graph.
 neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0144 = brick_000144, B0163 = brick_000163, B0124 = brick_000124, B0132 = brick_000132, B0131 = brick_000131, B0020 = brick_000020
Public evidence summary: 3 unique undirected pairs
Options: A: B0144; B: B0124; C: B0163; D: B0020; E: B0132
Reference output: {"choiceIds": ["A", "B", "E"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: pending required human review.
12. **ld2-omr-31031-graph-removal-1.** Delete B0131 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
 graph-removal; connector-graph; graph-internal; integer.
Operands: B0124 = brick_000124, B0131 = brick_000131, B0144 = brick_000144, B0163 = brick_000163, B0020 = brick_000020, B0132 = brick_000132
Public evidence summary: 3 unique undirected pairs; 6 nodes
Reference output: {"value": 5}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: pending required human review.
13. **ld2-omr-31031-neighbors-2.** Select every candidate directly adjacent to B0166 in the supplied connector graph.
 neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0166 = brick_000166, B0165 = brick_000165, B0202 = brick_000202, B0162 = brick_000162, B0094 = brick_000094
Public evidence summary: 3 unique undirected pairs
Options: A: B0202; B: B0094; C: B0162; D: B0165
Reference output: {"choiceIds": ["C", "D"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: pending required human review.
14. **ld2-omr-31031-graph-removal-2.** Delete B0166 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
 graph-removal; connector-graph; graph-internal; integer.
Operands: B0165 = brick_000165, B0162 = brick_000162, B0094 = brick_000094, B0202 = brick_000202, B0166 = brick_000166
Public evidence summary: 3 unique undirected pairs; 5 nodes
Reference output: {"value": 3}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: pending required human review.
15. **ld2-omr-31031-evidence-limit-1.** Only source geometry, connector matching and mesh intersection audits are available. Without mass, friction, material or dynamic tests, is gravitational stability established?
 evidence-limit; source-data; metacognitive; single-choice.
Operands: none
Public evidence summary: {"available": ["Original CAD", "Connector inference", "Inset-mesh intersection candidates"]}
Options: A: Not established by this evidence; B: Established
Reference output: {"choiceId": "A"}
Derivation: fixed epistemic control label
Equivalence: exact declared response
Review: machine checks passed; no human certification.

16. **ld2-omr-31031-source-step-1.** At which expanded author step does B0082 first appear in the supplied source-step table?
Source order is not a collision-free assembly proof.
source-step; source-data; procedural; single-choice.
Operands: B0001 = brick_000001, B0082 = brick_000082
Public evidence summary: 2 supplied author-step records; exact table in public JSON
Options: A: 9; B: 28; C: 45; D: 24
Reference output: {"choiceId": "C"}
Derivation: minimum public step index containing target
Equivalence: exact declared response
Review: machine checks passed; no human certification.
17. **ld2-omr-31031-source-step-2.** At which expanded author step does B0168 first appear in the supplied source-step table?
Source order is not a collision-free assembly proof.
source-step; source-data; procedural; single-choice.
Operands: B0001 = brick_000001, B0168 = brick_000168
Public evidence summary: 2 supplied author-step records; exact table in public JSON
Options: A: 93; B: 112; C: 4; D: 128
Reference output: {"choiceId": "A"}
Derivation: minimum public step index containing target
Equivalence: exact declared response
Review: machine checks passed; no human certification.
18. **ld2-omr-31031-source-sequence-1.** Reveal the supplied source-step window in author order. Actions change visibility only, without simulating insertion trajectories.
source-sequence; scene-edit; procedural; actions.
Operands: B0006 = brick_000006, B0002 = brick_000002, B0004 = brick_000004, B0005 = brick_000005, B0007 = brick_000007, B0001 = brick_000001, B0009 = brick_000009, B0003 = brick_000003, B0008 = brick_000008
Public evidence summary: 6 public actions; budget 6; {"initialFacts": ["start"], "goalFacts": ["done-6"], "absentFacts": []}
Reference output: {"actionIds": ["step-1", "step-2", "step-3", "step-4", "step-5", "step-6"]}
Derivation: public precondition solver
Equivalence: any legal sequence reaching all goals within budget; display-only semantics
Review: machine checks passed; no human certification.
19. **ld2-omr-31031-restore-instance-1.** The display copy hides B0182. Restore only that numbered instance, preserving all other instances and source poses.
restore-instance; scene-edit; procedural; actions.
Operands: B0182 = brick_000182, B0001 = brick_000001
Public evidence summary: 2 public actions; budget 1; {"initialFacts": ["missing-brick_000182", "present-brick_000001"], "goalFacts": ["present-brick_000182", "present-brick_000001"], "absentFacts": ["missing-brick_000182"]}
Reference output: {"actionIds": ["restore-B0182"]}
Derivation: public precondition solver
Equivalence: any legal sequence reaching all goals within budget; display-only semantics
Review: machine checks passed; no human certification.
20. **ld2-omr-31031-restore-instance-2.** The display copy hides B0111. Restore only that numbered instance, preserving all other instances and source poses.
restore-instance; scene-edit; procedural; actions.
Operands: B0001 = brick_000001, B0111 = brick_000111
Public evidence summary: 2 public actions; budget 1; {"initialFacts": ["missing-brick_000111", "present-brick_000001"], "goalFacts": ["present-brick_000111", "present-brick_000001"], "absentFacts": ["missing-brick_000111"]}

Reference output: {"actionIds": ["restore-B0111"]}

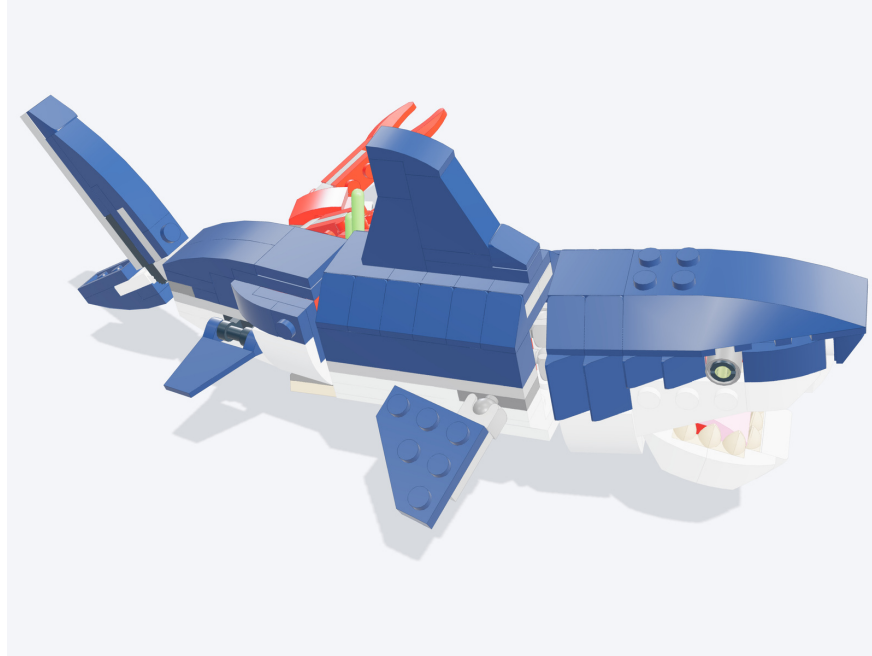
Derivation: public precondition solver

Equivalence: any legal sequence reaching all goals within budget; display-only semantics

Review: machine checks passed; no human certification.

19 31088: Deep Sea Creatures

230 source instances; 20 tasks; D2 size band. 71 type strings; 12 color codes; 4 unsupported instances; 4 intersection candidates. 98 expanded author steps. Independent reparse: passed.



Attribution. Marc Belanger [MonsieurPoulet]. CC BY 2.0.

<https://library.ldraw.org/library/omr/31088-1.mpd>

Source SHA256. c5cc0f95664d50244295783163d1ba3c73484f4b9bbfca89ada8bb1de0dbff5

Connector records. stud: 440; axle: 29; hinge: 2; fixed: 0; ball: 4

Public input lock. efb16992b05a5995159dd2148b6940c1bf6dc841eda128c442e2fbf037e121fd

Status. Source preservation does not certify stability. Reference outputs below are internal review material. Full model inputs are the versioned public JSON and numbered PNGs.

1. **ld2-omr-31088-color-1.** Identify the original color of B0176; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B0176 = brick_000176

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-31088-color-1.png; structured input = {}.

Options: A: Pearl Gold; B: Dark Blue; C: Red; D: Bright Pink

Reference output: {"choiceId": "B"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

2. **ld2-omr-31088-shape-match-1.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0176?

shape-match; visual; atomic; single-choice.

Operands: B0203 = brick_000203, B0176 = brick_000176, B0173 = brick_000173, B0158 = brick_000158, B0127 = brick_000127

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-31088-shape-match-1.png; structured input = {}.

Options: A: B0203; B: B0158; C: B0127; D: B0173

Reference output: {"choiceId": "D"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

3. **ld2-omr-31088-distance-1.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0176? distance; source-data; atomic; single-choice.

Operands: B0176 = brick_000176, B0219 = brick_000219, B0218 = brick_000218, B0134 = brick_000134

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0176": [-33.556, 26.7832, -46.9752], "B0134": [-22.2252, 26.8516, -9.6216], "B0218": [55.2016, 12, -67.3224], "B0219": [60.4518, 12, -67.413]}}

Options: A: B0134; B: B0219; C: B0218

Reference output: {"choiceId": "A"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

4. **ld2-omr-31088-color-2.** Identify the original color of B0038; numbered isolation is allowed. color; visual; atomic; single-choice.

Operands: B0038 = brick_000038

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-31088-color-2.png; structured input = {}.

Options: A: Reddish Brown; B: White; C: Trans Neon Green; D: Pearl Gold

Reference output: {"choiceId": "B"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

5. **ld2-omr-31088-distance-2.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0038? distance; source-data; atomic; single-choice.

Operands: B0038 = brick_000038, B0063 = brick_000063, B0170 = brick_000170, B0189 = brick_000189

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0038": [18.2528, 49.856, 114.62], "B0189": [22.964412, 16.89552, 11.27296], "B0170": [-48.90888, 71.1572, -64.25804], "B0063": [24.036997, 55.442263, 103.184794]}}

Options: A: B0189; B: B0170; C: B0063

Reference output: {"choiceId": "C"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

6. **ld2-omr-31088-interface-1.** Classify the supplied connector record for B0210 and B0211. This does not certify load capacity.

interface; connector-graph; metacognitive; single-choice.

Operands: B0210 = brick_000210, B0211 = brick_000211

Public evidence summary: {"connectorRecord": {"family": "axle", "aConnector": 1, "bConnector": 0}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}

Options: A: axle; B: stud; C: hinge; D: fixed; E: ball

Reference output: {"choiceId": "A"}

Derivation: public connectorRecord.family equality

Equivalence: exact declared response

Review: machine checks passed; no human certification.

7. **ld2-omr-31088-interface-2.** Classify the supplied connector record for B0089 and B0124. This does not certify load capacity.
 interface; connector-graph; metacognitive; single-choice.
Operands: B0124 = brick_000124, B0089 = brick_000089
Public evidence summary: {"connectorRecord": {"family": "ball", "aConnector": 3, "bConnector": 3}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
Options: A: hinge; B: axle; C: ball; D: fixed; E: stud
Reference output: {"choiceId": "C"}
Derivation: public connectorRecord.family equality
Equivalence: exact declared response
Review: machine checks passed; no human certification.
8. **ld2-omr-31088-interface-3.** Classify the supplied connector record for B0184 and B0180. This does not certify load capacity.
 interface; connector-graph; metacognitive; single-choice.
Operands: B0180 = brick_000180, B0184 = brick_000184
Public evidence summary: {"connectorRecord": {"family": "hinge", "aConnector": 6, "bConnector": 2}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
Options: A: ball; B: axle; C: stud; D: hinge; E: fixed
Reference output: {"choiceId": "D"}
Derivation: public connectorRecord.family equality
Equivalence: exact declared response
Review: machine checks passed; no human certification.
9. **ld2-omr-31088-interface-4.** Classify the supplied connector record for B0043 and B0044. This does not certify load capacity.
 interface; connector-graph; metacognitive; single-choice.
Operands: B0043 = brick_000043, B0044 = brick_000044
Public evidence summary: {"connectorRecord": {"family": "stud", "aConnector": 5, "bConnector": 2}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
Options: A: ball; B: axle; C: hinge; D: stud; E: fixed
Reference output: {"choiceId": "D"}
Derivation: public connectorRecord.family equality
Equivalence: exact declared response
Review: machine checks passed; no human certification.
10. **ld2-omr-31088-neighbors-1.** Select every candidate directly adjacent to B0045 in the supplied connector graph.
 neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0057 = brick_000057, B0061 = brick_000061, B0081 = brick_000081, B0039 = brick_000039, B0043 = brick_000043, B0045 = brick_000045
Public evidence summary: 4 unique undirected pairs
Options: A: B0057; B: B0061; C: B0043; D: B0081; E: B0039
Reference output: {"choiceIds": ["A", "B", "C"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: machine checks passed; no human certification.

11. **ld2-omr-31088-graph-removal-1.** Delete B0045 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
graph-removal; connector-graph; graph-internal; integer.
Operands: B0061 = brick_000061, B0043 = brick_000043, B0081 = brick_000081, B0057 = brick_000057, B0039 = brick_000039, B0045 = brick_000045
Public evidence summary: 4 unique undirected pairs; 6 nodes
Reference output: {"value": 4}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: machine checks passed; no human certification.
12. **ld2-omr-31088-neighbors-2.** Select every candidate directly adjacent to B0120 in the supplied connector graph.
neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0120 = brick_000120, B0094 = brick_000094, B0127 = brick_000127, B0117 = brick_000117, B0116 = brick_000116, B0072 = brick_000072
Public evidence summary: 3 unique undirected pairs
Options: A: B0127; B: B0116; C: B0072; D: B0117; E: B0094
Reference output: {"choiceIds": ["A", "B", "D"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: machine checks passed; no human certification.
13. **ld2-omr-31088-graph-removal-2.** Delete B0120 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
graph-removal; connector-graph; graph-internal; integer.
Operands: B0072 = brick_000072, B0116 = brick_000116, B0094 = brick_000094, B0117 = brick_000117, B0120 = brick_000120, B0127 = brick_000127
Public evidence summary: 3 unique undirected pairs; 6 nodes
Reference output: {"value": 5}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: machine checks passed; no human certification.
14. **ld2-omr-31088-coverage-1.** Which numbered instance lacks a connector definition in the coverage record, so this audit cannot establish that it is floating?
coverage; source-data; metacognitive; single-choice.
Operands: B0092 = brick_000092, B0181 = brick_000181, B0176 = brick_000176, B0038 = brick_000038
Public evidence summary: {"coverage": [{"label": "B0092", "supported": false}, {"label": "B0181", "supported": true}, {"label": "B0176", "supported": true}, {"label": "B0038", "supported": true}]}
Options: A: B0181; B: B0038; C: B0176; D: B0092
Reference output: {"choiceId": "D"}
Derivation: public supported=false record
Equivalence: exact declared response
Review: pending required human review.
15. **ld2-omr-31088-evidence-limit-1.** Only source geometry, connector matching and mesh intersection audits are available. Without mass, friction, material or dynamic tests, is gravitational stability established?
evidence-limit; source-data; metacognitive; single-choice.
Operands: none
Public evidence summary: {"available": ["Original CAD", "Connector inference", "Inset-mesh intersection candidates"]}

Options: A: Not established by this evidence; B: Established

Reference output: {"choiceId": "A"}

Derivation: fixed epistemic control label

Equivalence: exact declared response

Review: machine checks passed; no human certification.

16. **ld2-omr-31088-source-step-1.** At which expanded author step does B0184 first appear in the supplied source-step table?

Source order is not a collision-free assembly proof.

source-step; source-data; procedural; single-choice.

Operands: B0184 = brick_000184, B0001 = brick_000001

Public evidence summary: 2 supplied author-step records; exact table in public JSON

Options: A: 85; B: 91; C: 96; D: 97

Reference output: {"choiceId": "A"}

Derivation: minimum public step index containing target

Equivalence: exact declared response

Review: machine checks passed; no human certification.

17. **ld2-omr-31088-source-step-2.** At which expanded author step does B0083 first appear in the supplied source-step table?

Source order is not a collision-free assembly proof.

source-step; source-data; procedural; single-choice.

Operands: B0001 = brick_000001, B0083 = brick_000083

Public evidence summary: 2 supplied author-step records; exact table in public JSON

Options: A: 67; B: 29; C: 12; D: 44

Reference output: {"choiceId": "D"}

Derivation: minimum public step index containing target

Equivalence: exact declared response

Review: machine checks passed; no human certification.

18. **ld2-omr-31088-source-sequence-1.** Reveal the supplied source-step window in author order. Actions change visibility only, without simulating insertion trajectories.

source-sequence; scene-edit; procedural; actions.

Operands: B0001 = brick_000001, B0009 = brick_000009, B0006 = brick_000006, B0008 = brick_000008, B0005 = brick_000005, B0004 = brick_000004, B0002 = brick_000002, B0003 = brick_000003, B0007 = brick_000007

Public evidence summary: 6 public actions; budget 6; {"initialFacts": ["start"], "goalFacts": ["done-6"], "absentFacts": []}

Reference output: {"actionIds": ["step-1", "step-2", "step-3", "step-4", "step-5", "step-6"]}

Derivation: public precondition solver

Equivalence: any legal sequence reaching all goals within budget; display-only semantics

Review: machine checks passed; no human certification.

19. **ld2-omr-31088-restore-instance-1.** The display copy hides B0176. Restore only that numbered instance, preserving all other instances and source poses.

restore-instance; scene-edit; procedural; actions.

Operands: B0001 = brick_000001, B0176 = brick_000176

Public evidence summary: 2 public actions; budget 1; {"initialFacts": ["missing-brick_000176", "present-brick_000001"], "goalFacts": ["present-brick_000176", "present-brick_000001"], "absentFacts": ["missing-brick_000176"]}

Reference output: {"actionIds": ["restore-B0176"]}

Derivation: public precondition solver

Equivalence: any legal sequence reaching all goals within budget; display-only semantics

Review: machine checks passed; no human certification.

20. **ld2-omr-31088-restore-instance-2.** The display copy hides B0038. Restore only that numbered instance, preserving all other instances and source poses.

restore-instance; scene-edit; procedural; actions.

Operands: B0001 = brick_000001, B0038 = brick_000038

Public evidence summary: 2 public actions; budget 1; {"initialFacts": ["missing-brick_000038", "present-brick_000001"], "goalFacts": ["present-brick_000038", "present-brick_000001"], "absentFacts": ["missing-brick_000038"]}

Reference output: {"actionIds": ["restore-B0038"]}

Derivation: public precondition solver

Equivalence: any legal sequence reaching all goals within budget; display-only semantics

Review: machine checks passed; no human certification.

20 42004: Mini Backhoe Loader

246 source instances; 20 tasks; D2 size band. 58 type strings; 9 color codes; 0 unsupported instances; 4 intersection candidates. 86 expanded author steps. Independent reparse: passed.



Attribution. Tomas Kralicek [RabbiT_CZ]. CC BY 2.0.

<https://library.ldraw.org/library/omr/42004-1.mpd>

Source SHA256. a32b924f302a1c028c1131f78e438ad2d78e70a041999690a1e4285e851f7dbf

Connector records. stud: 3; axle: 333; hinge: 1; fixed: 16; ball: 0

Public input lock. 9554985a69f01e43f2836b31da3877455f1fa3d0f7f385b57b511812c06645d4

Status. Source preservation does not certify stability. Reference outputs below are internal review material. Full model inputs are the versioned public JSON and numbered PNGs.

1. **ld2-omr-42004-color-1.** Identify the original color of B0035; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B0035 = brick_000035

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-42004-color-1.png; structured input = {}.

Options: A: Blue; B: Tan; C: Rubber Black; D: Black

Reference output: {"choiceId": "D"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

2. **ld2-omr-42004-shape-match-1.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0035?

shape-match; visual; atomic; single-choice.

Operands: B0060 = brick_000060, B0116 = brick_000116, B0196 = brick_000196, B0035 = brick_000035, B0036 = brick_000036

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-42004-shape-match-1.png; structured input = {}.

Options: A: B0196; B: B0116; C: B0036; D: B0060

Reference output: {"choiceId": "C"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

3. **ld2-omr-42004-distance-1.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0035?

distance; source-data; atomic; single-choice.

Operands: B0233 = brick_000233, B0035 = brick_000035, B0116 = brick_000116, B0171 = brick_000171

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0035": [2, 32, 72], "B0171": [20, 57.456642, -92.888958], "B0233": [20.8, 22.3832, 119.5668], "B0116": [-18, 80, -4]}}

Options: A: B0171; B: B0116; C: B0233

Reference output: {"choiceId": "C"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

4. **ld2-omr-42004-color-2.** Identify the original color of B0191; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B0191 = brick_000191

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-42004-color-2.png; structured input = {}.

Options: A: Light Bluish Grey; B: Black; C: Yellow; D: Rubber Black

Reference output: {"choiceId": "B"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

5. **ld2-omr-42004-shape-match-2.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0191?

shape-match; visual; atomic; single-choice.

Operands: B0141 = brick_000141, B0039 = brick_000039, B0235 = brick_000235, B0191 = brick_000191, B0047 = brick_000047

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-42004-shape-match-2.png; structured input = {}.

Options: A: B0235; B: B0141; C: B0047; D: B0039

Reference output: {"choiceId": "B"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

6. **ld2-omr-42004-distance-2.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0191?

distance; source-data; atomic; single-choice.

Operands: B0003 = brick_000003, B0082 = brick_000082, B0172 = brick_000172, B0191 = brick_000191

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0191": [-16, 52.39, -28.712], "B0003": [8, 0, 24], "B0172": [32, 63.4256, -88], "B0082": [24, 16, 24]}}

Options: A: B0003; B: B0172; C: B0082

Reference output: {"choiceId": "C"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

7. **ld2-omr-42004-interface-1.** Classify the supplied connector record for B0047 and B0041. This does not certify load capacity.
- interface; connector-graph; metacognitive; single-choice.
- Operands:* B0041 = brick_000041, B0047 = brick_000047
- Public evidence summary:* {"connectorRecord": {"family": "axle", "aConnector": 0, "bConnector": 0}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
- Options:* A: axle; B: hinge; C: fixed; D: stud; E: ball
- Reference output:* {"choiceId": "A"}
- Derivation:* public connectorRecord.family equality
- Equivalence:* exact declared response
- Review:* machine checks passed; no human certification.
8. **ld2-omr-42004-interface-2.** Classify the supplied connector record for B0239 and B0240. This does not certify load capacity.
- interface; connector-graph; metacognitive; single-choice.
- Operands:* B0240 = brick_000240, B0239 = brick_000239
- Public evidence summary:* {"connectorRecord": {"family": "fixed", "aConnector": 2, "bConnector": 1}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
- Options:* A: hinge; B: stud; C: axle; D: ball; E: fixed
- Reference output:* {"choiceId": "E"}
- Derivation:* public connectorRecord.family equality
- Equivalence:* exact declared response
- Review:* machine checks passed; no human certification.
9. **ld2-omr-42004-interface-3.** Classify the supplied connector record for B0179 and B0178. This does not certify load capacity.
- interface; connector-graph; metacognitive; single-choice.
- Operands:* B0178 = brick_000178, B0179 = brick_000179
- Public evidence summary:* {"connectorRecord": {"family": "hinge", "aConnector": 1, "bConnector": 0}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
- Options:* A: stud; B: ball; C: hinge; D: fixed; E: axle
- Reference output:* {"choiceId": "C"}
- Derivation:* public connectorRecord.family equality
- Equivalence:* exact declared response
- Review:* machine checks passed; no human certification.
10. **ld2-omr-42004-interface-4.** Classify the supplied connector record for B0146 and B0145. This does not certify load capacity.
- interface; connector-graph; metacognitive; single-choice.
- Operands:* B0146 = brick_000146, B0145 = brick_000145
- Public evidence summary:* {"connectorRecord": {"family": "stud", "aConnector": 1, "bConnector": 5}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
- Options:* A: fixed; B: stud; C: axle; D: hinge; E: ball
- Reference output:* {"choiceId": "B"}
- Derivation:* public connectorRecord.family equality
- Equivalence:* exact declared response
- Review:* machine checks passed; no human certification.

11. **Id2-omr-42004-neighbors-1.** Select every candidate directly adjacent to B0192 in the supplied connector graph.
 neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0015 = brick_000015, B0191 = brick_000191, B0129 = brick_000129, B0099 = brick_000099, B0192 = brick_000192
Public evidence summary: 2 unique undirected pairs
Options: A: B0129; B: B0015; C: B0191; D: B0099
Reference output: {"choiceIds": ["A", "C"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: machine checks passed; no human certification.
12. **Id2-omr-42004-graph-removal-1.** Delete B0192 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
 graph-removal; connector-graph; graph-internal; integer.
Operands: B0099 = brick_000099, B0129 = brick_000129, B0192 = brick_000192, B0015 = brick_000015, B0191 = brick_000191
Public evidence summary: 2 unique undirected pairs; 5 nodes
Reference output: {"value": 4}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: machine checks passed; no human certification.
13. **Id2-omr-42004-neighbors-2.** Select every candidate directly adjacent to B0180 in the supplied connector graph.
 neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0176 = brick_000176, B0210 = brick_000210, B0173 = brick_000173, B0180 = brick_000180, B0011 = brick_000011
Public evidence summary: 2 unique undirected pairs
Options: A: B0176; B: B0011; C: B0173; D: B0210
Reference output: {"choiceIds": ["A", "C"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: machine checks passed; no human certification.
14. **Id2-omr-42004-graph-removal-2.** Delete B0180 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
 graph-removal; connector-graph; graph-internal; integer.
Operands: B0011 = brick_000011, B0210 = brick_000210, B0173 = brick_000173, B0176 = brick_000176, B0180 = brick_000180
Public evidence summary: 2 unique undirected pairs; 5 nodes
Reference output: {"value": 4}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: machine checks passed; no human certification.
15. **Id2-omr-42004-evidence-limit-1.** Only source geometry, connector matching and mesh intersection audits are available. Without mass, friction, material or dynamic tests, is gravitational stability established?
 evidence-limit; source-data; metacognitive; single-choice.
Operands: none
Public evidence summary: {"available": ["Original CAD", "Connector inference", "Inset-mesh intersection candidates"]}
Options: A: Established; B: Not established by this evidence
Reference output: {"choiceId": "B"}
Derivation: fixed epistemic control label
Equivalence: exact declared response
Review: machine checks passed; no human certification.

16. **Id2-omr-42004-source-step-1.** At which expanded author step does B0007 first appear in the supplied source-step table?
Source order is not a collision-free assembly proof.
source-step; source-data; procedural; single-choice.
Operands: B0001 = brick_000001, B0007 = brick_000007
Public evidence summary: 2 supplied author-step records; exact table in public JSON
Options: A: 73; B: 4; C: 11; D: 81
Reference output: {"choiceId": "B"}
Derivation: minimum public step index containing target
Equivalence: exact declared response
Review: machine checks passed; no human certification.
17. **Id2-omr-42004-source-step-2.** At which expanded author step does B0150 first appear in the supplied source-step table?
Source order is not a collision-free assembly proof.
source-step; source-data; procedural; single-choice.
Operands: B0150 = brick_000150, B0001 = brick_000001
Public evidence summary: 2 supplied author-step records; exact table in public JSON
Options: A: 80; B: 49; C: 20; D: 43
Reference output: {"choiceId": "B"}
Derivation: minimum public step index containing target
Equivalence: exact declared response
Review: pending required human review.
18. **Id2-omr-42004-source-sequence-1.** Reveal the supplied source-step window in author order. Actions change visibility only, without simulating insertion trajectories.
source-sequence; scene-edit; procedural; actions.
Operands: B0007 = brick_000007, B0010 = brick_000010, B0012 = brick_000012, B0002 = brick_000002, B0013 = brick_000013, B0003 = brick_000003, B0011 = brick_000011, B0001 = brick_000001, B0008 = brick_000008, B0004 = brick_000004, B0005 = brick_000005, B0006 = brick_000006, B0014 = brick_000014, B0009 = brick_000009, B0015 = brick_000015
Public evidence summary: 6 public actions; budget 6; {"initialFacts": ["start"], "goalFacts": ["done-6"], "absentFacts": []}
Reference output: {"actionIds": ["step-1", "step-2", "step-3", "step-4", "step-5", "step-6"]}
Derivation: public precondition solver
Equivalence: any legal sequence reaching all goals within budget; display-only semantics
Review: pending required human review.
19. **Id2-omr-42004-restore-instance-1.** The display copy hides B0035. Restore only that numbered instance, preserving all other instances and source poses.
restore-instance; scene-edit; procedural; actions.
Operands: B0001 = brick_000001, B0035 = brick_000035
Public evidence summary: 2 public actions; budget 1; {"initialFacts": ["missing-brick_000035", "present-brick_000001"], "goalFacts": ["present-brick_000035", "present-brick_000001"], "absentFacts": ["missing-brick_000035"]}
Reference output: {"actionIds": ["restore-B0035"]}
Derivation: public precondition solver
Equivalence: any legal sequence reaching all goals within budget; display-only semantics
Review: machine checks passed; no human certification.
20. **Id2-omr-42004-restore-instance-2.** The display copy hides B0191. Restore only that numbered instance, preserving all other instances and source poses.
restore-instance; scene-edit; procedural; actions.
Operands: B0001 = brick_000001, B0191 = brick_000191
Public evidence summary: 2 public actions; budget 1; {"initialFacts": ["missing-brick_000191", "present-brick_000001"], "goalFacts": ["present-brick_000191", "present-brick_000001"], "absentFacts": ["missing-brick_000191"]}

Reference output: {"actionIds": ["restore-B0191"]}

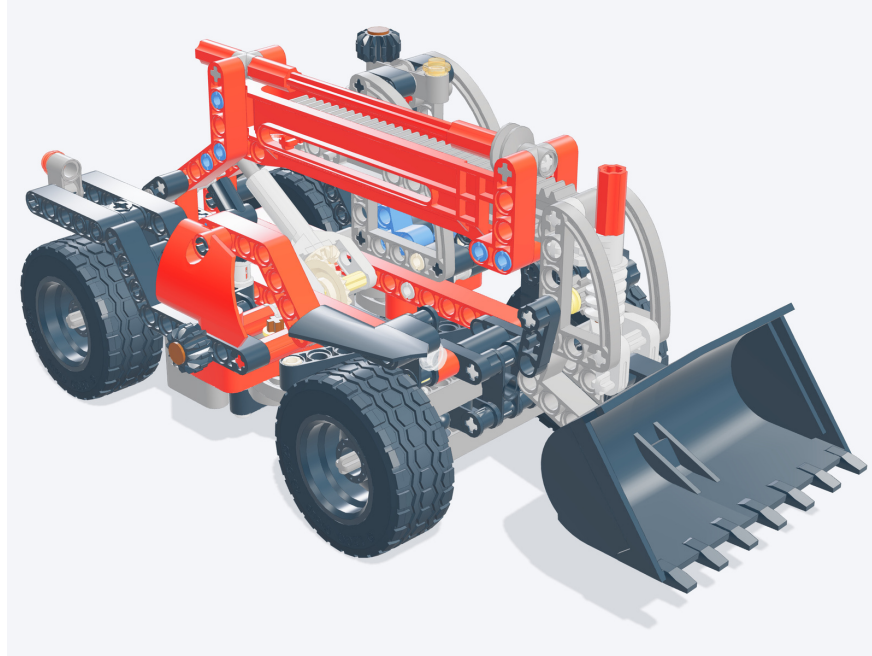
Derivation: public precondition solver

Equivalence: any legal sequence reaching all goals within budget; display-only semantics

Review: machine checks passed; no human certification.

21 42061: Telehandler

261 source instances; 20 tasks; D2 size band. 72 type strings; 14 color codes; 13 unsupported instances; 3 intersection candidates. 44 expanded author steps. Independent reparse: passed.



Attribution. Philippe Hurbain [Philo]. CC BY 2.0.

<https://library.ldraw.org/library/omr/42061-1.mpd>

Source SHA256. 040a7bd002747abebebc005d7d049c902ffd39d2b102c180458fcfdd94f369c

Connector records. stud: 5; axle: 317; hinge: 0; fixed: 16; ball: 0

Public input lock. e8bc90794cfa1e761053e31187ccc668dc86fd25fb6b674f9a29736f607c6dca

Status. Source preservation does not certify stability. Reference outputs below are internal review material. Full model inputs are the versioned public JSON and numbered PNGs.

1. **ld2-omr-42061-color-1.** Identify the original color of B0088; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B0088 = brick_000088

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-42061-color-1.png; structured input = {}.

Options: A: Blue; B: Tan; C: Rubber Black; D: Dark Bluish Grey

Reference output: {"choiceId": "A"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

2. **ld2-omr-42061-shape-match-1.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0088?

shape-match; visual; atomic; single-choice.

Operands: B0089 = brick_000089, B0088 = brick_000088, B0235 = brick_000235, B0064 = brick_000064, B0166 = brick_000166

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-42061-shape-match-1.png; structured input = {}.

Options: A: B0235; B: B0064; C: B0166; D: B0089

Reference output: {"choiceId": "D"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

3. **ld2-omr-42061-distance-1.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0088?

distance; source-data; atomic; single-choice.

Operands: B0088 = brick_000088, B0203 = brick_000203, B0194 = brick_000194, B0021 = brick_000021

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0088": [8, 92, 25.6], "B0021": [-8, 24.8, 76], "B0194": [8, 100, 129.6], "B0203": [39.9992, 48.8016, 56]}}

Options: A: B0021; B: B0194; C: B0203

Reference output: {"choiceId": "C"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

4. **ld2-omr-42061-color-2.** Identify the original color of B0104; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B0104 = brick_000104

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-42061-color-2.png; structured input = {}.

Options: A: Light Bluish Grey; B: Black; C: Red; D: Reddish Brown

Reference output: {"choiceId": "A"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

5. **ld2-omr-42061-shape-match-2.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0104?

shape-match; visual; atomic; single-choice.

Operands: B0174 = brick_000174, B0183 = brick_000183, B0151 = brick_000151, B0063 = brick_000063, B0104 = brick_000104

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-42061-shape-match-2.png; structured input = {}.

Options: A: B0063; B: B0183; C: B0151; D: B0174

Reference output: {"choiceId": "B"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

6. **ld2-omr-42061-distance-2.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0104?

distance; source-data; atomic; single-choice.

Operands: B0252 = brick_000252, B0104 = brick_000104, B0204 = brick_000204, B0134 = brick_000134

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0104": [8, 61.6, 6.4], "B0252": [40.0004, 88.9044, 48], "B0204": [39.9992, 48.8016, 88], "B0134": [-40, 50.381801, 52.818109]}}

Options: A: B0252; B: B0204; C: B0134

Reference output: {"choiceId": "A"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

7. **ld2-omr-42061-interface-1.** Classify the supplied connector record for B0237 and B0236. This does not certify load capacity.
- interface; connector-graph; metacognitive; single-choice.
- Operands:* B0237 = brick_000237, B0236 = brick_000236
- Public evidence summary:* {"connectorRecord": {"family": "axle", "aConnector": 1, "bConnector": 0}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
- Options:* A: stud; B: axle; C: fixed; D: ball; E: hinge
- Reference output:* {"choiceId": "B"}
- Derivation:* public connectorRecord.family equality
- Equivalence:* exact declared response
- Review:* machine checks passed; no human certification.
8. **ld2-omr-42061-interface-2.** Classify the supplied connector record for B0260 and B0261. This does not certify load capacity.
- interface; connector-graph; metacognitive; single-choice.
- Operands:* B0261 = brick_000261, B0260 = brick_000260
- Public evidence summary:* {"connectorRecord": {"family": "fixed", "aConnector": 2, "bConnector": 0}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
- Options:* A: hinge; B: ball; C: fixed; D: axle; E: stud
- Reference output:* {"choiceId": "C"}
- Derivation:* public connectorRecord.family equality
- Equivalence:* exact declared response
- Review:* machine checks passed; no human certification.
9. **ld2-omr-42061-interface-3.** Classify the supplied connector record for B0210 and B0211. This does not certify load capacity.
- interface; connector-graph; metacognitive; single-choice.
- Operands:* B0211 = brick_000211, B0210 = brick_000210
- Public evidence summary:* {"connectorRecord": {"family": "stud", "aConnector": 2, "bConnector": 0}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
- Options:* A: stud; B: axle; C: fixed; D: ball; E: hinge
- Reference output:* {"choiceId": "A"}
- Derivation:* public connectorRecord.family equality
- Equivalence:* exact declared response
- Review:* machine checks passed; no human certification.
10. **ld2-omr-42061-neighbors-1.** Select every candidate directly adjacent to B0130 in the supplied connector graph.
- neighbors; connector-graph; metacognitive; multiple-choice.
- Operands:* B0172 = brick_000172, B0128 = brick_000128, B0132 = brick_000132, B0130 = brick_000130, B0117 = brick_000117
- Public evidence summary:* 2 unique undirected pairs
- Options:* A: B0128; B: B0117; C: B0132; D: B0172
- Reference output:* {"choiceIds": ["A", "C"]}
- Derivation:* public incident edge endpoint set
- Equivalence:* unordered exact set; duplicate IDs invalid
- Review:* machine checks passed; no human certification.

11. **ld2-omr-42061-graph-removal-1.** Delete B0130 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
graph-removal; connector-graph; graph-internal; integer.
Operands: B0117 = brick_000117, B0172 = brick_000172, B0128 = brick_000128, B0132 = brick_000132, B0130 = brick_000130
Public evidence summary: 2 unique undirected pairs; 5 nodes
Reference output: {"value": 4}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: machine checks passed; no human certification.
12. **ld2-omr-42061-neighbors-2.** Select every candidate directly adjacent to B0075 in the supplied connector graph.
neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0026 = brick_000026, B0079 = brick_000079, B0076 = brick_000076, B0067 = brick_000067, B0075 = brick_000075
Public evidence summary: 2 unique undirected pairs
Options: A: B0067; B: B0026; C: B0076; D: B0079
Reference output: {"choiceIds": ["C", "D"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: machine checks passed; no human certification.
13. **ld2-omr-42061-graph-removal-2.** Delete B0075 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
graph-removal; connector-graph; graph-internal; integer.
Operands: B0067 = brick_000067, B0026 = brick_000026, B0076 = brick_000076, B0079 = brick_000079, B0075 = brick_000075
Public evidence summary: 2 unique undirected pairs; 5 nodes
Reference output: {"value": 4}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: machine checks passed; no human certification.
14. **ld2-omr-42061-coverage-1.** Which numbered instance lacks a connector definition in the coverage record, so this audit cannot establish that it is floating?
coverage; source-data; metacognitive; single-choice.
Operands: B0088 = brick_000088, B0066 = brick_000066, B0104 = brick_000104, B0212 = brick_000212
Public evidence summary: {"coverage": [{"label": "B0088", "supported": true}, {"label": "B0066", "supported": false}, {"label": "B0104", "supported": true}, {"label": "B0212", "supported": true}]}
Options: A: B0088; B: B0212; C: B0066; D: B0104
Reference output: {"choiceId": "C"}
Derivation: public supported=false record
Equivalence: exact declared response
Review: pending required human review.
15. **ld2-omr-42061-evidence-limit-1.** Only source geometry, connector matching and mesh intersection audits are available. Without mass, friction, material or dynamic tests, is gravitational stability established?
evidence-limit; source-data; metacognitive; single-choice.
Operands: none
Public evidence summary: {"available": ["Original CAD", "Connector inference", "Inset-mesh intersection candidates"]}
Options: A: Established; B: Not established by this evidence
Reference output: {"choiceId": "B"}
Derivation: fixed epistemic control label

Equivalence: exact declared response

Review: machine checks passed; no human certification.

16. **ld2-omr-42061-source-step-1.** At which expanded author step does B0041 first appear in the supplied source-step table?
Source order is not a collision-free assembly proof.

source-step; source-data; procedural; single-choice.

Operands: B0001 = brick_000001, B0041 = brick_000041

Public evidence summary: 2 supplied author-step records; exact table in public JSON

Options: A: 8; B: 7; C: 32; D: 5

Reference output: {"choiceId": "A"}

Derivation: minimum public step index containing target

Equivalence: exact declared response

Review: machine checks passed; no human certification.

17. **ld2-omr-42061-source-step-2.** At which expanded author step does B0245 first appear in the supplied source-step table?
Source order is not a collision-free assembly proof.

source-step; source-data; procedural; single-choice.

Operands: B0001 = brick_000001, B0245 = brick_000245

Public evidence summary: 2 supplied author-step records; exact table in public JSON

Options: A: 29; B: 43; C: 42; D: 2

Reference output: {"choiceId": "C"}

Derivation: minimum public step index containing target

Equivalence: exact declared response

Review: machine checks passed; no human certification.

18. **ld2-omr-42061-source-sequence-1.** Reveal the supplied source-step window in author order. Actions change visibility only, without simulating insertion trajectories.

source-sequence; scene-edit; procedural; actions.

Operands: B0022 = brick_000022, B0002 = brick_000002, B0031 = brick_000031, B0035 = brick_000035, B0028 = brick_000028, B0006 = brick_000006, B0024 = brick_000024, B0001 = brick_000001, B0034 = brick_000034, B0023 = brick_000023, B0007 = brick_000007, B0005 = brick_000005, B0003 = brick_000003, B0010 = brick_000010, B0025 = brick_000025, B0032 = brick_000032, B0011 = brick_000011, B0013 = brick_000013, B0012 = brick_000012, B0009 = brick_000009, B0020 = brick_000020, B0030 = brick_000030, B0019 = brick_000019, B0033 = brick_000033, B0029 = brick_000029, B0021 = brick_000021, B0017 = brick_000017, B0027 = brick_000027, B0026 = brick_000026, B0008 = brick_000008, B0016 = brick_000016, B0018 = brick_000018, B0015 = brick_000015, B0004 = brick_000004, B0014 = brick_000014

Public evidence summary: 6 public actions; budget 6; {"initialFacts": ["start"], "goalFacts": ["done-6"], "absentFacts": []}

Reference output: {"actionIds": ["step-1", "step-2", "step-3", "step-4", "step-5", "step-6"]}

Derivation: public precondition solver

Equivalence: any legal sequence reaching all goals within budget; display-only semantics

Review: machine checks passed; no human certification.

19. **ld2-omr-42061-restore-instance-1.** The display copy hides B0088. Restore only that numbered instance, preserving all other instances and source poses.

restore-instance; scene-edit; procedural; actions.

Operands: B0001 = brick_000001, B0088 = brick_000088

Public evidence summary: 2 public actions; budget 1; {"initialFacts": ["missing-brick_000088", "present-brick_000001"], "goalFacts": ["present-brick_000088", "present-brick_000001"], "absentFacts": ["missing-brick_000088"]}

Reference output: {"actionIds": ["restore-B0088"]}

Derivation: public precondition solver

Equivalence: any legal sequence reaching all goals within budget; display-only semantics

Review: machine checks passed; no human certification.

20. **ld2-omr-42061-restore-instance-2.** The display copy hides B0104. Restore only that numbered instance, preserving all other instances and source poses.
- restore-instance; scene-edit; procedural; actions.
- Operands:* B0001 = brick_000001, B0104 = brick_000104
- Public evidence summary:* 2 public actions; budget 1; {"initialFacts": ["missing-brick_000104", "present-brick_000001"], "goalFacts": ["present-brick_000104", "present-brick_000001"], "absentFacts": ["missing-brick_000104"]}
- Reference output:* {"actionIds": ["restore-B0104"]}
- Derivation:* public precondition solver
- Equivalence:* any legal sequence reaching all goals within budget; display-only semantics
- Review:* machine checks passed; no human certification.

22 31009: Small Cottage

270 source instances; 20 tasks; D2 size band. 65 type strings; 18 color codes; 8 unsupported instances; 0 intersection candidates. 79 expanded author steps. Independent reparse: passed.



Attribution. Stefan Frenz [smf]. CC BY 2.0.
<https://library.ldraw.org/omr/sets/1164>

Source SHA256. 97b83517b0e843f8071ea3e4e80fbe0c7897685a7f476d3ba83161bb1185333d

Connector records. stud: 772; axle: 3; hinge: 8; fixed: 2; ball: 0

Public input lock. 668f9531ce10e4f9cab3684f6b2297998cb509e814750d6f57aaf9bb0a21844e

Status. Source preservation does not certify stability. Reference outputs below are internal review material. Full model inputs are the versioned public JSON and numbered PNGs.

1. **ld2-31009-color-1.** Identify the original color of B0163; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B0163 = brick_000163

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-31009-color-1.png; structured input = {}.

Options: A: Trans Clear; B: Red; C: Medium Blue; D: Light Bluish Grey

Reference output: {"choiceId": "B"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

2. **ld2-31009-shape-match-1.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0163?
shape-match; visual; atomic; single-choice.

Operands: B0088 = brick_000088, B0163 = brick_000163, B0022 = brick_000022, B0118 = brick_000118, B0020 = brick_000020

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-31009-shape-match-1.png; structured input = {}.

Options: A: B0088; B: B0118; C: B0022; D: B0020

Reference output: {"choiceId": "C"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

3. **ld2-31009-distance-1.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0163?

distance; source-data; atomic; single-choice.

Operands: B0163 = brick_000163, B0145 = brick_000145, B0063 = brick_000063, B0200 = brick_000200

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0163": [4, 79.2, -16], "B0200": [-32, 96.8, 12], "B0063": [-68, 40.8, 0], "B0145": [12, 15.2, -52]}}

Options: A: B0063; B: B0200; C: B0145

Reference output: {"choiceId": "B"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

4. **ld2-31009-color-2.** Identify the original color of B0262; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B0262 = brick_000262

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-31009-color-2.png; structured input = {}.

Options: A: Trans Clear; B: Blue; C: Dark Blue; D: Light Bluish Grey

Reference output: {"choiceId": "B"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

5. **ld2-31009-distance-2.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0262?

distance; source-data; atomic; single-choice.

Operands: B0262 = brick_000262, B0235 = brick_000235, B0085 = brick_000085, B0099 = brick_000099

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0262": [-32, 28, 35.6], "B0099": [12, 67.12, -28], "B0085": [-72, 56.8, 16], "B0235": [-76, 12, 100]}}

Options: A: B0085; B: B0099; C: B0235

Reference output: {"choiceId": "A"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

6. **ld2-31009-interface-1.** Classify the supplied connector record for B0263 and B0262. This does not certify load capacity.

interface; connector-graph; metacognitive; single-choice.

Operands: B0262 = brick_000262, B0263 = brick_000263

Public evidence summary: {"connectorRecord": {"family": "axle", "aConnector": 1, "bConnector": 0}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}

Options: A: axle; B: hinge; C: stud; D: fixed; E: ball

Reference output: {"choiceId": "A"}

Derivation: public connectorRecord.family equality

Equivalence: exact declared response

Review: machine checks passed; no human certification.

7. **Id2-31009-interface-2.** Classify the supplied connector record for B0261 and B0262. This does not certify load capacity.
 interface; connector-graph; metacognitive; single-choice.
Operands: B0261 = brick_000261, B0262 = brick_000262
Public evidence summary: {"connectorRecord": {"family": "fixed", "aConnector": 1, "bConnector": 1}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
Options: A: stud; B: axle; C: ball; D: fixed; E: hinge
Reference output: {"choiceId": "D"}
Derivation: public connectorRecord.family equality
Equivalence: exact declared response
Review: machine checks passed; no human certification.
8. **Id2-31009-interface-3.** Classify the supplied connector record for B0259 and B0258. This does not certify load capacity.
 interface; connector-graph; metacognitive; single-choice.
Operands: B0259 = brick_000259, B0258 = brick_000258
Public evidence summary: {"connectorRecord": {"family": "hinge", "aConnector": 1, "bConnector": 2}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
Options: A: ball; B: fixed; C: axle; D: hinge; E: stud
Reference output: {"choiceId": "D"}
Derivation: public connectorRecord.family equality
Equivalence: exact declared response
Review: machine checks passed; no human certification.
9. **Id2-31009-interface-4.** Classify the supplied connector record for B0064 and B0074. This does not certify load capacity.
 interface; connector-graph; metacognitive; single-choice.
Operands: B0074 = brick_000074, B0064 = brick_000064
Public evidence summary: {"connectorRecord": {"family": "stud", "aConnector": 5, "bConnector": 0}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
Options: A: fixed; B: stud; C: ball; D: hinge; E: axle
Reference output: {"choiceId": "B"}
Derivation: public connectorRecord.family equality
Equivalence: exact declared response
Review: machine checks passed; no human certification.
10. **Id2-31009-neighbors-1.** Select every candidate directly adjacent to B0048 in the supplied connector graph.
 neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0165 = brick_000165, B0139 = brick_000139, B0048 = brick_000048, B0033 = brick_000033, B0057 = brick_000057
Public evidence summary: 2 unique undirected pairs
Options: A: B0165; B: B0033; C: B0057; D: B0139
Reference output: {"choiceIds": ["B", "C"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: machine checks passed; no human certification.
11. **Id2-31009-graph-removal-1.** Delete B0048 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
 graph-removal; connector-graph; graph-internal; integer.
Operands: B0139 = brick_000139, B0165 = brick_000165, B0048 = brick_000048, B0057 = brick_000057, B0033 = brick_000033

Public evidence summary: 2 unique undirected pairs; 5 nodes

Reference output: {"value": 4}

Derivation: union-find component count after public target deletion

Equivalence: exact declared response

Review: machine checks passed; no human certification.

12. **Id2-31009-neighbors-2.** Select every candidate directly adjacent to B0202 in the supplied connector graph.

neighbors; connector-graph; metacognitive; multiple-choice.

Operands: B0186 = brick_000186, B0202 = brick_000202, B0216 = brick_000216, B0195 = brick_000195, B0193 = brick_000193, B0218 = brick_000218, B0184 = brick_000184, B0217 = brick_000217, B0185 = brick_000185

Public evidence summary: 6 unique undirected pairs

Options: A: B0217; B: B0185; C: B0218; D: B0184; E: B0193; F: B0186; G: B0216; H: B0195

Reference output: {"choiceIds": ["A", "B", "C", "D", "E", "F"]}

Derivation: public incident edge endpoint set

Equivalence: unordered exact set; duplicate IDs invalid

Review: machine checks passed; no human certification.

13. **Id2-31009-graph-removal-2.** Delete B0202 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.

graph-removal; connector-graph; graph-internal; integer.

Operands: B0184 = brick_000184, B0216 = brick_000216, B0202 = brick_000202, B0218 = brick_000218, B0195 = brick_000195, B0217 = brick_000217, B0185 = brick_000185, B0193 = brick_000193, B0186 = brick_000186

Public evidence summary: 6 unique undirected pairs; 9 nodes

Reference output: {"value": 8}

Derivation: union-find component count after public target deletion

Equivalence: exact declared response

Review: machine checks passed; no human certification.

14. **Id2-31009-coverage-1.** Which numbered instance lacks a connector definition in the coverage record, so this audit cannot establish that it is floating?

coverage; source-data; metacognitive; single-choice.

Operands: B0163 = brick_000163, B0016 = brick_000016, B0068 = brick_000068, B0262 = brick_000262

Public evidence summary: {"coverage": [{"label": "B0163", "supported": true}, {"label": "B0016", "supported": true}, {"label": "B0262", "supported": true}, {"label": "B0068", "supported": false}]}

Options: A: B0016; B: B0163; C: B0068; D: B0262

Reference output: {"choiceId": "C"}

Derivation: public supported=false record

Equivalence: exact declared response

Review: pending required human review.

15. **Id2-31009-evidence-limit-1.** Only source geometry, connector matching and mesh intersection audits are available. Without mass, friction, material or dynamic tests, is gravitational stability established?

evidence-limit; source-data; metacognitive; single-choice.

Operands: none

Public evidence summary: {"available": ["Original CAD", "Connector inference", "Inset-mesh intersection candidates"]}

Options: A: Not established by this evidence; B: Established

Reference output: {"choiceId": "A"}

Derivation: fixed epistemic control label

Equivalence: exact declared response

Review: machine checks passed; no human certification.

16. **Id2-31009-source-step-1.** At which expanded author step does B0006 first appear in the supplied source-step table?
Source order is not a collision-free assembly proof.
source-step; source-data; procedural; single-choice.
Operands: B0006 = brick_000006, B0001 = brick_000001
Public evidence summary: 2 supplied author-step records; exact table in public JSON
Options: A: 25; B: 76; C: 3; D: 66
Reference output: {"choiceId": "C"}
Derivation: minimum public step index containing target
Equivalence: exact declared response
Review: machine checks passed; no human certification.
17. **Id2-31009-source-step-2.** At which expanded author step does B0107 first appear in the supplied source-step table?
Source order is not a collision-free assembly proof.
source-step; source-data; procedural; single-choice.
Operands: B0001 = brick_000001, B0107 = brick_000107
Public evidence summary: 2 supplied author-step records; exact table in public JSON
Options: A: 7; B: 6; C: 32; D: 15
Reference output: {"choiceId": "C"}
Derivation: minimum public step index containing target
Equivalence: exact declared response
Review: machine checks passed; no human certification.
18. **Id2-31009-source-sequence-1.** Reveal the supplied source-step window in author order. Actions change visibility only, without simulating insertion trajectories.
source-sequence; scene-edit; procedural; actions.
Operands: B0018 = brick_000018, B0021 = brick_000021, B0011 = brick_000011, B0022 = brick_000022, B0014 = brick_000014, B0007 = brick_000007, B0015 = brick_000015, B0020 = brick_000020, B0004 = brick_000004, B0002 = brick_000002, B0005 = brick_000005, B0013 = brick_000013, B0003 = brick_000003, B0010 = brick_000010, B0016 = brick_000016, B0001 = brick_000001, B0017 = brick_000017, B0012 = brick_000012, B0009 = brick_000009, B0019 = brick_000019, B0008 = brick_000008, B0006 = brick_000006
Public evidence summary: 6 public actions; budget 6; {"initialFacts": ["start"], "goalFacts": ["done-6"], "absentFacts": []}
Reference output: {"actionIds": ["step-1", "step-2", "step-3", "step-4", "step-5", "step-6"]}
Derivation: public precondition solver
Equivalence: any legal sequence reaching all goals within budget; display-only semantics
Review: machine checks passed; no human certification.
19. **Id2-31009-restore-instance-1.** The display copy hides B0163. Restore only that numbered instance, preserving all other instances and source poses.
restore-instance; scene-edit; procedural; actions.
Operands: B0001 = brick_000001, B0163 = brick_000163
Public evidence summary: 2 public actions; budget 1; {"initialFacts": ["missing-brick_000163", "present-brick_000001"], "goalFacts": ["present-brick_000163", "present-brick_000001"], "absentFacts": ["missing-brick_000163"]}
Reference output: {"actionIds": ["restore-B0163"]}
Derivation: public precondition solver
Equivalence: any legal sequence reaching all goals within budget; display-only semantics
Review: machine checks passed; no human certification.
20. **Id2-31009-restore-instance-2.** The display copy hides B0262. Restore only that numbered instance, preserving all other instances and source poses.
restore-instance; scene-edit; procedural; actions.
Operands: B0001 = brick_000001, B0262 = brick_000262

Public evidence summary: 2 public actions; budget 1; {"initialFacts": ["missing-brick_000262", "present-brick_000001"], "goalFacts": ["present-brick_000262", "present-brick_000001"], "absentFacts": ["missing-brick_000262"]}

Reference output: {"actionIds": ["restore-B0262"]}

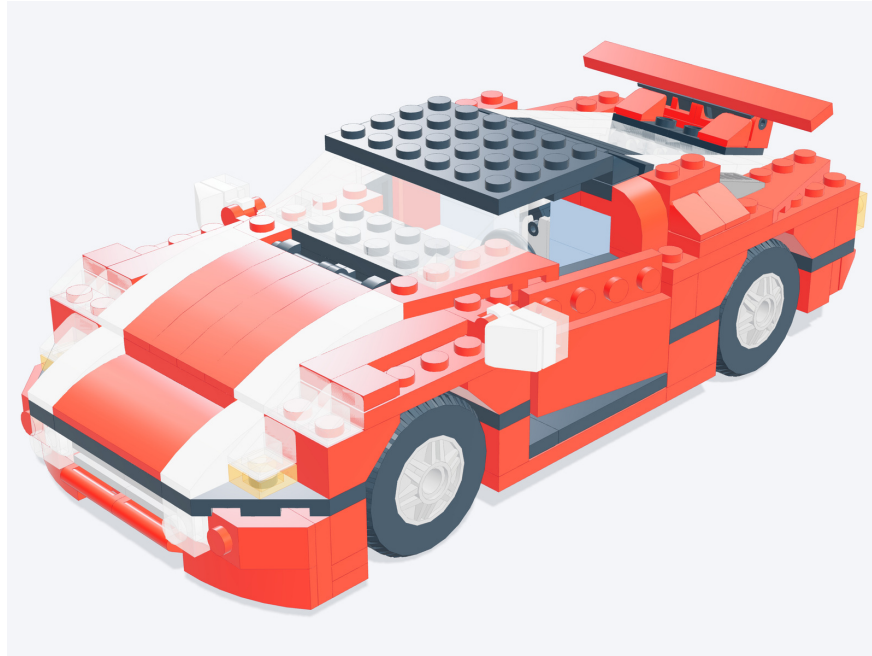
Derivation: public precondition solver

Equivalence: any legal sequence reaching all goals within budget; display-only semantics

Review: machine checks passed; no human certification.

23 5867: Super Speedster

278 source instances; 18 tasks; D2 size band. 66 type strings; 12 color codes; 1 unsupported instances; 0 intersection candidates. 0 expanded author steps. Independent reparse: passed.



Attribution. Takeshi Takahashi [RainbowDolphin]. CC BY 2.0.

<https://library.ldraw.org/omr/sets/1431>

Source SHA256. 48082e34b40a6ed5bbf2edabe9d0deb313fac7c20b1128970ba73f722b70216d

Connector records. stud: 724; axle: 32; hinge: 6; fixed: 16; ball: 0

Public input lock. 2e851c31a59697213e445bf901ecb0e4af598ed38bb69af2aee676148e0033a1

Status. Source preservation does not certify stability. Reference outputs below are internal review material. Full model inputs are the versioned public JSON and numbered PNGs.

1. **ld2-5867-color-1.** Identify the original color of B0126; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B0126 = brick_000126

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-5867-color-1.png; structured input = { }.

Options: A: Red; B: Trans Red; C: White; D: Light Bluish Grey

Reference output: {"choiceId": "B"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

2. **ld2-5867-shape-match-1.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0126?

shape-match; visual; atomic; single-choice.

Operands: B0126 = brick_000126, B0113 = brick_000113, B0067 = brick_000067, B0268 = brick_000268, B0011 = brick_000011

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-5867-shape-match-1.png; structured input = { }.

Options: A: B0113; B: B0011; C: B0268; D: B0067

Reference output: {"choiceId": "C"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

3. **ld2-5867-distance-1.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0126?

distance; source-data; atomic; single-choice.

Operands: B0127 = brick_000127, B0048 = brick_000048, B0126 = brick_000126, B0258 = brick_000258

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0126": [82.4, 21.6, -28], "B0258": [-56, 30.4, 4], "B0127": [82.4, 21.6, 28], "B0048": [-80.8, 8.8, 20]}}

Options: A: B0048; B: B0127; C: B0258

Reference output: {"choiceId": "B"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

4. **ld2-5867-color-2.** Identify the original color of B0232; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B0232 = brick_000232

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-5867-color-2.png; structured input = {}.

Options: A: Light Bluish Grey; B: Dark Bluish Grey; C: Black; D: Medium Blue

Reference output: {"choiceId": "C"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

5. **ld2-5867-shape-match-2.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0232?

shape-match; visual; atomic; single-choice.

Operands: B0086 = brick_000086, B0238 = brick_000238, B0138 = brick_000138, B0232 = brick_000232, B0114 = brick_000114

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-5867-shape-match-2.png; structured input = {}.

Options: A: B0138; B: B0238; C: B0114; D: B0086

Reference output: {"choiceId": "B"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

6. **ld2-5867-distance-2.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0232?

distance; source-data; atomic; single-choice.

Operands: B0079 = brick_000079, B0205 = brick_000205, B0232 = brick_000232, B0020 = brick_000020

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0232": [-7.648536, 24, 27.340845], "B0205": [-32, 21.6, -32], "B0020": [16, 8.8, -16], "B0079": [78.32, 0.8, 4]}}

Options: A: B0079; B: B0020; C: B0205

Reference output: {"choiceId": "B"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

7. **Id2-5867-interface-1.** Classify the supplied connector record for B0272 and B0017. This does not certify load capacity.
 interface; connector-graph; metacognitive; single-choice.
Operands: B0017 = brick_000017, B0272 = brick_000272
Public evidence summary: {"connectorRecord": {"family": "axle", "aConnector": 0, "bConnector": 1}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
Options: A: ball; B: hinge; C: stud; D: fixed; E: axle
Reference output: {"choiceId": "E"}
Derivation: public connectorRecord.family equality
Equivalence: exact declared response
Review: machine checks passed; no human certification.
8. **Id2-5867-interface-2.** Classify the supplied connector record for B0271 and B0275. This does not certify load capacity.
 interface; connector-graph; metacognitive; single-choice.
Operands: B0271 = brick_000271, B0275 = brick_000275
Public evidence summary: {"connectorRecord": {"family": "fixed", "aConnector": 2, "bConnector": 0}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
Options: A: ball; B: hinge; C: axle; D: fixed; E: stud
Reference output: {"choiceId": "D"}
Derivation: public connectorRecord.family equality
Equivalence: exact declared response
Review: machine checks passed; no human certification.
9. **Id2-5867-interface-3.** Classify the supplied connector record for B0109 and B0108. This does not certify load capacity.
 interface; connector-graph; metacognitive; single-choice.
Operands: B0108 = brick_000108, B0109 = brick_000109
Public evidence summary: {"connectorRecord": {"family": "hinge", "aConnector": 6, "bConnector": 2}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
Options: A: ball; B: stud; C: hinge; D: fixed; E: axle
Reference output: {"choiceId": "C"}
Derivation: public connectorRecord.family equality
Equivalence: exact declared response
Review: machine checks passed; no human certification.
10. **Id2-5867-interface-4.** Classify the supplied connector record for B0011 and B0033. This does not certify load capacity.
 interface; connector-graph; metacognitive; single-choice.
Operands: B0033 = brick_000033, B0011 = brick_000011
Public evidence summary: {"connectorRecord": {"family": "stud", "aConnector": 38, "bConnector": 3}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
Options: A: ball; B: stud; C: hinge; D: axle; E: fixed
Reference output: {"choiceId": "B"}
Derivation: public connectorRecord.family equality
Equivalence: exact declared response
Review: machine checks passed; no human certification.

11. **Id2-5867-neighbors-1.** Select every candidate directly adjacent to B0202 in the supplied connector graph.
- neighbors; connector-graph; metacognitive; multiple-choice.
- Operands:* B0245 = brick_000245, B0069 = brick_000069, B0021 = brick_000021, B0202 = brick_000202, B0215 = brick_000215, B0244 = brick_000244, B0056 = brick_000056
- Public evidence summary:* 4 unique undirected pairs
- Options:* A: B0056; B: B0069; C: B0245; D: B0244; E: B0021; F: B0215
- Reference output:* {"choiceIds": ["B", "C", "D", "F"]}
- Derivation:* public incident edge endpoint set
- Equivalence:* unordered exact set; duplicate IDs invalid
- Review:* machine checks passed; no human certification.
12. **Id2-5867-graph-removal-1.** Delete B0202 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
- graph-removal; connector-graph; graph-internal; integer.
- Operands:* B0056 = brick_000056, B0202 = brick_000202, B0215 = brick_000215, B0021 = brick_000021, B0069 = brick_000069, B0244 = brick_000244, B0245 = brick_000245
- Public evidence summary:* 4 unique undirected pairs; 7 nodes
- Reference output:* {"value": 6}
- Derivation:* union-find component count after public target deletion
- Equivalence:* exact declared response
- Review:* machine checks passed; no human certification.
13. **Id2-5867-neighbors-2.** Select every candidate directly adjacent to B0193 in the supplied connector graph.
- neighbors; connector-graph; metacognitive; multiple-choice.
- Operands:* B0228 = brick_000228, B0197 = brick_000197, B0193 = brick_000193, B0187 = brick_000187, B0031 = brick_000031, B0196 = brick_000196
- Public evidence summary:* 3 unique undirected pairs
- Options:* A: B0031; B: B0197; C: B0187; D: B0196; E: B0228
- Reference output:* {"choiceIds": ["C", "D", "E"]}
- Derivation:* public incident edge endpoint set
- Equivalence:* unordered exact set; duplicate IDs invalid
- Review:* machine checks passed; no human certification.
14. **Id2-5867-graph-removal-2.** Delete B0193 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
- graph-removal; connector-graph; graph-internal; integer.
- Operands:* B0196 = brick_000196, B0187 = brick_000187, B0228 = brick_000228, B0031 = brick_000031, B0197 = brick_000197, B0193 = brick_000193
- Public evidence summary:* 3 unique undirected pairs; 6 nodes
- Reference output:* {"value": 5}
- Derivation:* union-find component count after public target deletion
- Equivalence:* exact declared response
- Review:* machine checks passed; no human certification.
15. **Id2-5867-coverage-1.** Which numbered instance lacks a connector definition in the coverage record, so this audit cannot establish that it is floating?
- coverage; source-data; metacognitive; single-choice.
- Operands:* B0232 = brick_000232, B0194 = brick_000194, B0126 = brick_000126, B0109 = brick_000109
- Public evidence summary:* {"coverage": [{"label": "B0194", "supported": false}, {"label": "B0126", "supported": true}, {"label": "B0232", "supported": true}, {"label": "B0109", "supported": true}]}

Options: A: B0109; B: B0126; C: B0194; D: B0232

Reference output: {"choiceId": "C"}

Derivation: public supported=false record

Equivalence: exact declared response

Review: pending required human review.

16. **ld2-5867-evidence-limit-1.** Only source geometry, connector matching and mesh intersection audits are available. Without mass, friction, material or dynamic tests, is gravitational stability established?

evidence-limit; source-data; metacognitive; single-choice.

Operands: none

Public evidence summary: {"available": ["Original CAD", "Connector inference", "Inset-mesh intersection candidates"]}

Options: A: Not established by this evidence; B: Established

Reference output: {"choiceId": "A"}

Derivation: fixed epistemic control label

Equivalence: exact declared response

Review: machine checks passed; no human certification.

17. **ld2-5867-restore-instance-1.** The display copy hides B0126. Restore only that numbered instance, preserving all other instances and source poses.

restore-instance; scene-edit; procedural; actions.

Operands: B0001 = brick_000001, B0126 = brick_000126

Public evidence summary: 2 public actions; budget 1; {"initialFacts": ["missing-brick_000126", "present-brick_000001"], "goalFacts": ["present-brick_000126", "present-brick_000001"], "absentFacts": ["missing-brick_000126"]}

Reference output: {"actionIds": ["restore-B0126"]}

Derivation: public precondition solver

Equivalence: any legal sequence reaching all goals within budget; display-only semantics

Review: machine checks passed; no human certification.

18. **ld2-5867-restore-instance-2.** The display copy hides B0232. Restore only that numbered instance, preserving all other instances and source poses.

restore-instance; scene-edit; procedural; actions.

Operands: B0001 = brick_000001, B0232 = brick_000232

Public evidence summary: 2 public actions; budget 1; {"initialFacts": ["missing-brick_000232", "present-brick_000001"], "goalFacts": ["present-brick_000232", "present-brick_000001"], "absentFacts": ["missing-brick_000232"]}

Reference output: {"actionIds": ["restore-B0232"]}

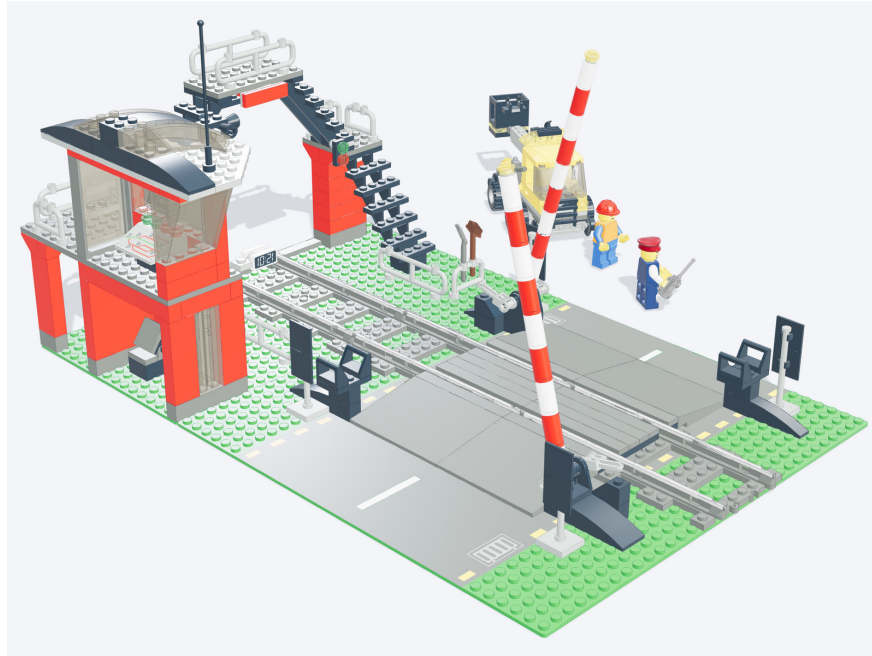
Derivation: public precondition solver

Equivalence: any legal sequence reaching all goals within budget; display-only semantics

Review: machine checks passed; no human certification.

24 10128: Train Level Crossing

336 source instances; 21 tasks; D2 size band. 111 type strings; 17 color codes; 2 unsupported instances; 1 intersection candidates. 87 expanded author steps. Independent reparse: passed.



Attribution. Robert Paciorek [bercik]. CC BY 2.0.

<https://library.ldraw.org/omr/sets/864>

Source SHA256. 7928f4e259455ec49cc02c62acb1653f7a002824c789f099845936d0a2e8a0b9

Connector records. stud: 963; axle: 89; hinge: 23; fixed: 20; ball: 0

Public input lock. 8bbe24f7936ea6cea2aea8fc7e4c0d3de2c086e4760a9f22631139beee0104e1

Status. Source preservation does not certify stability. Reference outputs below are internal review material. Full model inputs are the versioned public JSON and numbered PNGs.

1. **ld2-10128-color-1.** Identify the original color of B0075; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B0075 = brick_000075

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10128-color-1.png; structured input = {}.

Options: A: Blue; B: Trans Brown; C: Dark Blue; D: Dark Grey

Reference output: {"choiceId": "D"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

2. **ld2-10128-shape-match-1.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0075?

shape-match; visual; atomic; single-choice.

Operands: B0075 = brick_000075, B0313 = brick_000313, B0248 = brick_000248, B0125 = brick_000125, B0037 = brick_000037

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10128-shape-match-1.png; structured input = {}.

Options: A: B0125; B: B0037; C: B0248; D: B0313

Reference output: {"choiceId": "B"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

3. **ld2-10128-distance-1.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0075?

distance; source-data; atomic; single-choice.

Operands: B0310 = brick_000310, B0105 = brick_000105, B0031 = brick_000031, B0075 = brick_000075

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0075": [-72, 100, -60], "B0105": [-80, 133.6, 16], "B0310": [226.2552, 10.4, -29.7924], "B0031": [-44, 5.6, -28]}}

Options: A: B0031; B: B0310; C: B0105

Reference output: {"choiceId": "C"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

4. **ld2-10128-color-2.** Identify the original color of B0129; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B0129 = brick_000129

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10128-color-2.png; structured input = {}.

Options: A: Green; B: Dark Red; C: Light Grey; D: Trans Red

Reference output: {"choiceId": "C"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

5. **ld2-10128-shape-match-2.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0129?

shape-match; visual; atomic; single-choice.

Operands: B0129 = brick_000129, B0145 = brick_000145, B0074 = brick_000074, B0161 = brick_000161, B0144 = brick_000144

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10128-shape-match-2.png; structured input = {}.

Options: A: B0074; B: B0161; C: B0144; D: B0145

Reference output: {"choiceId": "A"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

6. **ld2-10128-distance-2.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0129?

distance; source-data; atomic; single-choice.

Operands: B0095 = brick_000095, B0082 = brick_000082, B0141 = brick_000141, B0129 = brick_000129

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0129": [56, 141.6, -108], "B0141": [24, 2.4, 96], "B0082": [120.4, 76, -84], "B0095": [-52, 104.8, 12]}}

Options: A: B0082; B: B0141; C: B0095

Reference output: {"choiceId": "A"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

7. **ld2-10128-interface-1.** Classify the supplied connector record for B0260 and B0251. This does not certify load capacity.
interface; connector-graph; metacognitive; single-choice.
Operands: B0260 = brick_000260, B0251 = brick_000251
Public evidence summary: {"connectorRecord": {"family": "axle", "aConnector": 1, "bConnector": 0}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
Options: A: axle; B: fixed; C: hinge; D: ball; E: stud
Reference output: {"choiceId": "A"}
Derivation: public connectorRecord.family equality
Equivalence: exact declared response
Review: machine checks passed; no human certification.
8. **ld2-10128-interface-2.** Classify the supplied connector record for B0327 and B0326. This does not certify load capacity.
interface; connector-graph; metacognitive; single-choice.
Operands: B0327 = brick_000327, B0326 = brick_000326
Public evidence summary: {"connectorRecord": {"family": "fixed", "aConnector": 0, "bConnector": 2}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
Options: A: stud; B: axle; C: fixed; D: ball; E: hinge
Reference output: {"choiceId": "C"}
Derivation: public connectorRecord.family equality
Equivalence: exact declared response
Review: machine checks passed; no human certification.
9. **ld2-10128-interface-3.** Classify the supplied connector record for B0328 and B0326. This does not certify load capacity.
interface; connector-graph; metacognitive; single-choice.
Operands: B0326 = brick_000326, B0328 = brick_000328
Public evidence summary: {"connectorRecord": {"family": "hinge", "aConnector": 0, "bConnector": 6}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
Options: A: ball; B: fixed; C: axle; D: stud; E: hinge
Reference output: {"choiceId": "E"}
Derivation: public connectorRecord.family equality
Equivalence: exact declared response
Review: machine checks passed; no human certification.
10. **ld2-10128-interface-4.** Classify the supplied connector record for B0249 and B0270. This does not certify load capacity.
interface; connector-graph; metacognitive; single-choice.
Operands: B0249 = brick_000249, B0270 = brick_000270
Public evidence summary: {"connectorRecord": {"family": "stud", "aConnector": 84, "bConnector": 5}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
Options: A: hinge; B: fixed; C: ball; D: axle; E: stud
Reference output: {"choiceId": "E"}
Derivation: public connectorRecord.family equality
Equivalence: exact declared response
Review: machine checks passed; no human certification.

11. **Id2-10128-neighbors-1.** Select every candidate directly adjacent to B0188 in the supplied connector graph.
neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0199 = brick_000199, B0139 = brick_000139, B0055 = brick_000055, B0190 = brick_000190, B0188 = brick_000188
Public evidence summary: 2 unique undirected pairs
Options: A: B0199; B: B0190; C: B0055; D: B0139
Reference output: {"choiceIds": ["B", "D"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: machine checks passed; no human certification.
12. **Id2-10128-graph-removal-1.** Delete B0188 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
graph-removal; connector-graph; graph-internal; integer.
Operands: B0188 = brick_000188, B0055 = brick_000055, B0190 = brick_000190, B0139 = brick_000139, B0199 = brick_000199
Public evidence summary: 2 unique undirected pairs; 5 nodes
Reference output: {"value": 4}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: machine checks passed; no human certification.
13. **Id2-10128-neighbors-2.** Select every candidate directly adjacent to B0163 in the supplied connector graph.
neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0154 = brick_000154, B0163 = brick_000163, B0165 = brick_000165, B0148 = brick_000148, B0242 = brick_000242
Public evidence summary: 4 unique undirected pairs
Options: A: B0165; B: B0154; C: B0148; D: B0242
Reference output: {"choiceIds": ["B", "C"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: machine checks passed; no human certification.
14. **Id2-10128-graph-removal-2.** Delete B0163 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
graph-removal; connector-graph; graph-internal; integer.
Operands: B0242 = brick_000242, B0165 = brick_000165, B0154 = brick_000154, B0163 = brick_000163, B0148 = brick_000148
Public evidence summary: 4 unique undirected pairs; 5 nodes
Reference output: {"value": 2}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: machine checks passed; no human certification.
15. **Id2-10128-coverage-1.** Which numbered instance lacks a connector definition in the coverage record, so this audit cannot establish that it is floating?
coverage; source-data; metacognitive; single-choice.
Operands: B0038 = brick_000038, B0128 = brick_000128, B0129 = brick_000129, B0075 = brick_000075
Public evidence summary: {"coverage": [{"label": "B0038", "supported": false}, {"label": "B0129", "supported": true}, {"label": "B0075", "supported": true}, {"label": "B0128", "supported": true}]}
Options: A: B0129; B: B0038; C: B0128; D: B0075
Reference output: {"choiceId": "B"}
Derivation: public supported=false record
Equivalence: exact declared response
Review: pending required human review.

16. **ld2-10128-evidence-limit-1.** Only source geometry, connector matching and mesh intersection audits are available. Without mass, friction, material or dynamic tests, is gravitational stability established?
evidence-limit; source-data; metacognitive; single-choice.
Operands: none
Public evidence summary: {"available": ["Original CAD", "Connector inference", "Inset-mesh intersection candidates"]}
Options: A: Established; B: Not established by this evidence
Reference output: {"choiceId": "B"}
Derivation: fixed epistemic control label
Equivalence: exact declared response
Review: machine checks passed; no human certification.
17. **ld2-10128-source-step-1.** At which expanded author step does B0189 first appear in the supplied source-step table? Source order is not a collision-free assembly proof.
source-step; source-data; procedural; single-choice.
Operands: B0001 = brick_000001, B0189 = brick_000189
Public evidence summary: 2 supplied author-step records; exact table in public JSON
Options: A: 65; B: 40; C: 13; D: 18
Reference output: {"choiceId": "B"}
Derivation: minimum public step index containing target
Equivalence: exact declared response
Review: machine checks passed; no human certification.
18. **ld2-10128-source-step-2.** At which expanded author step does B0297 first appear in the supplied source-step table? Source order is not a collision-free assembly proof.
source-step; source-data; procedural; single-choice.
Operands: B0297 = brick_000297, B0001 = brick_000001
Public evidence summary: 2 supplied author-step records; exact table in public JSON
Options: A: 80; B: 6; C: 63; D: 81
Reference output: {"choiceId": "A"}
Derivation: minimum public step index containing target
Equivalence: exact declared response
Review: machine checks passed; no human certification.
19. **ld2-10128-source-sequence-1.** Reveal the supplied source-step window in author order. Actions change visibility only, without simulating insertion trajectories.
source-sequence; scene-edit; procedural; actions.
Operands: B0021 = brick_000021, B0002 = brick_000002, B0009 = brick_000009, B0012 = brick_000012, B0005 = brick_000005, B0018 = brick_000018, B0023 = brick_000023, B0022 = brick_000022, B0003 = brick_000003, B0013 = brick_000013, B0014 = brick_000014, B0024 = brick_000024, B0016 = brick_000016, B0010 = brick_000010, B0001 = brick_000001, B0004 = brick_000004, B0006 = brick_000006, B0019 = brick_000019, B0007 = brick_000007, B0011 = brick_000011, B0020 = brick_000020, B0017 = brick_000017, B0015 = brick_000015, B0008 = brick_000008
Public evidence summary: 6 public actions; budget 6; {"initialFacts": ["start"], "goalFacts": ["done-6"], "absentFacts": []}
Reference output: {"actionIds": ["step-1", "step-2", "step-3", "step-4", "step-5", "step-6"]}
Derivation: public precondition solver
Equivalence: any legal sequence reaching all goals within budget; display-only semantics
Review: machine checks passed; no human certification.

20. **ld2-10128-restore-instance-1.** The display copy hides B0075. Restore only that numbered instance, preserving all other instances and source poses.
- restore-instance; scene-edit; procedural; actions.
- Operands:* B0001 = brick_000001, B0075 = brick_000075
- Public evidence summary:* 2 public actions; budget 1; {"initialFacts": ["missing-brick_000075", "present-brick_000001"], "goalFacts": ["present-brick_000075", "present-brick_000001"], "absentFacts": ["missing-brick_000075"]}
- Reference output:* {"actionIds": ["restore-B0075"]}
- Derivation:* public precondition solver
- Equivalence:* any legal sequence reaching all goals within budget; display-only semantics
- Review:* machine checks passed; no human certification.
21. **ld2-10128-restore-instance-2.** The display copy hides B0129. Restore only that numbered instance, preserving all other instances and source poses.
- restore-instance; scene-edit; procedural; actions.
- Operands:* B0129 = brick_000129, B0001 = brick_000001
- Public evidence summary:* 2 public actions; budget 1; {"initialFacts": ["missing-brick_000129", "present-brick_000001"], "goalFacts": ["present-brick_000129", "present-brick_000001"], "absentFacts": ["missing-brick_000129"]}
- Reference output:* {"actionIds": ["restore-B0129"]}
- Derivation:* public precondition solver
- Equivalence:* any legal sequence reaching all goals within budget; display-only semantics
- Review:* machine checks passed; no human certification.

25 10159: City Airport

592 source instances; 25 tasks; D3 size band. 160 type strings; 14 color codes; 30 unsupported instances; 26 intersection candidates. 0 expanded author steps. Independent reparse: passed.



Attribution. Zoltan Keri [kzoltan82]. CC BY 2.0.

<https://library.ldraw.org/omr/sets/660>

Source SHA256. ef0b08af1e69059813e8965d3aa8fdd6a4aadb04a21d92723cc4f9ae26bc1ec

Connector records. stud: 1109; axle: 31; hinge: 76; fixed: 62; ball: 1

Public input lock. 47fdd0a148a08b129d77bd41f06ab35dca4bb74db3c43da8359e585139153494

Status. Source preservation does not certify stability. Reference outputs below are internal review material. Full model inputs are the versioned public JSON and numbered PNGs.

1. **ld2-10159-color-1.** Identify the original color of B0132; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B0132 = brick_000132

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10159-color-1.png; structured input = {}.

Options: A: Black; B: Brown; C: Green; D: White

Reference output: {"choiceId": "D"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

2. **ld2-10159-shape-match-1.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0132?
shape-match; visual; atomic; single-choice.

Operands: B0332 = brick_000332, B0374 = brick_000374, B0131 = brick_000131, B0354 = brick_000354, B0132 = brick_000132

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10159-shape-match-1.png; structured input = {}.

Options: A: B0332; B: B0374; C: B0354; D: B0131

Reference output: {"choiceId": "D"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

3. **ld2-10159-distance-1.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0132?

distance; source-data; atomic; single-choice.

Operands: B0510 = brick_000510, B0234 = brick_000234, B0132 = brick_000132, B0138 = brick_000138

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0132": [124, 53.6, -224], "B0234": [93.534, 213.4228, -211.256], "B0138": [-116, 53.6, -224], "B0510": [255.005523, 23.458353, 71.775633]}}

Options: A: B0234; B: B0510; C: B0138

Reference output: {"choiceId": "A"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

4. **ld2-10159-color-2.** Identify the original color of B0181; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B0181 = brick_000181

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10159-color-2.png; structured input = {}.

Options: A: Trans Yellow; B: Blue; C: Trans Light Blue; D: Light Grey

Reference output: {"choiceId": "A"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

5. **ld2-10159-shape-match-2.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0181?

shape-match; visual; atomic; single-choice.

Operands: B0181 = brick_000181, B0356 = brick_000356, B0367 = brick_000367, B0530 = brick_000530, B0380 = brick_000380

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10159-shape-match-2.png; structured input = {}.

Options: A: B0380; B: B0356; C: B0530; D: B0367

Reference output: {"choiceId": "A"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

6. **ld2-10159-distance-2.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0181?

distance; source-data; atomic; single-choice.

Operands: B0228 = brick_000228, B0181 = brick_000181, B0297 = brick_000297, B0245 = brick_000245

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0181": [-4, 82.4, -232], "B0297": [288, 36, -48], "B0245": [-43.4404, 15.721947, -167.7786], "B0228": [96, 200, -224]}}

Options: A: B0297; B: B0228; C: B0245

Reference output: {"choiceId": "C"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

7. **ld2-10159-color-3.** Identify the original color of B0485; numbered isolation is allowed.
 color; visual; atomic; single-choice.
Operands: B0485 = brick_000485
Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10159-color-3.png; structured input = {}.
Options: A: Trans Red; B: Light Grey; C: Yellow; D: Red
Reference output: {"choiceId": "D"}
Derivation: source colorName by rendered B-label; source oracle only
Equivalence: exact declared response
Review: pending required human review.
8. **ld2-10159-shape-match-3.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0485?
 shape-match; visual; atomic; single-choice.
Operands: B0254 = brick_000254, B0555 = brick_000555, B0250 = brick_000250, B0425 = brick_000425, B0485 = brick_000485
Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10159-shape-match-3.png; structured input = {}.
Options: A: B0254; B: B0425; C: B0555; D: B0250
Reference output: {"choiceId": "B"}
Derivation: source partNumber equality across labeled candidates; source oracle only
Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending
Review: pending required human review.
9. **ld2-10159-distance-3.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0485?
 distance; source-data; atomic; single-choice.
Operands: B0485 = brick_000485, B0560 = brick_000560, B0372 = brick_000372, B0206 = brick_000206
Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0485": [93.602, 14.8, 1.614], "B0372": [290.4, 48, -80], "B0560": [171.7044, 22.058761, 34.162], "B0206": [112, 169.6, -204]}}
Options: A: B0206; B: B0372; C: B0560
Reference output: {"choiceId": "C"}
Derivation: Euclidean norm from public rounded centers
Equivalence: exact declared response
Review: pending required human review.
10. **ld2-10159-interface-1.** Classify the supplied connector record for B0230 and B0231. This does not certify load capacity.
 interface; connector-graph; metacognitive; single-choice.
Operands: B0230 = brick_000230, B0231 = brick_000231
Public evidence summary: {"connectorRecord": {"family": "axle", "aConnector": 1, "bConnector": 1}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
Options: A: ball; B: hinge; C: stud; D: axle; E: fixed
Reference output: {"choiceId": "D"}
Derivation: public connectorRecord.family equality
Equivalence: exact declared response
Review: machine checks passed; no human certification.
11. **ld2-10159-interface-2.** Classify the supplied connector record for B0486 and B0454. This does not certify load capacity.
 interface; connector-graph; metacognitive; single-choice.
Operands: B0486 = brick_000486, B0454 = brick_000454
Public evidence summary: {"connectorRecord": {"family": "ball", "aConnector": 4, "bConnector": 0}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
Options: A: axle; B: stud; C: ball; D: fixed; E: hinge

Reference output: {"choiceId": "C"}

Derivation: public connectorRecord.family equality

Equivalence: exact declared response

Review: machine checks passed; no human certification.

12. **ld2-10159-interface-3.** Classify the supplied connector record for B0383 and B0387. This does not certify load capacity.
interface; connector-graph; metacognitive; single-choice.

Operands: B0383 = brick_000383, B0387 = brick_000387

Public evidence summary: {"connectorRecord": {"family": "fixed", "aConnector": 1, "bConnector": 1}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}

Options: A: fixed; B: axle; C: stud; D: hinge; E: ball

Reference output: {"choiceId": "A"}

Derivation: public connectorRecord.family equality

Equivalence: exact declared response

Review: pending required human review.

13. **ld2-10159-interface-4.** Classify the supplied connector record for B0539 and B0545. This does not certify load capacity.
interface; connector-graph; metacognitive; single-choice.

Operands: B0539 = brick_000539, B0545 = brick_000545

Public evidence summary: {"connectorRecord": {"family": "hinge", "aConnector": 3, "bConnector": 1}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}

Options: A: axle; B: stud; C: hinge; D: fixed; E: ball

Reference output: {"choiceId": "C"}

Derivation: public connectorRecord.family equality

Equivalence: exact declared response

Review: pending required human review.

14. **ld2-10159-interface-5.** Classify the supplied connector record for B0089 and B0122. This does not certify load capacity.
interface; connector-graph; metacognitive; single-choice.

Operands: B0089 = brick_000089, B0122 = brick_000122

Public evidence summary: {"connectorRecord": {"family": "stud", "aConnector": 14, "bConnector": 4}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}

Options: A: hinge; B: fixed; C: axle; D: stud; E: ball

Reference output: {"choiceId": "D"}

Derivation: public connectorRecord.family equality

Equivalence: exact declared response

Review: machine checks passed; no human certification.

15. **ld2-10159-neighbors-1.** Select every candidate directly adjacent to B0079 in the supplied connector graph.
neighbors; connector-graph; metacognitive; multiple-choice.

Operands: B0283 = brick_000283, B0525 = brick_000525, B0079 = brick_000079, B0078 = brick_000078, B0081 = brick_000081, B0080 = brick_000080

Public evidence summary: 3 unique undirected pairs

Options: A: B0078; B: B0081; C: B0525; D: B0283; E: B0080

Reference output: {"choiceIds": ["A", "B", "E"]}

Derivation: public incident edge endpoint set

Equivalence: unordered exact set; duplicate IDs invalid

Review: pending required human review.

16. **ld2-10159-graph-removal-1.** Delete B0079 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
graph-removal; connector-graph; graph-internal; integer.
Operands: B0080 = brick_000080, B0283 = brick_000283, B0078 = brick_000078, B0081 = brick_000081, B0525 = brick_000525, B0079 = brick_000079
Public evidence summary: 3 unique undirected pairs; 6 nodes
Reference output: {"value": 5}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: pending required human review.
17. **ld2-10159-neighbors-2.** Select every candidate directly adjacent to B0353 in the supplied connector graph.
neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0336 = brick_000336, B0353 = brick_000353, B0112 = brick_000112, B0186 = brick_000186, B0375 = brick_000375
Public evidence summary: 2 unique undirected pairs
Options: A: B0336; B: B0375; C: B0112; D: B0186
Reference output: {"choiceIds": ["A", "B"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: machine checks passed; no human certification.
18. **ld2-10159-graph-removal-2.** Delete B0353 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
graph-removal; connector-graph; graph-internal; integer.
Operands: B0112 = brick_000112, B0375 = brick_000375, B0353 = brick_000353, B0336 = brick_000336, B0186 = brick_000186
Public evidence summary: 2 unique undirected pairs; 5 nodes
Reference output: {"value": 4}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: machine checks passed; no human certification.
19. **ld2-10159-neighbors-3.** Select every candidate directly adjacent to B0328 in the supplied connector graph.
neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0328 = brick_000328, B0185 = brick_000185, B0351 = brick_000351, B0560 = brick_000560, B0325 = brick_000325
Public evidence summary: 2 unique undirected pairs
Options: A: B0325; B: B0185; C: B0560; D: B0351
Reference output: {"choiceIds": ["A", "D"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: machine checks passed; no human certification.
20. **ld2-10159-graph-removal-3.** Delete B0328 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
graph-removal; connector-graph; graph-internal; integer.
Operands: B0325 = brick_000325, B0351 = brick_000351, B0328 = brick_000328, B0185 = brick_000185, B0560 = brick_000560
Public evidence summary: 2 unique undirected pairs; 5 nodes
Reference output: {"value": 4}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: machine checks passed; no human certification.

21. **Id2-10159-coverage-1.** Which numbered instance lacks a connector definition in the coverage record, so this audit cannot establish that it is floating?
 coverage; source-data; metacognitive; single-choice.
Operands: B0485 = brick_000485, B0238 = brick_000238, B0132 = brick_000132, B0181 = brick_000181
Public evidence summary: {"coverage": [{"label": "B0485", "supported": true}, {"label": "B0238", "supported": false}, {"label": "B0132", "supported": true}, {"label": "B0181", "supported": true}]}
Options: A: B0238; B: B0181; C: B0485; D: B0132
Reference output: {"choiceId": "A"}
Derivation: public supported=false record
Equivalence: exact declared response
Review: pending required human review.
22. **Id2-10159-evidence-limit-1.** Only source geometry, connector matching and mesh intersection audits are available. Without mass, friction, material or dynamic tests, is gravitational stability established?
 evidence-limit; source-data; metacognitive; single-choice.
Operands: none
Public evidence summary: {"available": ["Original CAD", "Connector inference", "Inset-mesh intersection candidates"]}
Options: A: Established; B: Not established by this evidence
Reference output: {"choiceId": "B"}
Derivation: fixed epistemic control label
Equivalence: exact declared response
Review: machine checks passed; no human certification.
23. **Id2-10159-restore-instance-1.** The display copy hides B0132. Restore only that numbered instance, preserving all other instances and source poses.
 restore-instance; scene-edit; procedural; actions.
Operands: B0132 = brick_000132, B0001 = brick_000001
Public evidence summary: 2 public actions; budget 1; {"initialFacts": ["missing-brick_000132", "present-brick_000001"], "goalFacts": ["present-brick_000132", "present-brick_000001"], "absentFacts": ["missing-brick_000132"]}
Reference output: {"actionIds": ["restore-B0132"]}
Derivation: public precondition solver
Equivalence: any legal sequence reaching all goals within budget; display-only semantics
Review: machine checks passed; no human certification.
24. **Id2-10159-restore-instance-2.** The display copy hides B0181. Restore only that numbered instance, preserving all other instances and source poses.
 restore-instance; scene-edit; procedural; actions.
Operands: B0181 = brick_000181, B0001 = brick_000001
Public evidence summary: 2 public actions; budget 1; {"initialFacts": ["missing-brick_000181", "present-brick_000001"], "goalFacts": ["present-brick_000181", "present-brick_000001"], "absentFacts": ["missing-brick_000181"]}
Reference output: {"actionIds": ["restore-B0181"]}
Derivation: public precondition solver
Equivalence: any legal sequence reaching all goals within budget; display-only semantics
Review: machine checks passed; no human certification.
25. **Id2-10159-restore-instance-3.** The display copy hides B0485. Restore only that numbered instance, preserving all other instances and source poses.
 restore-instance; scene-edit; procedural; actions.
Operands: B0485 = brick_000485, B0001 = brick_000001
Public evidence summary: 2 public actions; budget 1; {"initialFacts": ["missing-brick_000485", "present-brick_000001"], "goalFacts": ["present-brick_000485", "present-brick_000001"], "absentFacts": ["missing-brick_000485"]}

Reference output: {"actionIds": ["restore-B0485"]}

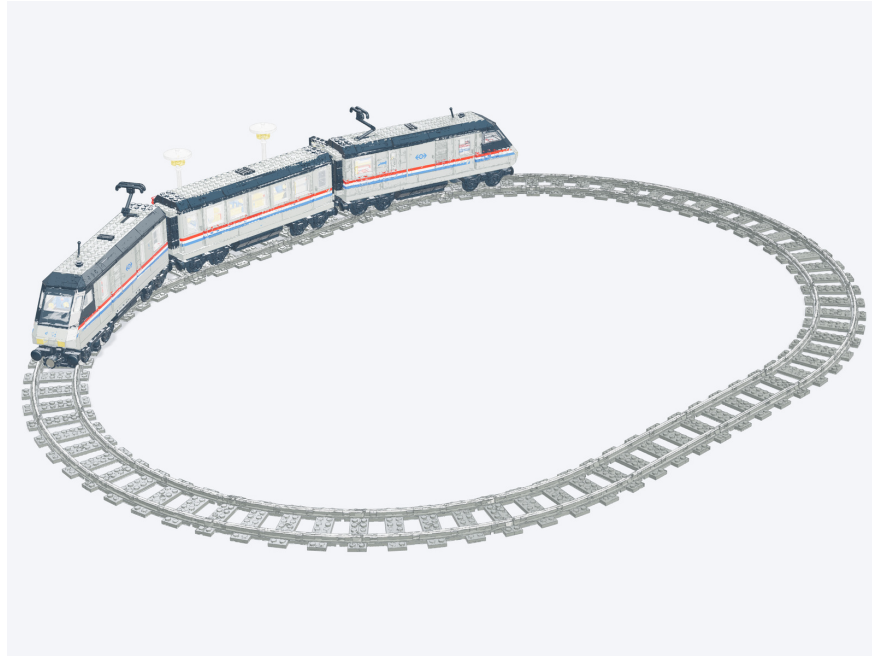
Derivation: public precondition solver

Equivalence: any legal sequence reaching all goals within budget; display-only semantics

Review: machine checks passed; no human certification.

26 10001: Metroliner

848 source instances; 34 tasks; D3 size band. 139 type strings; 10 color codes; 48 unsupported instances; 75 intersection candidates. 26 expanded author steps. Independent reparse: passed.



Attribution. Zoltan Keri [kzoltan82]. CC BY 2.0.
<https://library.ldraw.org/omr/sets/657>

Source SHA256. 684a8032488dc2c5cb72a9ef32befb1399914eb32423e5b0cd91b6acfd3ab0e3

Connector records. stud: 2248; axle: 24; hinge: 96; fixed: 24; ball: 0

Public input lock. 99cb64d39956d2ba1af371943780e3539e60539786c0938ebd1f8292099d11a0

Status. Source preservation does not certify stability. Reference outputs below are internal review material. Full model inputs are the versioned public JSON and numbered PNGs.

1. **ld2-10001-color-1.** Identify the original color of B0115; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B0115 = brick_000115

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10001-color-1.png; structured input = {}.

Options: A: Black; B: Trans Yellow; C: White; D: Trans Clear

Reference output: {"choiceId": "A"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

2. **ld2-10001-shape-match-1.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0115?

shape-match; visual; atomic; single-choice.

Operands: B0446 = brick_000446, B0710 = brick_000710, B0115 = brick_000115, B0720 = brick_000720, B0004 = brick_000004

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10001-shape-match-1.png; structured input = {}.

Options: A: B0004; B: B0710; C: B0720; D: B0446

Reference output: {"choiceId": "B"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

3. **ld2-10001-distance-1.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0115?

distance; source-data; atomic; single-choice.

Operands: B0115 = brick_000115, B0272 = brick_000272, B0050 = brick_000050, B0720 = brick_000720

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0115": [-74.436799, 20.8, 3.551199], "B0720": [343.856, 20.8, 30.516], "B0050": [-39.756, 13.2228, -92], "B0272": [-109.031199, 68.8, 0.431199]}}

Options: A: B0272; B: B0720; C: B0050

Reference output: {"choiceId": "A"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

4. **ld2-10001-color-2.** Identify the original color of B0159; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B0159 = brick_000159

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10001-color-2.png; structured input = {}.

Options: A: White; B: Light Grey; C: Black; D: Brown

Reference output: {"choiceId": "B"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

5. **ld2-10001-shape-match-2.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0159?

shape-match; visual; atomic; single-choice.

Operands: B0192 = brick_000192, B0794 = brick_000794, B0165 = brick_000165, B0163 = brick_000163, B0159 = brick_000159

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10001-shape-match-2.png; structured input = {}.

Options: A: B0192; B: B0794; C: B0165; D: B0163

Reference output: {"choiceId": "D"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

6. **ld2-10001-distance-2.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0159?

distance; source-data; atomic; single-choice.

Operands: B0350 = brick_000350, B0018 = brick_000018, B0159 = brick_000159, B0637 = brick_000637

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0159": [-157.527199, 40, 28.431199], "B0350": [-140.599199, 88, 41.751199], "B0637": [370.82, 59.2, 41.768], "B0018": [-178.751951, 1.6, 570.031873]}}

Options: A: B0637; B: B0018; C: B0350

Reference output: {"choiceId": "C"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

7. **ld2-10001-color-3.** Identify the original color of B0739; numbered isolation is allowed.
 color; visual; atomic; single-choice.
Operands: B0739 = brick_000739
Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10001-color-3.png; structured input = {}.
Options: A: Light Grey; B: Trans Yellow; C: Black; D: White
Reference output: {"choiceId": "C"}
Derivation: source colorName by rendered B-label; source oracle only
Equivalence: exact declared response
Review: pending required human review.
8. **ld2-10001-shape-match-3.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0739?
 shape-match; visual; atomic; single-choice.
Operands: B0739 = brick_000739, B0712 = brick_000712, B0776 = brick_000776, B0546 = brick_000546, B0023 = brick_000023
Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10001-shape-match-3.png; structured input = {}.
Options: A: B0546; B: B0712; C: B0776; D: B0023
Reference output: {"choiceId": "B"}
Derivation: source partNumber equality across labeled candidates; source oracle only
Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending
Review: pending required human review.
9. **ld2-10001-distance-3.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0739?
 distance; source-data; atomic; single-choice.
Operands: B0816 = brick_000816, B0474 = brick_000474, B0739 = brick_000739, B0770 = brick_000770
Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0739": [275.744, 23.2, -21.2716], "B0816": [316.144, 59.2, -10.808], "B0474": [64, 40, -20], "B0770": [274.873286, 99.670801, -5.26347]}}
Options: A: B0816; B: B0770; C: B0474
Reference output: {"choiceId": "A"}
Derivation: Euclidean norm from public rounded centers
Equivalence: exact declared response
Review: machine checks passed; no human certification.
10. **ld2-10001-color-4.** Identify the original color of B0826; numbered isolation is allowed.
 color; visual; atomic; single-choice.
Operands: B0826 = brick_000826
Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10001-color-4.png; structured input = {}.
Options: A: Trans Clear; B: Black; C: White; D: Brown
Reference output: {"choiceId": "B"}
Derivation: source colorName by rendered B-label; source oracle only
Equivalence: exact declared response
Review: pending required human review.
11. **ld2-10001-shape-match-4.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0826?
 shape-match; visual; atomic; single-choice.
Operands: B0826 = brick_000826, B0079 = brick_000079, B0121 = brick_000121, B0581 = brick_000581, B0137 = brick_000137
Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10001-shape-match-4.png; structured input = {}.
Options: A: B0137; B: B0079; C: B0121; D: B0581
Reference output: {"choiceId": "C"}
Derivation: source partNumber equality across labeled candidates; source oracle only
Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending
Review: pending required human review.

12. **ld2-10001-distance-4.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0826?
distance; source-data; atomic; single-choice.
Operands: B0826 = brick_000826, B0373 = brick_000373, B0666 = brick_000666, B0599 = brick_000599
Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0826": [230.90065, 19.05, -16.2808], "B0373": [208, 56, 0], "B0666": [369.2836, 78.4, -1.4436], "B0599": [254.82, 40, 11.724]}}
- Options:* A: B0373; B: B0666; C: B0599
Reference output: {"choiceId": "C"}
Derivation: Euclidean norm from public rounded centers
Equivalence: exact declared response
Review: pending required human review.
13. **ld2-10001-interface-1.** Classify the supplied connector record for B0476 and B0477. This does not certify load capacity.
interface; connector-graph; metacognitive; single-choice.
Operands: B0477 = brick_000477, B0476 = brick_000476
Public evidence summary: {"connectorRecord": {"family": "axle", "aConnector": 0, "bConnector": 0}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
Options: A: hinge; B: ball; C: fixed; D: stud; E: axle
Reference output: {"choiceId": "E"}
Derivation: public connectorRecord.family equality
Equivalence: exact declared response
Review: machine checks passed; no human certification.
14. **ld2-10001-interface-2.** Classify the supplied connector record for B0101 and B0103. This does not certify load capacity.
interface; connector-graph; metacognitive; single-choice.
Operands: B0101 = brick_000101, B0103 = brick_000103
Public evidence summary: {"connectorRecord": {"family": "fixed", "aConnector": 3, "bConnector": 1}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
Options: A: fixed; B: hinge; C: stud; D: axle; E: ball
Reference output: {"choiceId": "A"}
Derivation: public connectorRecord.family equality
Equivalence: exact declared response
Review: pending required human review.
15. **ld2-10001-interface-3.** Classify the supplied connector record for B0840 and B0845. This does not certify load capacity.
interface; connector-graph; metacognitive; single-choice.
Operands: B0845 = brick_000845, B0840 = brick_000840
Public evidence summary: {"connectorRecord": {"family": "hinge", "aConnector": 0, "bConnector": 6}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
Options: A: stud; B: hinge; C: fixed; D: ball; E: axle
Reference output: {"choiceId": "B"}
Derivation: public connectorRecord.family equality
Equivalence: exact declared response
Review: pending required human review.

16. **ld2-10001-interface-4.** Classify the supplied connector record for B0592 and B0787. This does not certify load capacity.
 interface; connector-graph; metacognitive; single-choice.
Operands: B0592 = brick_000592, B0787 = brick_000787
Public evidence summary: {"connectorRecord": {"family": "stud", "aConnector": 225, "bConnector": 7}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
Options: A: stud; B: fixed; C: hinge; D: ball; E: axle
Reference output: {"choiceId": "A"}
Derivation: public connectorRecord.family equality
Equivalence: exact declared response
Review: pending required human review.
17. **ld2-10001-neighbors-1.** Select every candidate directly adjacent to B0058 in the supplied connector graph.
 neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0484 = brick_000484, B0053 = brick_000053, B0052 = brick_000052, B0058 = brick_000058, B0076 = brick_000076, B0653 = brick_000653
Public evidence summary: 3 unique undirected pairs
Options: A: B0052; B: B0653; C: B0076; D: B0053; E: B0484
Reference output: {"choiceIds": ["A", "C", "D"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: pending required human review.
18. **ld2-10001-graph-removal-1.** Delete B0058 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
 graph-removal; connector-graph; graph-internal; integer.
Operands: B0058 = brick_000058, B0053 = brick_000053, B0484 = brick_000484, B0052 = brick_000052, B0076 = brick_000076, B0653 = brick_000653
Public evidence summary: 3 unique undirected pairs; 6 nodes
Reference output: {"value": 5}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: pending required human review.
19. **ld2-10001-neighbors-2.** Select every candidate directly adjacent to B0734 in the supplied connector graph.
 neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0233 = brick_000233, B0733 = brick_000733, B0729 = brick_000729, B0013 = brick_000013, B0734 = brick_000734, B0731 = brick_000731
Public evidence summary: 3 unique undirected pairs
Options: A: B0013; B: B0733; C: B0729; D: B0233; E: B0731
Reference output: {"choiceIds": ["B", "C", "E"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: pending required human review.
20. **ld2-10001-graph-removal-2.** Delete B0734 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
 graph-removal; connector-graph; graph-internal; integer.
Operands: B0731 = brick_000731, B0733 = brick_000733, B0013 = brick_000013, B0734 = brick_000734, B0233 = brick_000233, B0729 = brick_000729

Public evidence summary: 3 unique undirected pairs; 6 nodes

Reference output: {"value": 5}

Derivation: union-find component count after public target deletion

Equivalence: exact declared response

Review: pending required human review.

21. **ld2-10001-neighbors-3.** Select every candidate directly adjacent to B0828 in the supplied connector graph.

neighbors; connector-graph; metacognitive; multiple-choice.

Operands: B0836 = brick_000836, B0677 = brick_000677, B0833 = brick_000833, B0834 = brick_000834, B0549 = brick_000549, B0828 = brick_000828

Public evidence summary: 3 unique undirected pairs

Options: A: B0836; B: B0833; C: B0834; D: B0677; E: B0549

Reference output: {"choiceIds": ["A", "B", "C"]}

Derivation: public incident edge endpoint set

Equivalence: unordered exact set; duplicate IDs invalid

Review: pending required human review.

22. **ld2-10001-graph-removal-3.** Delete B0828 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.

graph-removal; connector-graph; graph-internal; integer.

Operands: B0677 = brick_000677, B0834 = brick_000834, B0549 = brick_000549, B0833 = brick_000833, B0836 = brick_000836, B0828 = brick_000828

Public evidence summary: 3 unique undirected pairs; 6 nodes

Reference output: {"value": 5}

Derivation: union-find component count after public target deletion

Equivalence: exact declared response

Review: pending required human review.

23. **ld2-10001-neighbors-4.** Select every candidate directly adjacent to B0447 in the supplied connector graph.

neighbors; connector-graph; metacognitive; multiple-choice.

Operands: B0593 = brick_000593, B0447 = brick_000447, B0442 = brick_000442, B0444 = brick_000444, B0012 = brick_000012, B0445 = brick_000445, B0439 = brick_000439, B0440 = brick_000440

Public evidence summary: 5 unique undirected pairs

Options: A: B0445; B: B0444; C: B0442; D: B0593; E: B0012; F: B0439; G: B0440

Reference output: {"choiceIds": ["A", "B", "C", "F", "G"]}

Derivation: public incident edge endpoint set

Equivalence: unordered exact set; duplicate IDs invalid

Review: pending required human review.

24. **ld2-10001-graph-removal-4.** Delete B0447 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.

graph-removal; connector-graph; graph-internal; integer.

Operands: B0440 = brick_000440, B0445 = brick_000445, B0439 = brick_000439, B0444 = brick_000444, B0442 = brick_000442, B0447 = brick_000447, B0012 = brick_000012, B0593 = brick_000593

Public evidence summary: 5 unique undirected pairs; 8 nodes

Reference output: {"value": 7}

Derivation: union-find component count after public target deletion

Equivalence: exact declared response

Review: pending required human review.

25. **Id2-10001-coverage-1.** Which numbered instance lacks a connector definition in the coverage record, so this audit cannot establish that it is floating?
 coverage; source-data; metacognitive; single-choice.
Operands: B0739 = brick_000739, B0159 = brick_000159, B0115 = brick_000115, B0092 = brick_000092
Public evidence summary: {"coverage": [{"label": "B0739", "supported": true}, {"label": "B0115", "supported": true}, {"label": "B0159", "supported": true}, {"label": "B0092", "supported": false}]}
Options: A: B0739; B: B0159; C: B0092; D: B0115
Reference output: {"choiceId": "C"}
Derivation: public supported=false record
Equivalence: exact declared response
Review: pending required human review.
26. **Id2-10001-evidence-limit-1.** Only source geometry, connector matching and mesh intersection audits are available. Without mass, friction, material or dynamic tests, is gravitational stability established?
 evidence-limit; source-data; metacognitive; single-choice.
Operands: none
Public evidence summary: {"available": ["Original CAD", "Connector inference", "Inset-mesh intersection candidates"]}
Options: A: Established; B: Not established by this evidence
Reference output: {"choiceId": "B"}
Derivation: fixed epistemic control label
Equivalence: exact declared response
Review: machine checks passed; no human certification.
27. **Id2-10001-source-step-1.** At which expanded author step does B0037 first appear in the supplied source-step table? Source order is not a collision-free assembly proof.
 source-step; source-data; procedural; single-choice.
Operands: B0001 = brick_000001, B0037 = brick_000037
Public evidence summary: 2 supplied author-step records; exact table in public JSON
Options: A: 4; B: 3; C: 2; D: 1
Reference output: {"choiceId": "A"}
Derivation: minimum public step index containing target
Equivalence: exact declared response
Review: pending required human review.
28. **Id2-10001-source-step-2.** At which expanded author step does B0079 first appear in the supplied source-step table? Source order is not a collision-free assembly proof.
 source-step; source-data; procedural; single-choice.
Operands: B0079 = brick_000079, B0001 = brick_000001
Public evidence summary: 2 supplied author-step records; exact table in public JSON
Options: A: 25; B: 7; C: 26; D: 13
Reference output: {"choiceId": "B"}
Derivation: minimum public step index containing target
Equivalence: exact declared response
Review: pending required human review.
29. **Id2-10001-source-step-3.** At which expanded author step does B0052 first appear in the supplied source-step table? Source order is not a collision-free assembly proof.
 source-step; source-data; procedural; single-choice.
Operands: B0052 = brick_000052, B0001 = brick_000001
Public evidence summary: 2 supplied author-step records; exact table in public JSON

Options: A: 1; B: 5; C: 4; D: 3

Reference output: {"choiceId": "B"}

Derivation: minimum public step index containing target

Equivalence: exact declared response

Review: pending required human review.

30. **ld2-10001-source-step-4.** At which expanded author step does B0104 first appear in the supplied source-step table? Source order is not a collision-free assembly proof.

source-step; source-data; procedural; single-choice.

Operands: B0104 = brick_000104, B0001 = brick_000001

Public evidence summary: 2 supplied author-step records; exact table in public JSON

Options: A: 8; B: 9; C: 18; D: 6

Reference output: {"choiceId": "B"}

Derivation: minimum public step index containing target

Equivalence: exact declared response

Review: pending required human review.

31. **ld2-10001-source-sequence-1.** Reveal the supplied source-step window in author order. Actions change visibility only, without simulating insertion trajectories.

source-sequence; scene-edit; procedural; actions.

Operands: B0021 = brick_000021, B0046 = brick_000046, B0059 = brick_000059, B0073 = brick_000073, B0066 = brick_000066, B0078 = brick_000078, B0038 = brick_000038, B0003 = brick_000003, B0029 = brick_000029, B0005 = brick_000005, B0024 = brick_000024, B0012 = brick_000012, B0036 = brick_000036, B0014 = brick_000014, B0034 = brick_000034, B0065 = brick_000065, B0067 = brick_000067, B0020 = brick_000020, B0060 = brick_000060, B0007 = brick_000007, B0011 = brick_000011, B0040 = brick_000040, B0070 = brick_000070, B0010 = brick_000010, B0042 = brick_000042, B0015 = brick_000015, B0023 = brick_000023, B0043 = brick_000043, B0049 = brick_000049, B0008 = brick_000008, B0063 = brick_000063, B0045 = brick_000045, B0033 = brick_000033, B0002 = brick_000002, B0064 = brick_000064, B0051 = brick_000051, B0035 = brick_000035, B0047 = brick_000047, B0072 = brick_000072, B0048 = brick_000048, B0006 = brick_000006, B0053 = brick_000053, B0018 = brick_000018, B0077 = brick_000077, B0013 = brick_000013, B0016 = brick_000016, B0030 = brick_000030, B0069 = brick_000069, B0041 = brick_000041, B0052 = brick_000052, B0056 = brick_000056, B0058 = brick_000058, B0031 = brick_000031, B0032 = brick_000032, B0004 = brick_000004, B0074 = brick_000074, B0050 = brick_000050, B0037 = brick_000037, B0009 = brick_000009, B0068 = brick_000068, B0022 = brick_000022, B0062 = brick_000062, B0057 = brick_000057, B0061 = brick_000061, B0019 = brick_000019, B0001 = brick_000001, B0017 = brick_000017, B0075 = brick_000075, B0076 = brick_000076, B0039 = brick_000039, B0025 = brick_000025, B0028 = brick_000028, B0044 = brick_000044, B0071 = brick_000071, B0026 = brick_000026, B0054 = brick_000054, B0027 = brick_000027, B0055 = brick_000055

Public evidence summary: 6 public actions; budget 6; {"initialFacts": ["start"], "goalFacts": ["done-6"], "absentFacts": []}

Reference output: {"actionIds": ["step-1", "step-2", "step-3", "step-4", "step-5", "step-6"]}

Derivation: public precondition solver

Equivalence: any legal sequence reaching all goals within budget; display-only semantics

Review: pending required human review.

32. **ld2-10001-restore-instance-1.** The display copy hides B0115. Restore only that numbered instance, preserving all other instances and source poses.

restore-instance; scene-edit; procedural; actions.

Operands: B0001 = brick_000001, B0115 = brick_000115

Public evidence summary: 2 public actions; budget 1; {"initialFacts": ["missing-brick_000115", "present-brick_000001"], "goalFacts": ["present-brick_000115", "present-brick_000001"], "absentFacts": ["missing-brick_000115"]}

Reference output: {"actionIds": ["restore-B0115"]}

Derivation: public precondition solver

Equivalence: any legal sequence reaching all goals within budget; display-only semantics

Review: pending required human review.

33. **ld2-10001-restore-instance-2.** The display copy hides B0159. Restore only that numbered instance, preserving all other instances and source poses.
- restore-instance; scene-edit; procedural; actions.
- Operands:* B0159 = brick_000159, B0001 = brick_000001
- Public evidence summary:* 2 public actions; budget 1; {"initialFacts": ["missing-brick_000159", "present-brick_000001"], "goalFacts": ["present-brick_000159", "present-brick_000001"], "absentFacts": ["missing-brick_000159"]}
- Reference output:* {"actionIds": ["restore-B0159"]}
- Derivation:* public precondition solver
- Equivalence:* any legal sequence reaching all goals within budget; display-only semantics
- Review:* pending required human review.
34. **ld2-10001-restore-instance-3.** The display copy hides B0739. Restore only that numbered instance, preserving all other instances and source poses.
- restore-instance; scene-edit; procedural; actions.
- Operands:* B0001 = brick_000001, B0739 = brick_000739
- Public evidence summary:* 2 public actions; budget 1; {"initialFacts": ["missing-brick_000739", "present-brick_000001"], "goalFacts": ["present-brick_000739", "present-brick_000001"], "absentFacts": ["missing-brick_000739"]}
- Reference output:* {"actionIds": ["restore-B0739"]}
- Derivation:* public precondition solver
- Equivalence:* any legal sequence reaching all goals within budget; display-only semantics
- Review:* pending required human review.

27 10269: Harley-Davidson Fat Boy

1029 source instances; 34 tasks; D4 size band. 225 type strings; 16 color codes; 12 unsupported instances; 40 intersection candidates. 340 expanded author steps. Independent reparse: passed.



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<https://library.ldraw.org/library/omr/10269-1.mpd>

Source SHA256. 777f57f34cc8f4bddcee0ff94d81bfefd615b7ad4e92ab668cedb66d4b29f476

Connector records. stud: 1638; axle: 437; hinge: 50; fixed: 0; ball: 4

Public input lock. 25ba21e39ed4d4933199c3b30ecf244b1b5edefb976b26b72dd1dc3c76bf89ff

Status. Source preservation does not certify stability. Reference outputs below are internal review material. Full model inputs are the versioned public JSON and numbered PNGs.

1. **ld2-omr-10269-color-1.** Identify the original color of B0802; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B0802 = brick_000802

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-10269-color-1.png; structured input = {}.

Options: A: Reddish Brown; B: Rubber Black; C: Light Bluish Grey; D: Sand Blue

Reference output: {"choiceId": "C"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

2. **ld2-omr-10269-shape-match-1.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0802?

shape-match; visual; atomic; single-choice.

Operands: B0136 = brick_000136, B0636 = brick_000636, B0979 = brick_000979, B0056 = brick_000056, B0802 = brick_000802

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-10269-shape-match-1.png; structured input = {}.

Options: A: B0056; B: B0979; C: B0636; D: B0136

Reference output: {"choiceId": "D"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

3. **ld2-omr-10269-distance-1.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0802?

distance; source-data; atomic; single-choice.

Operands: B0407 = brick_000407, B0796 = brick_000796, B0950 = brick_000950, B0802 = brick_000802

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0802": [32.8, 64.264, -107.36], "B0407": [24, 58.380604, -65.432031], "B0950": [12, 149.6982, 72.237453], "B0796": [-48, 29.1214, 10.6146]}}

Options: A: B0796; B: B0950; C: B0407

Reference output: {"choiceId": "C"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

4. **ld2-omr-10269-color-2.** Identify the original color of B0626; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B0626 = brick_000626

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-10269-color-2.png; structured input = {}.

Options: A: Yellow; B: Trans Red; C: Dark Bluish Grey; D: Dark Red

Reference output: {"choiceId": "A"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

5. **ld2-omr-10269-shape-match-2.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0626?

shape-match; visual; atomic; single-choice.

Operands: B0970 = brick_000970, B0626 = brick_000626, B0359 = brick_000359, B0405 = brick_000405, B0474 = brick_000474

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-10269-shape-match-2.png; structured input = {}.

Options: A: B0359; B: B0405; C: B0970; D: B0474

Reference output: {"choiceId": "D"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

6. **ld2-omr-10269-distance-2.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0626?

distance; source-data; atomic; single-choice.

Operands: B0492 = brick_000492, B0911 = brick_000911, B0626 = brick_000626, B0388 = brick_000388

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0626": [0, 133.4356, -8.7248], "B0388": [24, 26.5669, -98.571706], "B0911": [12, 44.1108, 118.6532], "B0492": [20, 95.7316, -99.5424]}}

Options: A: B0911; B: B0492; C: B0388

Reference output: {"choiceId": "B"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

7. **ld2-omr-10269-color-3.** Identify the original color of B0487; numbered isolation is allowed.
 color; visual; atomic; single-choice.
Operands: B0487 = brick_000487
Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-10269-color-3.png; structured input = {}.
Options: A: Dark Red; B: Black; C: Light Bluish Grey; D: Blue
Reference output: {"choiceId": "A"}
Derivation: source colorName by rendered B-label; source oracle only
Equivalence: exact declared response
Review: pending required human review.
8. **ld2-omr-10269-shape-match-3.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0487?
 shape-match; visual; atomic; single-choice.
Operands: B0255 = brick_000255, B0586 = brick_000586, B0483 = brick_000483, B0487 = brick_000487, B0608 = brick_000608
Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-10269-shape-match-3.png; structured input = {}.
Options: A: B0586; B: B0608; C: B0255; D: B0483
Reference output: {"choiceId": "D"}
Derivation: source partNumber equality across labeled candidates; source oracle only
Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending
Review: pending required human review.
9. **ld2-omr-10269-distance-3.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0487?
 distance; source-data; atomic; single-choice.
Operands: B0242 = brick_000242, B0487 = brick_000487, B0865 = brick_000865, B0317 = brick_000317
Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0487": [4, 108.5316, -129.1424], "B0317": [8, 95.208418, 52.719016], "B0242": [0, 98.6624, 21.9632], "B0865": [-24, 141.1188, 61.0124]}}
Options: A: B0242; B: B0865; C: B0317
Reference output: {"choiceId": "A"}
Derivation: Euclidean norm from public rounded centers
Equivalence: exact declared response
Review: machine checks passed; no human certification.
10. **ld2-omr-10269-color-4.** Identify the original color of B0033; numbered isolation is allowed.
 color; visual; atomic; single-choice.
Operands: B0033 = brick_000033
Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-10269-color-4.png; structured input = {}.
Options: A: Dark Bluish Grey; B: Red; C: Light Bluish Grey; D: Trans Orange
Reference output: {"choiceId": "C"}
Derivation: source colorName by rendered B-label; source oracle only
Equivalence: exact declared response
Review: pending required human review.
11. **ld2-omr-10269-shape-match-4.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0033?
 shape-match; visual; atomic; single-choice.
Operands: B0468 = brick_000468, B0033 = brick_000033, B0685 = brick_000685, B0314 = brick_000314, B0990 = brick_000990
Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-10269-shape-match-4.png; structured input = {}.
Options: A: B0685; B: B0468; C: B0314; D: B0990

Reference output: {"choiceId": "B"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

12. **ld2-omr-10269-distance-4.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0033?

distance; source-data; atomic; single-choice.

Operands: B0003 = brick_000003, B0147 = brick_000147, B0374 = brick_000374, B0033 = brick_000033

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0033": [20, 28, 12], "B0003": [12, 21.6, -60], "B0374": [-8, 120.2692, 26.2124], "B0147": [-12, 34.4, 28]}}

Options: A: B0003; B: B0374; C: B0147

Reference output: {"choiceId": "C"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

13. **ld2-omr-10269-interface-1.** Classify the supplied connector record for B0090 and B0091. This does not certify load capacity.

interface; connector-graph; metacognitive; single-choice.

Operands: B0091 = brick_000091, B0090 = brick_000090

Public evidence summary: {"connectorRecord": {"family": "axle", "aConnector": 22, "bConnector": 1}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}

Options: A: axle; B: fixed; C: stud; D: ball; E: hinge

Reference output: {"choiceId": "A"}

Derivation: public connectorRecord.family equality

Equivalence: exact declared response

Review: machine checks passed; no human certification.

14. **ld2-omr-10269-interface-2.** Classify the supplied connector record for B0045 and B0433. This does not certify load capacity.

interface; connector-graph; metacognitive; single-choice.

Operands: B0433 = brick_000433, B0045 = brick_000045

Public evidence summary: {"connectorRecord": {"family": "ball", "aConnector": 3, "bConnector": 0}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}

Options: A: ball; B: fixed; C: axle; D: stud; E: hinge

Reference output: {"choiceId": "A"}

Derivation: public connectorRecord.family equality

Equivalence: exact declared response

Review: machine checks passed; no human certification.

15. **ld2-omr-10269-interface-3.** Classify the supplied connector record for B0384 and B0383. This does not certify load capacity.

interface; connector-graph; metacognitive; single-choice.

Operands: B0383 = brick_000383, B0384 = brick_000384

Public evidence summary: {"connectorRecord": {"family": "hinge", "aConnector": 0, "bConnector": 1}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}

Options: A: fixed; B: hinge; C: ball; D: stud; E: axle

Reference output: {"choiceId": "B"}

Derivation: public connectorRecord.family equality

Equivalence: exact declared response

Review: pending required human review.

16. **ld2-omr-10269-interface-4.** Classify the supplied connector record for B0454 and B0456. This does not certify load capacity.

interface; connector-graph; metacognitive; single-choice.

Operands: B0456 = brick_000456, B0454 = brick_000454

Public evidence summary: {"connectorRecord": {"family": "stud", "aConnector": 10, "bConnector": 2}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}

Options: A: fixed; B: axle; C: hinge; D: ball; E: stud

Reference output: {"choiceId": "E"}

Derivation: public connectorRecord.family equality

Equivalence: exact declared response

Review: pending required human review.

17. **ld2-omr-10269-neighbors-1.** Select every candidate directly adjacent to B0282 in the supplied connector graph.

neighbors; connector-graph; metacognitive; multiple-choice.

Operands: B0282 = brick_000282, B0283 = brick_000283, B0265 = brick_000265, B0413 = brick_000413, B0284 = brick_000284, B0618 = brick_000618, B0280 = brick_000280

Public evidence summary: 4 unique undirected pairs

Options: A: B0280; B: B0283; C: B0265; D: B0618; E: B0284; F: B0413

Reference output: {"choiceIds": ["A", "B", "C", "E"]}

Derivation: public incident edge endpoint set

Equivalence: unordered exact set; duplicate IDs invalid

Review: pending required human review.

18. **ld2-omr-10269-graph-removal-1.** Delete B0282 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.

graph-removal; connector-graph; graph-internal; integer.

Operands: B0413 = brick_000413, B0283 = brick_000283, B0284 = brick_000284, B0265 = brick_000265, B0280 = brick_000280, B0618 = brick_000618, B0282 = brick_000282

Public evidence summary: 4 unique undirected pairs; 7 nodes

Reference output: {"value": 6}

Derivation: union-find component count after public target deletion

Equivalence: exact declared response

Review: pending required human review.

19. **ld2-omr-10269-neighbors-2.** Select every candidate directly adjacent to B0764 in the supplied connector graph.

neighbors; connector-graph; metacognitive; multiple-choice.

Operands: B0766 = brick_000766, B0762 = brick_000762, B0053 = brick_000053, B0763 = brick_000763, B0422 = brick_000422, B0764 = brick_000764, B0765 = brick_000765

Public evidence summary: 5 unique undirected pairs

Options: A: B0762; B: B0765; C: B0422; D: B0763; E: B0766; F: B0053

Reference output: {"choiceIds": ["A", "B", "D", "E"]}

Derivation: public incident edge endpoint set

Equivalence: unordered exact set; duplicate IDs invalid

Review: machine checks passed; no human certification.

20. **ld2-omr-10269-graph-removal-2.** Delete B0764 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
graph-removal; connector-graph; graph-internal; integer.
Operands: B0053 = brick_000053, B0766 = brick_000766, B0764 = brick_000764, B0422 = brick_000422, B0765 = brick_000765, B0763 = brick_000763, B0762 = brick_000762
Public evidence summary: 5 unique undirected pairs; 7 nodes
Reference output: {"value": 5}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: machine checks passed; no human certification.
21. **ld2-omr-10269-neighbors-3.** Select every candidate directly adjacent to B0973 in the supplied connector graph.
neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0836 = brick_000836, B0972 = brick_000972, B0624 = brick_000624, B0973 = brick_000973, B0975 = brick_000975
Public evidence summary: 2 unique undirected pairs
Options: A: B0975; B: B0836; C: B0972; D: B0624
Reference output: {"choiceIds": ["A", "C"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: machine checks passed; no human certification.
22. **ld2-omr-10269-graph-removal-3.** Delete B0973 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
graph-removal; connector-graph; graph-internal; integer.
Operands: B0975 = brick_000975, B0836 = brick_000836, B0973 = brick_000973, B0624 = brick_000624, B0972 = brick_000972
Public evidence summary: 2 unique undirected pairs; 5 nodes
Reference output: {"value": 4}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: machine checks passed; no human certification.
23. **ld2-omr-10269-neighbors-4.** Select every candidate directly adjacent to B0127 in the supplied connector graph.
neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0128 = brick_000128, B0126 = brick_000126, B0127 = brick_000127, B0129 = brick_000129, B0537 = brick_000537, B0130 = brick_000130, B0681 = brick_000681
Public evidence summary: 4 unique undirected pairs
Options: A: B0129; B: B0126; C: B0681; D: B0130; E: B0128; F: B0537
Reference output: {"choiceIds": ["A", "B", "D", "E"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: pending required human review.
24. **ld2-omr-10269-graph-removal-4.** Delete B0127 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
graph-removal; connector-graph; graph-internal; integer.
Operands: B0128 = brick_000128, B0130 = brick_000130, B0126 = brick_000126, B0129 = brick_000129, B0127 = brick_000127, B0681 = brick_000681, B0537 = brick_000537
Public evidence summary: 4 unique undirected pairs; 7 nodes
Reference output: {"value": 6}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: pending required human review.

25. **ld2-omr-10269-coverage-1.** Which numbered instance lacks a connector definition in the coverage record, so this audit cannot establish that it is floating?
 coverage; source-data; metacognitive; single-choice.
Operands: B0487 = brick_000487, B0802 = brick_000802, B0626 = brick_000626, B0256 = brick_000256
Public evidence summary: {"coverage": [{"label": "B0802", "supported": true}, {"label": "B0626", "supported": true}, {"label": "B0487", "supported": true}, {"label": "B0256", "supported": false}]}
Options: A: B0802; B: B0487; C: B0256; D: B0626
Reference output: {"choiceId": "C"}
Derivation: public supported=false record
Equivalence: exact declared response
Review: pending required human review.
26. **ld2-omr-10269-evidence-limit-1.** Only source geometry, connector matching and mesh intersection audits are available. Without mass, friction, material or dynamic tests, is gravitational stability established?
 evidence-limit; source-data; metacognitive; single-choice.
Operands: none
Public evidence summary: {"available": ["Original CAD", "Connector inference", "Inset-mesh intersection candidates"]}
Options: A: Established; B: Not established by this evidence
Reference output: {"choiceId": "B"}
Derivation: fixed epistemic control label
Equivalence: exact declared response
Review: machine checks passed; no human certification.
27. **ld2-omr-10269-source-step-1.** At which expanded author step does B0533 first appear in the supplied source-step table? Source order is not a collision-free assembly proof.
 source-step; source-data; procedural; single-choice.
Operands: B0001 = brick_000001, B0533 = brick_000533
Public evidence summary: 2 supplied author-step records; exact table in public JSON
Options: A: 324; B: 335; C: 269; D: 152
Reference output: {"choiceId": "D"}
Derivation: minimum public step index containing target
Equivalence: exact declared response
Review: machine checks passed; no human certification.
28. **ld2-omr-10269-source-step-2.** At which expanded author step does B0037 first appear in the supplied source-step table? Source order is not a collision-free assembly proof.
 source-step; source-data; procedural; single-choice.
Operands: B0037 = brick_000037, B0001 = brick_000001
Public evidence summary: 2 supplied author-step records; exact table in public JSON
Options: A: 328; B: 10; C: 2; D: 131
Reference output: {"choiceId": "B"}
Derivation: minimum public step index containing target
Equivalence: exact declared response
Review: machine checks passed; no human certification.
29. **ld2-omr-10269-source-step-3.** At which expanded author step does B0667 first appear in the supplied source-step table? Source order is not a collision-free assembly proof.
 source-step; source-data; procedural; single-choice.
Operands: B0667 = brick_000667, B0001 = brick_000001
Public evidence summary: 2 supplied author-step records; exact table in public JSON

Options: A: 182; B: 67; C: 204; D: 29

Reference output: {"choiceId": "C"}

Derivation: minimum public step index containing target

Equivalence: exact declared response

Review: machine checks passed; no human certification.

30. **ld2-omr-10269-source-step-4.** At which expanded author step does B0767 first appear in the supplied source-step table? Source order is not a collision-free assembly proof.

source-step; source-data; procedural; single-choice.

Operands: B0001 = brick_000001, B0767 = brick_000767

Public evidence summary: 2 supplied author-step records; exact table in public JSON

Options: A: 253; B: 318; C: 243; D: 133

Reference output: {"choiceId": "C"}

Derivation: minimum public step index containing target

Equivalence: exact declared response

Review: machine checks passed; no human certification.

31. **ld2-omr-10269-source-sequence-1.** Reveal the supplied source-step window in author order. Actions change visibility only, without simulating insertion trajectories.

source-sequence; scene-edit; procedural; actions.

Operands: B0016 = brick_000016, B0003 = brick_000003, B0006 = brick_000006, B0017 = brick_000017, B0009 = brick_000009, B0001 = brick_000001, B0011 = brick_000011, B0004 = brick_000004, B0005 = brick_000005, B0002 = brick_000002, B0013 = brick_000013, B0012 = brick_000012, B0015 = brick_000015, B0022 = brick_000022, B0020 = brick_000020, B0018 = brick_000018, B0008 = brick_000008, B0007 = brick_000007, B0019 = brick_000019, B0021 = brick_000021, B0010 = brick_000010, B0023 = brick_000023, B0014 = brick_000014

Public evidence summary: 6 public actions; budget 6; {"initialFacts": ["start"], "goalFacts": ["done-6"], "absentFacts": []}

Reference output: {"actionIds": ["step-1", "step-2", "step-3", "step-4", "step-5", "step-6"]}

Derivation: public precondition solver

Equivalence: any legal sequence reaching all goals within budget; display-only semantics

Review: machine checks passed; no human certification.

32. **ld2-omr-10269-restore-instance-1.** The display copy hides B0802. Restore only that numbered instance, preserving all other instances and source poses.

restore-instance; scene-edit; procedural; actions.

Operands: B0001 = brick_000001, B0802 = brick_000802

Public evidence summary: 2 public actions; budget 1; {"initialFacts": ["missing-brick_000802", "present-brick_000001"], "goalFacts": ["present-brick_000802", "present-brick_000001"], "absentFacts": ["missing-brick_000802"]}

Reference output: {"actionIds": ["restore-B0802"]}

Derivation: public precondition solver

Equivalence: any legal sequence reaching all goals within budget; display-only semantics

Review: machine checks passed; no human certification.

33. **ld2-omr-10269-restore-instance-2.** The display copy hides B0626. Restore only that numbered instance, preserving all other instances and source poses.

restore-instance; scene-edit; procedural; actions.

Operands: B0626 = brick_000626, B0001 = brick_000001

Public evidence summary: 2 public actions; budget 1; {"initialFacts": ["missing-brick_000626", "present-brick_000001"], "goalFacts": ["present-brick_000626", "present-brick_000001"], "absentFacts": ["missing-brick_000626"]}

Reference output: {"actionIds": ["restore-B0626"]}

Derivation: public precondition solver

Equivalence: any legal sequence reaching all goals within budget; display-only semantics

Review: machine checks passed; no human certification.

34. **ld2-omr-10269-restore-instance-3.** The display copy hides B0487. Restore only that numbered instance, preserving all other instances and source poses.
- restore-instance; scene-edit; procedural; actions.
- Operands:* B0001 = brick_000001, B0487 = brick_000487
- Public evidence summary:* 2 public actions; budget 1; {"initialFacts": ["missing-brick_000487", "present-brick_000001"], "goalFacts": ["present-brick_000487", "present-brick_000001"], "absentFacts": ["missing-brick_000487"]}
- Reference output:* {"actionIds": ["restore-B0487"]}
- Derivation:* public precondition solver
- Equivalence:* any legal sequence reaching all goals within budget; display-only semantics
- Review:* machine checks passed; no human certification.

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1086 source instances; 34 tasks; D4 size band. 136 type strings; 22 color codes; 10 unsupported instances; 17 intersection candidates. 233 expanded author steps. Independent reparse: passed.



Attribution. Magnus Forsberg [MagFors]. CC BY 2.0.

<https://library.ldraw.org/library/omr/10242-1.mpd>

Source SHA256. d79fd06503b145cc5ff3c5bcfe71a1c402910dee78424f54077faba9cf402b74

Connector records. stud: 2293; axle: 161; hinge: 7; fixed: 20; ball: 0

Public input lock. 9e9a4464b790765d8611cd98f178ffe0da91c11d977e386c0c91f3be7dd0f599

Status. Source preservation does not certify stability. Reference outputs below are internal review material. Full model inputs are the versioned public JSON and numbered PNGs.

1. **ld2-omr-10242-color-1.** Identify the original color of B0591; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B0591 = brick_000591

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-10242-color-1.png; structured input = {}.

Options: A: Reddish Brown; B: White; C: Trans Light Blue; D: Pearl Gold

Reference output: {"choiceId": "A"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

2. **ld2-omr-10242-shape-match-1.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0591?

shape-match; visual; atomic; single-choice.

Operands: B0591 = brick_000591, B0304 = brick_000304, B0281 = brick_000281, B0512 = brick_000512, B0073 = brick_000073

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-10242-shape-match-1.png; structured input = {}.

Options: A: B0281; B: B0304; C: B0073; D: B0512

Reference output: {"choiceId": "C"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

3. **ld2-omr-10242-distance-1.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0591?

distance; source-data; atomic; single-choice.

Operands: B0669 = brick_000669, B0840 = brick_000840, B0950 = brick_000950, B0591 = brick_000591

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0591": [44, 34.4, 64], "B0669": [-12, 20.726885, 90.813115], "B0950": [60.928, 31.2, 42.122316], "B0840": [8, 24.059068, -4.9024]}}

Options: A: B0669; B: B0840; C: B0950

Reference output: {"choiceId": "C"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: pending required human review.

4. **ld2-omr-10242-color-2.** Identify the original color of B0837; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B0837 = brick_000837

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-10242-color-2.png; structured input = {}.

Options: A: Trans Clear; B: Yellow; C: White; D: Tan

Reference output: {"choiceId": "D"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

5. **ld2-omr-10242-shape-match-2.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0837?

shape-match; visual; atomic; single-choice.

Operands: B0166 = brick_000166, B0341 = brick_000341, B0837 = brick_000837, B0845 = brick_000845, B0303 = brick_000303

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-10242-shape-match-2.png; structured input = {}.

Options: A: B0166; B: B0303; C: B0845; D: B0341

Reference output: {"choiceId": "B"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

6. **ld2-omr-10242-distance-2.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0837?

distance; source-data; atomic; single-choice.

Operands: B0103 = brick_000103, B0465 = brick_000465, B0837 = brick_000837, B0511 = brick_000511

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0837": [40, 31.6984, -0.1184], "B0465": [-36, 21.6, 44], "B0511": [0, 37.6, 38.4], "B0103": [0, 5.6, 116]}}

Options: A: B0103; B: B0465; C: B0511

Reference output: {"choiceId": "C"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: pending required human review.

7. **ld2-omr-10242-color-3.** Identify the original color of B0962; numbered isolation is allowed.
 color; visual; atomic; single-choice.
Operands: B0962 = brick_000962
Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-10242-color-3.png; structured input = {}.
Options: A: Bright Light Orange; B: Trans Brown; C: Medium Nougat; D: Black
Reference output: {"choiceId": "D"}
Derivation: source colorName by rendered B-label; source oracle only
Equivalence: exact declared response
Review: pending required human review.
8. **ld2-omr-10242-shape-match-3.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0962?
 shape-match; visual; atomic; single-choice.
Operands: B0969 = brick_000969, B0962 = brick_000962, B0654 = brick_000654, B0610 = brick_000610, B0111 = brick_000111
Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-10242-shape-match-3.png; structured input = {}.
Options: A: B0111; B: B0654; C: B0610; D: B0969
Reference output: {"choiceId": "B"}
Derivation: source partNumber equality across labeled candidates; source oracle only
Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending
Review: pending required human review.
9. **ld2-omr-10242-distance-3.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0962?
 distance; source-data; atomic; single-choice.
Operands: B0714 = brick_000714, B0029 = brick_000029, B0962 = brick_000962, B0312 = brick_000312
Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0962": [58.7176, 44, 45.1072], "B0714": [36, 21.6, -60], "B0029": [24, -4, -40], "B0312": [-36, 24.8, -80]}}
Options: A: B0714; B: B0312; C: B0029
Reference output: {"choiceId": "C"}
Derivation: Euclidean norm from public rounded centers
Equivalence: exact declared response
Review: machine checks passed; no human certification.
10. **ld2-omr-10242-color-4.** Identify the original color of B0975; numbered isolation is allowed.
 color; visual; atomic; single-choice.
Operands: B0975 = brick_000975
Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-10242-color-4.png; structured input = {}.
Options: A: Dark Green; B: Trans Brown; C: Medium Nougat; D: Yellow
Reference output: {"choiceId": "A"}
Derivation: source colorName by rendered B-label; source oracle only
Equivalence: exact declared response
Review: pending required human review.
11. **ld2-omr-10242-shape-match-4.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0975?
 shape-match; visual; atomic; single-choice.
Operands: B0485 = brick_000485, B0180 = brick_000180, B0975 = brick_000975, B1004 = brick_001004, B0657 = brick_000657
Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-10242-shape-match-4.png; structured input = {}.
Options: A: B0657; B: B0485; C: B1004; D: B0180

Reference output: {"choiceId": "B"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

12. **Id2-omr-10242-distance-4.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0975?

distance; source-data; atomic; single-choice.

Operands: B0857 = brick_000857, B0379 = brick_000379, B0265 = brick_000265, B0975 = brick_000975

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0975": [-51.2, 28, 28], "B0857": [24, 59.534, 21.9088], "B0379": [-40, 37.6, -68], "B0265": [48, 5.6, -44]}}

Options: A: B0379; B: B0265; C: B0857

Reference output: {"choiceId": "C"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

13. **Id2-omr-10242-interface-1.** Classify the supplied connector record for B0028 and B0032. This does not certify load capacity.

interface; connector-graph; metacognitive; single-choice.

Operands: B0032 = brick_000032, B0028 = brick_000028

Public evidence summary: {"connectorRecord": {"family": "axle", "aConnector": 12, "bConnector": 0}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}

Options: A: hinge; B: axle; C: ball; D: stud; E: fixed

Reference output: {"choiceId": "B"}

Derivation: public connectorRecord.family equality

Equivalence: exact declared response

Review: machine checks passed; no human certification.

14. **Id2-omr-10242-interface-2.** Classify the supplied connector record for B0153 and B0154. This does not certify load capacity.

interface; connector-graph; metacognitive; single-choice.

Operands: B0153 = brick_000153, B0154 = brick_000154

Public evidence summary: {"connectorRecord": {"family": "fixed", "aConnector": 1, "bConnector": 1}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}

Options: A: axle; B: stud; C: fixed; D: hinge; E: ball

Reference output: {"choiceId": "C"}

Derivation: public connectorRecord.family equality

Equivalence: exact declared response

Review: machine checks passed; no human certification.

15. **Id2-omr-10242-interface-3.** Classify the supplied connector record for B0578 and B0981. This does not certify load capacity.

interface; connector-graph; metacognitive; single-choice.

Operands: B0578 = brick_000578, B0981 = brick_000981

Public evidence summary: {"connectorRecord": {"family": "hinge", "aConnector": 1, "bConnector": 2}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}

Options: A: ball; B: fixed; C: hinge; D: axle; E: stud

Reference output: {"choiceId": "C"}

Derivation: public connectorRecord.family equality

Equivalence: exact declared response

Review: machine checks passed; no human certification.

16. **ld2-omr-10242-interface-4.** Classify the supplied connector record for B1027 and B1034. This does not certify load capacity.

interface; connector-graph; metacognitive; single-choice.

Operands: B1027 = brick_001027, B1034 = brick_001034

Public evidence summary: {"connectorRecord": {"family": "stud", "aConnector": 203, "bConnector": 5}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}

Options: A: stud; B: axle; C: hinge; D: fixed; E: ball

Reference output: {"choiceId": "A"}

Derivation: public connectorRecord.family equality

Equivalence: exact declared response

Review: machine checks passed; no human certification.

17. **ld2-omr-10242-neighbors-1.** Select every candidate directly adjacent to B0732 in the supplied connector graph.

neighbors; connector-graph; metacognitive; multiple-choice.

Operands: B0729 = brick_000729, B0550 = brick_000550, B0380 = brick_000380, B0720 = brick_000720, B0733 = brick_000733, B0732 = brick_000732

Public evidence summary: 4 unique undirected pairs

Options: A: B0380; B: B0733; C: B0729; D: B0550; E: B0720

Reference output: {"choiceIds": ["A", "B", "E"]}

Derivation: public incident edge endpoint set

Equivalence: unordered exact set; duplicate IDs invalid

Review: machine checks passed; no human certification.

18. **ld2-omr-10242-graph-removal-1.** Delete B0732 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.

graph-removal; connector-graph; graph-internal; integer.

Operands: B0720 = brick_000720, B0732 = brick_000732, B0550 = brick_000550, B0380 = brick_000380, B0729 = brick_000729, B0733 = brick_000733

Public evidence summary: 4 unique undirected pairs; 6 nodes

Reference output: {"value": 4}

Derivation: union-find component count after public target deletion

Equivalence: exact declared response

Review: machine checks passed; no human certification.

19. **ld2-omr-10242-neighbors-2.** Select every candidate directly adjacent to B0711 in the supplied connector graph.

neighbors; connector-graph; metacognitive; multiple-choice.

Operands: B0469 = brick_000469, B0679 = brick_000679, B0719 = brick_000719, B0711 = brick_000711, B0030 = brick_000030

Public evidence summary: 2 unique undirected pairs

Options: A: B0469; B: B0679; C: B0030; D: B0719

Reference output: {"choiceIds": ["B", "D"]}

Derivation: public incident edge endpoint set

Equivalence: unordered exact set; duplicate IDs invalid

Review: machine checks passed; no human certification.

20. **ld2-omr-10242-graph-removal-2.** Delete B0711 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
graph-removal; connector-graph; graph-internal; integer.
Operands: B0679 = brick_000679, B0719 = brick_000719, B0469 = brick_000469, B0711 = brick_000711, B0030 = brick_000030
Public evidence summary: 2 unique undirected pairs; 5 nodes
Reference output: {"value": 4}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: machine checks passed; no human certification.
21. **ld2-omr-10242-neighbors-3.** Select every candidate directly adjacent to B0074 in the supplied connector graph.
neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0349 = brick_000349, B0294 = brick_000294, B0704 = brick_000704, B0231 = brick_000231, B0074 = brick_000074, B0043 = brick_000043
Public evidence summary: 3 unique undirected pairs
Options: A: B0349; B: B0704; C: B0294; D: B0231; E: B0043
Reference output: {"choiceIds": ["C", "D", "E"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: pending required human review.
22. **ld2-omr-10242-graph-removal-3.** Delete B0074 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
graph-removal; connector-graph; graph-internal; integer.
Operands: B0349 = brick_000349, B0231 = brick_000231, B0043 = brick_000043, B0074 = brick_000074, B0294 = brick_000294, B0704 = brick_000704
Public evidence summary: 3 unique undirected pairs; 6 nodes
Reference output: {"value": 5}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: pending required human review.
23. **ld2-omr-10242-neighbors-4.** Select every candidate directly adjacent to B0402 in the supplied connector graph.
neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0917 = brick_000917, B0384 = brick_000384, B0402 = brick_000402, B0188 = brick_000188, B0415 = brick_000415
Public evidence summary: 2 unique undirected pairs
Options: A: B0188; B: B0917; C: B0415; D: B0384
Reference output: {"choiceIds": ["C", "D"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: machine checks passed; no human certification.
24. **ld2-omr-10242-graph-removal-4.** Delete B0402 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
graph-removal; connector-graph; graph-internal; integer.
Operands: B0402 = brick_000402, B0188 = brick_000188, B0917 = brick_000917, B0384 = brick_000384, B0415 = brick_000415
Public evidence summary: 2 unique undirected pairs; 5 nodes
Reference output: {"value": 4}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: machine checks passed; no human certification.

25. **ld2-omr-10242-coverage-1.** Which numbered instance lacks a connector definition in the coverage record, so this audit cannot establish that it is floating?
 coverage; source-data; metacognitive; single-choice.
Operands: B0591 = brick_000591, B0962 = brick_000962, B0014 = brick_000014, B0837 = brick_000837
Public evidence summary: {"coverage": [{"label": "B0962", "supported": true}, {"label": "B0591", "supported": true}, {"label": "B0837", "supported": true}, {"label": "B0014", "supported": false}]}
Options: A: B0837; B: B0591; C: B0014; D: B0962
Reference output: {"choiceId": "C"}
Derivation: public supported=false record
Equivalence: exact declared response
Review: pending required human review.
26. **ld2-omr-10242-evidence-limit-1.** Only source geometry, connector matching and mesh intersection audits are available. Without mass, friction, material or dynamic tests, is gravitational stability established?
 evidence-limit; source-data; metacognitive; single-choice.
Operands: none
Public evidence summary: {"available": ["Original CAD", "Connector inference", "Inset-mesh intersection candidates"]}
Options: A: Established; B: Not established by this evidence
Reference output: {"choiceId": "B"}
Derivation: fixed epistemic control label
Equivalence: exact declared response
Review: machine checks passed; no human certification.
27. **ld2-omr-10242-source-step-1.** At which expanded author step does B0110 first appear in the supplied source-step table? Source order is not a collision-free assembly proof.
 source-step; source-data; procedural; single-choice.
Operands: B0001 = brick_000001, B0110 = brick_000110
Public evidence summary: 2 supplied author-step records; exact table in public JSON
Options: A: 121; B: 31; C: 216; D: 18
Reference output: {"choiceId": "B"}
Derivation: minimum public step index containing target
Equivalence: exact declared response
Review: machine checks passed; no human certification.
28. **ld2-omr-10242-source-step-2.** At which expanded author step does B0762 first appear in the supplied source-step table? Source order is not a collision-free assembly proof.
 source-step; source-data; procedural; single-choice.
Operands: B0001 = brick_000001, B0762 = brick_000762
Public evidence summary: 2 supplied author-step records; exact table in public JSON
Options: A: 209; B: 159; C: 198; D: 195
Reference output: {"choiceId": "B"}
Derivation: minimum public step index containing target
Equivalence: exact declared response
Review: machine checks passed; no human certification.
29. **ld2-omr-10242-source-step-3.** At which expanded author step does B0332 first appear in the supplied source-step table? Source order is not a collision-free assembly proof.
 source-step; source-data; procedural; single-choice.
Operands: B0001 = brick_000001, B0332 = brick_000332
Public evidence summary: 2 supplied author-step records; exact table in public JSON

Options: A: 43; B: 169; C: 74; D: 69

Reference output: {"choiceId": "C"}

Derivation: minimum public step index containing target

Equivalence: exact declared response

Review: machine checks passed; no human certification.

30. **ld2-omr-10242-source-step-4.** At which expanded author step does B1032 first appear in the supplied source-step table?

Source order is not a collision-free assembly proof.

source-step; source-data; procedural; single-choice.

Operands: B0001 = brick_000001, B1032 = brick_001032

Public evidence summary: 2 supplied author-step records; exact table in public JSON

Options: A: 229; B: 231; C: 232; D: 233

Reference output: {"choiceId": "A"}

Derivation: minimum public step index containing target

Equivalence: exact declared response

Review: machine checks passed; no human certification.

31. **ld2-omr-10242-source-sequence-1.** Reveal the supplied source-step window in author order. Actions change visibility only, without simulating insertion trajectories.

source-sequence; scene-edit; procedural; actions.

Operands: B0010 = brick_000010, B0013 = brick_000013, B0002 = brick_000002, B0001 = brick_000001, B0009 = brick_000009, B0007 = brick_000007, B0011 = brick_000011, B0012 = brick_000012, B0017 = brick_000017, B0016 = brick_000016, B0014 = brick_000014, B0005 = brick_000005, B0004 = brick_000004, B0006 = brick_000006, B0003 = brick_000003, B0019 = brick_000019, B0015 = brick_000015, B0018 = brick_000018, B0008 = brick_000008

Public evidence summary: 6 public actions; budget 6; {"initialFacts": ["start"], "goalFacts": ["done-6"], "absentFacts": []}

Reference output: {"actionIds": ["step-1", "step-2", "step-3", "step-4", "step-5", "step-6"]}

Derivation: public precondition solver

Equivalence: any legal sequence reaching all goals within budget; display-only semantics

Review: pending required human review.

32. **ld2-omr-10242-restore-instance-1.** The display copy hides B0591. Restore only that numbered instance, preserving all other instances and source poses.

restore-instance; scene-edit; procedural; actions.

Operands: B0591 = brick_000591, B0001 = brick_000001

Public evidence summary: 2 public actions; budget 1; {"initialFacts": ["missing-brick_000591", "present-brick_000001"], "goalFacts": ["present-brick_000591", "present-brick_000001"], "absentFacts": ["missing-brick_000591"]}

Reference output: {"actionIds": ["restore-B0591"]}

Derivation: public precondition solver

Equivalence: any legal sequence reaching all goals within budget; display-only semantics

Review: machine checks passed; no human certification.

33. **ld2-omr-10242-restore-instance-2.** The display copy hides B0837. Restore only that numbered instance, preserving all other instances and source poses.

restore-instance; scene-edit; procedural; actions.

Operands: B0001 = brick_000001, B0837 = brick_000837

Public evidence summary: 2 public actions; budget 1; {"initialFacts": ["missing-brick_000837", "present-brick_000001"], "goalFacts": ["present-brick_000837", "present-brick_000001"], "absentFacts": ["missing-brick_000837"]}

Reference output: {"actionIds": ["restore-B0837"]}

Derivation: public precondition solver

Equivalence: any legal sequence reaching all goals within budget; display-only semantics

Review: machine checks passed; no human certification.

34. **ld2-omr-10242-restore-instance-3.** The display copy hides B0962. Restore only that numbered instance, preserving all other instances and source poses.
- restore-instance; scene-edit; procedural; actions.
- Operands:* B0962 = brick_000962, B0001 = brick_000001
- Public evidence summary:* 2 public actions; budget 1; {"initialFacts": ["missing-brick_000962", "present-brick_000001"], "goalFacts": ["present-brick_000962", "present-brick_000001"], "absentFacts": ["missing-brick_000962"]}
- Reference output:* {"actionIds": ["restore-B0962"]}
- Derivation:* public precondition solver
- Equivalence:* any legal sequence reaching all goals within budget; display-only semantics
- Review:* machine checks passed; no human certification.

29 42066: Air Race Jet

1244 source instances; 41 tasks; D4 size band. 127 type strings; 15 color codes; 114 unsupported instances; 90 intersection candidates. 141 expanded author steps. Independent reparse: passed.



Attribution. Philippe Hurbain [Philo]. CC BY 2.0.

<https://library.ldraw.org/library/omr/42066-1.mpd>

Source SHA256. d1463b1464a83d54a6ed372fba732eabaf79b3857185f4ab60b314e741caa067

Connector records. stud: 32; axle: 1534; hinge: 4; fixed: 8; ball: 8

Public input lock. 45366a7ebcd1ddbb144cda3c8ebd441e3d177e2893d9f7a41c43be8d4cfc61fb

Status. Source preservation does not certify stability. Reference outputs below are internal review material. Full model inputs are the versioned public JSON and numbered PNGs.

1. **ld2-omr-42066-color-1.** Identify the original color of B1144; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B1144 = brick_001144

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-42066-color-1.png; structured input = {}.

Options: A: Blue; B: White; C: Yellow; D: Black

Reference output: {"choiceId": "D"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

2. **ld2-omr-42066-shape-match-1.** Ignoring color and pose, which numbered instance has the same LDraw part type as B1144?

shape-match; visual; atomic; single-choice.

Operands: B0739 = brick_000739, B1144 = brick_001144, B0576 = brick_000576, B0405 = brick_000405, B0546 = brick_000546

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-42066-shape-match-1.png; structured input = {}.

Options: A: B0576; B: B0739; C: B0405; D: B0546

Reference output: {"choiceId": "C"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

3. **ld2-omr-42066-distance-1.** Using the supplied geometry bounding-box centers, which candidate is nearest to B1144?

distance; source-data; atomic; single-choice.

Operands: B1226 = brick_001226, B1144 = brick_001144, B0947 = brick_000947, B0240 = brick_000240

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B1144": [-47.588, 105.202992, -266.945856], "B1226": [58.8468, 166.972, -189.987204], "B0947": [0, 64.148442, 284.320131], "B0240": [20, 85.6348, 20.876756]}}

Options: A: B0240; B: B1226; C: B0947

Reference output: {"choiceId": "B"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

4. **ld2-omr-42066-color-2.** Identify the original color of B0286; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B0286 = brick_000286

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-42066-color-2.png; structured input = {}.

Options: A: Black; B: Dark Brown; C: Trans Green; D: Trans Red

Reference output: {"choiceId": "A"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

5. **ld2-omr-42066-shape-match-2.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0286?

shape-match; visual; atomic; single-choice.

Operands: B0175 = brick_000175, B0867 = brick_000867, B0286 = brick_000286, B0604 = brick_000604, B0128 = brick_000128

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-42066-shape-match-2.png; structured input = {}.

Options: A: B0128; B: B0175; C: B0867; D: B0604

Reference output: {"choiceId": "A"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

6. **ld2-omr-42066-distance-2.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0286?

distance; source-data; atomic; single-choice.

Operands: B0964 = brick_000964, B0757 = brick_000757, B0286 = brick_000286, B0584 = brick_000584

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0286": [-26, 101.822608, 2.184334], "B0757": [15.9892, 61.5388, 20.5948], "B0964": [15.800047, 98.829629, 261.097606], "B0584": [-16, 131.207349, 129.781993]}}

Options: A: B0757; B: B0964; C: B0584

Reference output: {"choiceId": "A"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

7. **ld2-omr-42066-color-3.** Identify the original color of B0606; numbered isolation is allowed.
 color; visual; atomic; single-choice.
Operands: B0606 = brick_000606
Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-42066-color-3.png; structured input = {}.
Options: A: Rubber Black; B: Dark Bluish Grey; C: Yellow; D: Blue
Reference output: {"choiceId": "D"}
Derivation: source colorName by rendered B-label; source oracle only
Equivalence: exact declared response
Review: pending required human review.
8. **ld2-omr-42066-shape-match-3.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0606?
 shape-match; visual; atomic; single-choice.
Operands: B0606 = brick_000606, B0435 = brick_000435, B1052 = brick_001052, B0302 = brick_000302, B1082 = brick_001082
Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-42066-shape-match-3.png; structured input = {}.
Options: A: B0302; B: B1082; C: B1052; D: B0435
Reference output: {"choiceId": "B"}
Derivation: source partNumber equality across labeled candidates; source oracle only
Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending
Review: pending required human review.
9. **ld2-omr-42066-distance-3.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0606?
 distance; source-data; atomic; single-choice.
Operands: B0606 = brick_000606, B1010 = brick_001010, B0658 = brick_000658, B1003 = brick_001003
Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0606": [12.1, 106.503744, 166.636178], "B1010": [-31.6688, 124.275493, 152.61352], "B0658": [40, 85.5388, 28.8768], "B1003": [29.695036, 112.555599, 194.485323]}}
Options: A: B1003; B: B0658; C: B1010
Reference output: {"choiceId": "A"}
Derivation: Euclidean norm from public rounded centers
Equivalence: exact declared response
Review: machine checks passed; no human certification.
10. **ld2-omr-42066-color-4.** Identify the original color of B0478; numbered isolation is allowed.
 color; visual; atomic; single-choice.
Operands: B0478 = brick_000478
Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-42066-color-4.png; structured input = {}.
Options: A: Tan; B: Light Bluish Grey; C: White; D: Blue
Reference output: {"choiceId": "D"}
Derivation: source colorName by rendered B-label; source oracle only
Equivalence: exact declared response
Review: pending required human review.
11. **ld2-omr-42066-shape-match-4.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0478?
 shape-match; visual; atomic; single-choice.
Operands: B0305 = brick_000305, B0402 = brick_000402, B1073 = brick_001073, B0947 = brick_000947, B0478 = brick_000478
Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-42066-shape-match-4.png; structured input = {}.

Options: A: B1073; B: B0402; C: B0947; D: B0305

Reference output: {"choiceId": "C"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

12. **ld2-omr-42066-distance-4.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0478?

distance; source-data; atomic; single-choice.

Operands: B0478 = brick_000478, B0379 = brick_000379, B0569 = brick_000569, B0103 = brick_000103

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0478": [16, 59.4268, 204.5888], "B0569": [28, 116.1948, 141.2608], "B0379": [-24, 110.8828, -82.8352], "B0103": [-29.50087, 89.069647, -28.56805]}}

Options: A: B0569; B: B0103; C: B0379

Reference output: {"choiceId": "A"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: pending required human review.

13. **ld2-omr-42066-color-5.** Identify the original color of B0644; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B0644 = brick_000644

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-42066-color-5.png; structured input = {}.

Options: A: Medium Blue; B: Dark Brown; C: Light Bluish Grey; D: Yellow

Reference output: {"choiceId": "C"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

14. **ld2-omr-42066-shape-match-5.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0644?

shape-match; visual; atomic; single-choice.

Operands: B0352 = brick_000352, B0566 = brick_000566, B0744 = brick_000744, B0644 = brick_000644, B0945 = brick_000945

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-42066-shape-match-5.png; structured input = {}.

Options: A: B0945; B: B0352; C: B0566; D: B0744

Reference output: {"choiceId": "A"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

15. **ld2-omr-42066-distance-5.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0644?

distance; source-data; atomic; single-choice.

Operands: B1071 = brick_001071, B0306 = brick_000306, B0644 = brick_000644, B0074 = brick_000074

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0644": [24, 59.2348, 216.398877], "B0306": [-44.4, 79.183139, -81.276917], "B1071": [0.0008, 133.7308, 13.4528], "B0074": [-24, 68.845746, -90.005559]}}

Options: A: B0074; B: B0306; C: B1071

Reference output: {"choiceId": "C"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: pending required human review.

16. **ld2-omr-42066-interface-1.** Classify the supplied connector record for B0785 and B0786. This does not certify load capacity.
- interface; connector-graph; metacognitive; single-choice.
- Operands:* B0785 = brick_000785, B0786 = brick_000786
- Public evidence summary:* {"connectorRecord": {"family": "axle", "aConnector": 0, "bConnector": 0}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
- Options:* A: hinge; B: stud; C: ball; D: fixed; E: axle
- Reference output:* {"choiceId": "E"}
- Derivation:* public connectorRecord.family equality
- Equivalence:* exact declared response
- Review:* machine checks passed; no human certification.
17. **ld2-omr-42066-interface-2.** Classify the supplied connector record for B0692 and B0694. This does not certify load capacity.
- interface; connector-graph; metacognitive; single-choice.
- Operands:* B0694 = brick_000694, B0692 = brick_000692
- Public evidence summary:* {"connectorRecord": {"family": "ball", "aConnector": 1, "bConnector": 0}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
- Options:* A: hinge; B: fixed; C: ball; D: axle; E: stud
- Reference output:* {"choiceId": "C"}
- Derivation:* public connectorRecord.family equality
- Equivalence:* exact declared response
- Review:* machine checks passed; no human certification.
18. **ld2-omr-42066-interface-3.** Classify the supplied connector record for B0413 and B0414. This does not certify load capacity.
- interface; connector-graph; metacognitive; single-choice.
- Operands:* B0413 = brick_000413, B0414 = brick_000414
- Public evidence summary:* {"connectorRecord": {"family": "fixed", "aConnector": 1, "bConnector": 1}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
- Options:* A: axle; B: hinge; C: fixed; D: ball; E: stud
- Reference output:* {"choiceId": "C"}
- Derivation:* public connectorRecord.family equality
- Equivalence:* exact declared response
- Review:* machine checks passed; no human certification.
19. **ld2-omr-42066-interface-4.** Classify the supplied connector record for B0553 and B0551. This does not certify load capacity.
- interface; connector-graph; metacognitive; single-choice.
- Operands:* B0553 = brick_000553, B0551 = brick_000551
- Public evidence summary:* {"connectorRecord": {"family": "hinge", "aConnector": 0, "bConnector": 1}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
- Options:* A: stud; B: fixed; C: hinge; D: axle; E: ball
- Reference output:* {"choiceId": "C"}
- Derivation:* public connectorRecord.family equality
- Equivalence:* exact declared response
- Review:* machine checks passed; no human certification.

20. **ld2-omr-42066-interface-5.** Classify the supplied connector record for B0988 and B0994. This does not certify load capacity.
 interface; connector-graph; metacognitive; single-choice.
Operands: B0994 = brick_000994, B0988 = brick_000988
Public evidence summary: {"connectorRecord": {"family": "stud", "aConnector": 8, "bConnector": 2}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
Options: A: ball; B: axle; C: stud; D: fixed; E: hinge
Reference output: {"choiceId": "C"}
Derivation: public connectorRecord.family equality
Equivalence: exact declared response
Review: machine checks passed; no human certification.
21. **ld2-omr-42066-neighbors-1.** Select every candidate directly adjacent to B1156 in the supplied connector graph.
 neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0368 = brick_000368, B1156 = brick_001156, B1123 = brick_001123, B1157 = brick_001157, B1145 = brick_001145
Public evidence summary: 2 unique undirected pairs
Options: A: B0368; B: B1157; C: B1123; D: B1145
Reference output: {"choiceIds": ["B", "D"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: machine checks passed; no human certification.
22. **ld2-omr-42066-graph-removal-1.** Delete B1156 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
 graph-removal; connector-graph; graph-internal; integer.
Operands: B1145 = brick_001145, B1123 = brick_001123, B0368 = brick_000368, B1156 = brick_001156, B1157 = brick_001157
Public evidence summary: 2 unique undirected pairs; 5 nodes
Reference output: {"value": 4}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: machine checks passed; no human certification.
23. **ld2-omr-42066-neighbors-2.** Select every candidate directly adjacent to B0916 in the supplied connector graph.
 neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0176 = brick_000176, B0426 = brick_000426, B0916 = brick_000916, B0824 = brick_000824, B0915 = brick_000915
Public evidence summary: 2 unique undirected pairs
Options: A: B0824; B: B0426; C: B0176; D: B0915
Reference output: {"choiceIds": ["B", "D"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: pending required human review.
24. **ld2-omr-42066-graph-removal-2.** Delete B0916 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
 graph-removal; connector-graph; graph-internal; integer.
Operands: B0915 = brick_000915, B0426 = brick_000426, B0916 = brick_000916, B0824 = brick_000824, B0176 = brick_000176
Public evidence summary: 2 unique undirected pairs; 5 nodes
Reference output: {"value": 4}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: pending required human review.

25. **ld2-omr-42066-neighbors-3.** Select every candidate directly adjacent to B1244 in the supplied connector graph.
 neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B1243 = brick_001243, B0583 = brick_000583, B1242 = brick_001242, B1244 = brick_001244, B0013 = brick_000013
Public evidence summary: 2 unique undirected pairs
Options: A: B0583; B: B0013; C: B1242; D: B1243
Reference output: {"choiceIds": ["C", "D"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: machine checks passed; no human certification.
26. **ld2-omr-42066-graph-removal-3.** Delete B1244 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
 graph-removal; connector-graph; graph-internal; integer.
Operands: B1243 = brick_001243, B1242 = brick_001242, B1244 = brick_001244, B0013 = brick_000013, B0583 = brick_000583
Public evidence summary: 2 unique undirected pairs; 5 nodes
Reference output: {"value": 4}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: machine checks passed; no human certification.
27. **ld2-omr-42066-neighbors-4.** Select every candidate directly adjacent to B0962 in the supplied connector graph.
 neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0960 = brick_000960, B0847 = brick_000847, B1087 = brick_001087, B0962 = brick_000962, B0961 = brick_000961
Public evidence summary: 2 unique undirected pairs
Options: A: B0961; B: B0960; C: B0847; D: B1087
Reference output: {"choiceIds": ["A", "B"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: machine checks passed; no human certification.
28. **ld2-omr-42066-graph-removal-4.** Delete B0962 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
 graph-removal; connector-graph; graph-internal; integer.
Operands: B0847 = brick_000847, B1087 = brick_001087, B0961 = brick_000961, B0960 = brick_000960, B0962 = brick_000962
Public evidence summary: 2 unique undirected pairs; 5 nodes
Reference output: {"value": 4}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: machine checks passed; no human certification.
29. **ld2-omr-42066-neighbors-5.** Select every candidate directly adjacent to B1003 in the supplied connector graph.
 neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B1005 = brick_001005, B0769 = brick_000769, B1003 = brick_001003, B0741 = brick_000741, B1001 = brick_001001
Public evidence summary: 2 unique undirected pairs
Options: A: B1005; B: B1001; C: B0741; D: B0769
Reference output: {"choiceIds": ["A", "B"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: machine checks passed; no human certification.

30. **ld2-omr-42066-graph-removal-5.** Delete B1003 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
graph-removal; connector-graph; graph-internal; integer.
Operands: B1001 = brick_001001, B0769 = brick_000769, B1005 = brick_001005, B0741 = brick_000741, B1003 = brick_001003
Public evidence summary: 2 unique undirected pairs; 5 nodes
Reference output: {"value": 4}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: machine checks passed; no human certification.
31. **ld2-omr-42066-coverage-1.** Which numbered instance lacks a connector definition in the coverage record, so this audit cannot establish that it is floating?
coverage; source-data; metacognitive; single-choice.
Operands: B0025 = brick_000025, B0286 = brick_000286, B1144 = brick_001144, B0606 = brick_000606
Public evidence summary: {"coverage": [{"label": "B1144", "supported": true}, {"label": "B0025", "supported": false}, {"label": "B0606", "supported": true}, {"label": "B0286", "supported": true}]}
Options: A: B0606; B: B0286; C: B1144; D: B0025
Reference output: {"choiceId": "D"}
Derivation: public supported=false record
Equivalence: exact declared response
Review: pending required human review.
32. **ld2-omr-42066-evidence-limit-1.** Only source geometry, connector matching and mesh intersection audits are available. Without mass, friction, material or dynamic tests, is gravitational stability established?
evidence-limit; source-data; metacognitive; single-choice.
Operands: none
Public evidence summary: {"available": ["Original CAD", "Connector inference", "Inset-mesh intersection candidates"]}
Options: A: Established; B: Not established by this evidence
Reference output: {"choiceId": "B"}
Derivation: fixed epistemic control label
Equivalence: exact declared response
Review: machine checks passed; no human certification.
33. **ld2-omr-42066-source-step-1.** At which expanded author step does B0163 first appear in the supplied source-step table? Source order is not a collision-free assembly proof.
source-step; source-data; procedural; single-choice.
Operands: B0163 = brick_000163, B0001 = brick_000001
Public evidence summary: 2 supplied author-step records; exact table in public JSON
Options: A: 1; B: 2; C: 13; D: 40
Reference output: {"choiceId": "C"}
Derivation: minimum public step index containing target
Equivalence: exact declared response
Review: pending required human review.
34. **ld2-omr-42066-source-step-2.** At which expanded author step does B0709 first appear in the supplied source-step table? Source order is not a collision-free assembly proof.
source-step; source-data; procedural; single-choice.
Operands: B0709 = brick_000709, B0001 = brick_000001
Public evidence summary: 2 supplied author-step records; exact table in public JSON
Options: A: 9; B: 57; C: 73; D: 14

Reference output: {"choiceId": "C"}

Derivation: minimum public step index containing target

Equivalence: exact declared response

Review: machine checks passed; no human certification.

35. **Id2-omr-42066-source-step-3.** At which expanded author step does B0804 first appear in the supplied source-step table?
Source order is not a collision-free assembly proof.

source-step; source-data; procedural; single-choice.

Operands: B0804 = brick_000804, B0001 = brick_000001

Public evidence summary: 2 supplied author-step records; exact table in public JSON

Options: A: 112; B: 84; C: 130; D: 91

Reference output: {"choiceId": "B"}

Derivation: minimum public step index containing target

Equivalence: exact declared response

Review: machine checks passed; no human certification.

36. **Id2-omr-42066-source-step-4.** At which expanded author step does B1237 first appear in the supplied source-step table?
Source order is not a collision-free assembly proof.

source-step; source-data; procedural; single-choice.

Operands: B0001 = brick_000001, B1237 = brick_001237

Public evidence summary: 2 supplied author-step records; exact table in public JSON

Options: A: 95; B: 77; C: 141; D: 134

Reference output: {"choiceId": "C"}

Derivation: minimum public step index containing target

Equivalence: exact declared response

Review: machine checks passed; no human certification.

37. **Id2-omr-42066-source-step-5.** At which expanded author step does B1173 first appear in the supplied source-step table?
Source order is not a collision-free assembly proof.

source-step; source-data; procedural; single-choice.

Operands: B0001 = brick_000001, B1173 = brick_001173

Public evidence summary: 2 supplied author-step records; exact table in public JSON

Options: A: 76; B: 28; C: 8; D: 133

Reference output: {"choiceId": "D"}

Derivation: minimum public step index containing target

Equivalence: exact declared response

Review: machine checks passed; no human certification.

38. **Id2-omr-42066-source-sequence-1.** Reveal the supplied source-step window in author order. Actions change visibility only, without simulating insertion trajectories.

source-sequence; scene-edit; procedural; actions.

Operands: B0045 = brick_000045, B0044 = brick_000044, B0092 = brick_000092, B0040 = brick_000040, B0024 = brick_000024, B0008 = brick_000008, B0031 = brick_000031, B0015 = brick_000015, B0116 = brick_000116, B0025 = brick_000025, B0039 = brick_000039, B0041 = brick_000041, B0109 = brick_000109, B0078 = brick_000078, B0080 = brick_000080, B0023 = brick_000023, B0014 = brick_000014, B0048 = brick_000048, B0006 = brick_000006, B0065 = brick_000065, B0067 = brick_000067, B0118 = brick_000118, B0114 = brick_000114, B0121 = brick_000121, B0087 = brick_000087, B0091 = brick_000091, B0020 = brick_000020, B0032 = brick_000032, B0062 = brick_000062, B0098 = brick_000098, B0051 = brick_000051, B0016 = brick_000016, B0117 = brick_000117, B0103 = brick_000103, B0113 = brick_000113, B0094 = brick_000094, B0110 = brick_000110, B0104 = brick_000104, B0002 = brick_000002, B0012 = brick_000012, B0111 = brick_000111, B0079 = brick_000079, B0095 = brick_000095, B0090 = brick_000090, B0030 = brick_000030, B0082 = brick_000082, B0052 = brick_000052, B0066 = brick_000066, B0077 = brick_000077, B0021 = brick_000021, B0057 = brick_000057, B0055 = brick_000055, B0083 = brick_000083, B0099 = brick_000099, B0035 = brick_000035, B0033 =

brick_000033, B0072 = brick_000072, B0013 = brick_000013, B0043 = brick_000043, B0038 = brick_000038, B0120 = brick_000120, B0076 = brick_000076, B0019 = brick_000019, B0100 = brick_000100, B0108 = brick_000108, B0101 = brick_000101, B0017 = brick_000017, B0027 = brick_000027, B0074 = brick_000074, B0018 = brick_000018, B0059 = brick_000059, B0119 = brick_000119, B0004 = brick_000004, B0029 = brick_000029, B0084 = brick_000084, B0093 = brick_000093, B0046 = brick_000046, B0028 = brick_000028, B0070 = brick_000070, B0063 = brick_000063, B0056 = brick_000056, B0005 = brick_000005, B0096 = brick_000096, B0088 = brick_000088, B0064 = brick_000064, B0053 = brick_000053, B0047 = brick_000047, B0003 = brick_000003, B0037 = brick_000037, B0097 = brick_000097, B0011 = brick_000011, B0107 = brick_000107, B0081 = brick_000081, B0102 = brick_000102, B0115 = brick_000115, B0026 = brick_000026, B0001 = brick_000001, B0061 = brick_000061, B0071 = brick_000071, B0042 = brick_000042, B0112 = brick_000112, B0105 = brick_000105, B0075 = brick_000075, B0007 = brick_000007, B0010 = brick_000010, B0069 = brick_000069, B0073 = brick_000073, B0036 = brick_000036, B0009 = brick_000009, B0068 = brick_000068, B0054 = brick_000054, B0060 = brick_000060, B0089 = brick_000089, B0034 = brick_000034, B0049 = brick_000049, B0022 = brick_000022, B0085 = brick_000085, B0058 = brick_000058, B0106 = brick_000106, B0086 = brick_000086, B0050 = brick_000050

Public evidence summary: 6 public actions; budget 6; {"initialFacts": ["start"], "goalFacts": ["done-6"], "absentFacts": []}

Reference output: {"actionIds": ["step-1", "step-2", "step-3", "step-4", "step-5", "step-6"]}

Derivation: public precondition solver

Equivalence: any legal sequence reaching all goals within budget; display-only semantics

Review: pending required human review.

39. **ld2-omr-42066-restore-instance-1.** The display copy hides B1144. Restore only that numbered instance, preserving all other instances and source poses.

restore-instance; scene-edit; procedural; actions.

Operands: B0001 = brick_000001, B1144 = brick_001144

Public evidence summary: 2 public actions; budget 1; {"initialFacts": ["missing-brick_001144", "present-brick_000001"], "goalFacts": ["present-brick_001144", "present-brick_000001"], "absentFacts": ["missing-brick_001144"]}

Reference output: {"actionIds": ["restore-B1144"]}

Derivation: public precondition solver

Equivalence: any legal sequence reaching all goals within budget; display-only semantics

Review: machine checks passed; no human certification.

40. **ld2-omr-42066-restore-instance-2.** The display copy hides B0286. Restore only that numbered instance, preserving all other instances and source poses.

restore-instance; scene-edit; procedural; actions.

Operands: B0001 = brick_000001, B0286 = brick_000286

Public evidence summary: 2 public actions; budget 1; {"initialFacts": ["missing-brick_000286", "present-brick_000001"], "goalFacts": ["present-brick_000286", "present-brick_000001"], "absentFacts": ["missing-brick_000286"]}

Reference output: {"actionIds": ["restore-B0286"]}

Derivation: public precondition solver

Equivalence: any legal sequence reaching all goals within budget; display-only semantics

Review: machine checks passed; no human certification.

41. **ld2-omr-42066-restore-instance-3.** The display copy hides B0606. Restore only that numbered instance, preserving all other instances and source poses.

restore-instance; scene-edit; procedural; actions.

Operands: B0001 = brick_000001, B0606 = brick_000606

Public evidence summary: 2 public actions; budget 1; {"initialFacts": ["missing-brick_000606", "present-brick_000001"], "goalFacts": ["present-brick_000606", "present-brick_000001"], "absentFacts": ["missing-brick_000606"]}

Reference output: {"actionIds": ["restore-B0606"]}

Derivation: public precondition solver

Equivalence: any legal sequence reaching all goals within budget; display-only semantics

Review: machine checks passed; no human certification.

30 10213: Shuttle Adventure

1254 source instances; 34 tasks; D4 size band. 191 type strings; 21 color codes; 8 unsupported instances; 63 intersection candidates. 0 expanded author steps. Independent reparse: passed.



Attribution. Edward Carman [EdmanZA]. CC BY 2.0.
<https://library.ldraw.org/omr/sets/870>

Source SHA256. e295434c8fc8c067e6cd01c398dccd3059a38799f03a6cecd9319ecf16d909

Connector records. stud: 3403; axle: 278; hinge: 35; fixed: 46; ball: 0

Public input lock. 4507454adaa976aa47da6667cc340c69548b3b8358a21db12a1c31b2d33ae60e

Status. Source preservation does not certify stability. Reference outputs below are internal review material. Full model inputs are the versioned public JSON and numbered PNGs.

1. **ld2-10213-color-1.** Identify the original color of B0957; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B0957 = brick_000957

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10213-color-1.png; structured input = {}.

Options: A: Trans Brown; B: White; C: Dark Blue; D: Black

Reference output: {"choiceId": "D"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

2. **ld2-10213-shape-match-1.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0957?
shape-match; visual; atomic; single-choice.

Operands: B0137 = brick_000137, B0893 = brick_000893, B0957 = brick_000957, B0249 = brick_000249, B0782 = brick_000782

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10213-shape-match-1.png; structured input = {}.

Options: A: B0782; B: B0893; C: B0137; D: B0249

Reference output: {"choiceId": "C"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

3. **ld2-10213-distance-1.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0957?

distance; source-data; atomic; single-choice.

Operands: B0762 = brick_000762, B0957 = brick_000957, B0706 = brick_000706, B0713 = brick_000713

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0957": [-4, 297.6, 23.2], "B0713": [-16.1, 237.6, 7.2], "B0706": [0, 113.452, 13.3412], "B0762": [-20, 237.6, 21.6]}}

Options: A: B0762; B: B0706; C: B0713

Reference output: {"choiceId": "A"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: pending required human review.

4. **ld2-10213-color-2.** Identify the original color of B0396; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B0396 = brick_000396

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10213-color-2.png; structured input = {}.

Options: A: Dark Blue; B: Reddish Brown; C: Dark Bluish Grey; D: Light Bluish Grey

Reference output: {"choiceId": "C"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

5. **ld2-10213-shape-match-2.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0396?

shape-match; visual; atomic; single-choice.

Operands: B0388 = brick_000388, B1086 = brick_001086, B0268 = brick_000268, B0396 = brick_000396, B0469 = brick_000469

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10213-shape-match-2.png; structured input = {}.

Options: A: B1086; B: B0268; C: B0469; D: B0388

Reference output: {"choiceId": "D"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

6. **ld2-10213-distance-2.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0396?

distance; source-data; atomic; single-choice.

Operands: B0106 = brick_000106, B0394 = brick_000394, B0396 = brick_000396, B1116 = brick_001116

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0396": [36, 325.6, -34.4], "B0394": [36, 277.6, -34.4], "B0106": [130.111595, 62.1388, -141.423595], "B1116": [-21.0432, 41.9484, 41.6]}}

Options: A: B0106; B: B1116; C: B0394

Reference output: {"choiceId": "C"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

7. **ld2-10213-color-3.** Identify the original color of B0228; numbered isolation is allowed.
 color; visual; atomic; single-choice.
Operands: B0228 = brick_000228
Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10213-color-3.png; structured input = {}.
Options: A: Trans Clear; B: Tan; C: Dark Brown; D: Reddish Brown
Reference output: {"choiceId": "D"}
Derivation: source colorName by rendered B-label; source oracle only
Equivalence: exact declared response
Review: pending required human review.
8. **ld2-10213-shape-match-3.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0228?
 shape-match; visual; atomic; single-choice.
Operands: B0921 = brick_000921, B0237 = brick_000237, B0228 = brick_000228, B0303 = brick_000303, B1084 = brick_001084
Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10213-shape-match-3.png; structured input = {}.
Options: A: B0237; B: B0921; C: B0303; D: B1084
Reference output: {"choiceId": "C"}
Derivation: source partNumber equality across labeled candidates; source oracle only
Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending
Review: pending required human review.
9. **ld2-10213-distance-3.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0228?
 distance; source-data; atomic; single-choice.
Operands: B0228 = brick_000228, B0150 = brick_000150, B0732 = brick_000732, B1155 = brick_001155
Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0228": [0, 266.4, -40], "B0150": [-12, 109.6, -15.2], "B0732": [0, 69.7, 23.2], "B1155": [4, 209.6, 15.2]}}
Options: A: B0732; B: B1155; C: B0150
Reference output: {"choiceId": "B"}
Derivation: Euclidean norm from public rounded centers
Equivalence: exact declared response
Review: machine checks passed; no human certification.
10. **ld2-10213-color-4.** Identify the original color of B0987; numbered isolation is allowed.
 color; visual; atomic; single-choice.
Operands: B0987 = brick_000987
Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10213-color-4.png; structured input = {}.
Options: A: Blue; B: White; C: Metallic Gold; D: Trans Clear
Reference output: {"choiceId": "B"}
Derivation: source colorName by rendered B-label; source oracle only
Equivalence: exact declared response
Review: pending required human review.
11. **ld2-10213-shape-match-4.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0987?
 shape-match; visual; atomic; single-choice.
Operands: B0987 = brick_000987, B0580 = brick_000580, B0189 = brick_000189, B0348 = brick_000348, B0260 = brick_000260
Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10213-shape-match-4.png; structured input = {}.
Options: A: B0260; B: B0580; C: B0348; D: B0189
Reference output: {"choiceId": "C"}
Derivation: source partNumber equality across labeled candidates; source oracle only
Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending
Review: pending required human review.

12. **ld2-10213-distance-4.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0987?
distance; source-data; atomic; single-choice.
Operands: B0987 = brick_000987, B0260 = brick_000260, B0588 = brick_000588, B0146 = brick_000146
Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0987": [-12, 117.6, 16], "B0588": [44, 109.6, -2.4], "B0146": [-24.8, 109.6, -52], "B0260": [20, 349.6, -20]}}
Options: A: B0588; B: B0146; C: B0260
Reference output: {"choiceId": "A"}
Derivation: Euclidean norm from public rounded centers
Equivalence: exact declared response
Review: pending required human review.
13. **ld2-10213-color-5.** Identify the original color of B0984; numbered isolation is allowed.
color; visual; atomic; single-choice.
Operands: B0984 = brick_000984
Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10213-color-5.png; structured input = {}.
Options: A: Metallic Gold; B: Yellow; C: Trans Red; D: Black
Reference output: {"choiceId": "D"}
Derivation: source colorName by rendered B-label; source oracle only
Equivalence: exact declared response
Review: pending required human review.
14. **ld2-10213-shape-match-5.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0984?
shape-match; visual; atomic; single-choice.
Operands: B1072 = brick_001072, B0984 = brick_000984, B0926 = brick_000926, B0356 = brick_000356, B0881 = brick_000881
Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10213-shape-match-5.png; structured input = {}.
Options: A: B1072; B: B0356; C: B0881; D: B0926
Reference output: {"choiceId": "D"}
Derivation: source partNumber equality across labeled candidates; source oracle only
Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending
Review: pending required human review.
15. **ld2-10213-distance-5.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0984?
distance; source-data; atomic; single-choice.
Operands: B1060 = brick_001060, B0984 = brick_000984, B0273 = brick_000273, B1233 = brick_001233
Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0984": [-20, 301.6, 4], "B1060": [-20, 300, 22.4], "B0273": [20, 349.6, -28], "B1233": [-30.6, 5.8, 170.8]}}
Options: A: B1060; B: B0273; C: B1233
Reference output: {"choiceId": "A"}
Derivation: Euclidean norm from public rounded centers
Equivalence: exact declared response
Review: machine checks passed; no human certification.
16. **ld2-10213-interface-1.** Classify the supplied connector record for B0645 and B0800. This does not certify load capacity.
interface; connector-graph; metacognitive; single-choice.
Operands: B0645 = brick_000645, B0800 = brick_000800
Public evidence summary: {"connectorRecord": {"family": "axle", "aConnector": 1, "bConnector": 0}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
Options: A: axle; B: fixed; C: stud; D: ball; E: hinge
Reference output: {"choiceId": "A"}

Derivation: public connectorRecord.family equality

Equivalence: exact declared response

Review: machine checks passed; no human certification.

17. **ld2-10213-interface-2.** Classify the supplied connector record for B0723 and B0724. This does not certify load capacity. interface; connector-graph; metacognitive; single-choice.

Operands: B0723 = brick_000723, B0724 = brick_000724

Public evidence summary: {"connectorRecord": {"family": "fixed", "aConnector": 1, "bConnector": 0}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}

Options: A: axle; B: fixed; C: hinge; D: ball; E: stud

Reference output: {"choiceId": "B"}

Derivation: public connectorRecord.family equality

Equivalence: exact declared response

Review: machine checks passed; no human certification.

18. **ld2-10213-interface-3.** Classify the supplied connector record for B1248 and B1244. This does not certify load capacity. interface; connector-graph; metacognitive; single-choice.

Operands: B1248 = brick_001248, B1244 = brick_001244

Public evidence summary: {"connectorRecord": {"family": "hinge", "aConnector": 0, "bConnector": 5}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}

Options: A: axle; B: hinge; C: fixed; D: ball; E: stud

Reference output: {"choiceId": "B"}

Derivation: public connectorRecord.family equality

Equivalence: exact declared response

Review: machine checks passed; no human certification.

19. **ld2-10213-interface-4.** Classify the supplied connector record for B0375 and B0391. This does not certify load capacity. interface; connector-graph; metacognitive; single-choice.

Operands: B0375 = brick_000375, B0391 = brick_000391

Public evidence summary: {"connectorRecord": {"family": "stud", "aConnector": 68, "bConnector": 9}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}

Options: A: fixed; B: stud; C: axle; D: hinge; E: ball

Reference output: {"choiceId": "B"}

Derivation: public connectorRecord.family equality

Equivalence: exact declared response

Review: machine checks passed; no human certification.

20. **ld2-10213-neighbors-1.** Select every candidate directly adjacent to B0219 in the supplied connector graph. neighbors; connector-graph; metacognitive; multiple-choice.

Operands: B0446 = brick_000446, B0178 = brick_000178, B0191 = brick_000191, B0984 = brick_000984, B0219 = brick_000219, B0196 = brick_000196, B0224 = brick_000224, B0181 = brick_000181, B0187 = brick_000187

Public evidence summary: 6 unique undirected pairs

Options: A: B0191; B: B0181; C: B0984; D: B0196; E: B0187; F: B0446; G: B0224; H: B0178

Reference output: {"choiceIds": ["A", "B", "D", "E", "G", "H"]}

Derivation: public incident edge endpoint set

Equivalence: unordered exact set; duplicate IDs invalid

Review: pending required human review.

21. **ld2-10213-graph-removal-1.** Delete B0219 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
graph-removal; connector-graph; graph-internal; integer.
Operands: B0446 = brick_000446, B0196 = brick_000196, B0224 = brick_000224, B0984 = brick_000984, B0178 = brick_000178, B0191 = brick_000191, B0181 = brick_000181, B0219 = brick_000219, B0187 = brick_000187
Public evidence summary: 6 unique undirected pairs; 9 nodes
Reference output: {"value": 8}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: pending required human review.
22. **ld2-10213-neighbors-2.** Select every candidate directly adjacent to B0487 in the supplied connector graph.
neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0476 = brick_000476, B0487 = brick_000487, B0488 = brick_000488, B1002 = brick_001002, B0310 = brick_000310
Public evidence summary: 2 unique undirected pairs
Options: A: B0310; B: B1002; C: B0476; D: B0488
Reference output: {"choiceIds": ["C", "D"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: machine checks passed; no human certification.
23. **ld2-10213-graph-removal-2.** Delete B0487 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
graph-removal; connector-graph; graph-internal; integer.
Operands: B0487 = brick_000487, B0310 = brick_000310, B1002 = brick_001002, B0488 = brick_000488, B0476 = brick_000476
Public evidence summary: 2 unique undirected pairs; 5 nodes
Reference output: {"value": 4}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: machine checks passed; no human certification.
24. **ld2-10213-neighbors-3.** Select every candidate directly adjacent to B0445 in the supplied connector graph.
neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0445 = brick_000445, B0570 = brick_000570, B0449 = brick_000449, B0440 = brick_000440, B0601 = brick_000601
Public evidence summary: 2 unique undirected pairs
Options: A: B0440; B: B0570; C: B0601; D: B0449
Reference output: {"choiceIds": ["A", "D"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: machine checks passed; no human certification.
25. **ld2-10213-graph-removal-3.** Delete B0445 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
graph-removal; connector-graph; graph-internal; integer.
Operands: B0601 = brick_000601, B0570 = brick_000570, B0440 = brick_000440, B0449 = brick_000449, B0445 = brick_000445
Public evidence summary: 2 unique undirected pairs; 5 nodes
Reference output: {"value": 4}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: machine checks passed; no human certification.

26. **Id2-10213-neighbors-4.** Select every candidate directly adjacent to B0668 in the supplied connector graph.
 neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0759 = brick_000759, B0418 = brick_000418, B0672 = brick_000672, B0669 = brick_000669, B0668 = brick_000668
Public evidence summary: 2 unique undirected pairs
Options: A: B0672; B: B0418; C: B0759; D: B0669
Reference output: {"choiceIds": ["A", "D"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: pending required human review.
27. **Id2-10213-graph-removal-4.** Delete B0668 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
 graph-removal; connector-graph; graph-internal; integer.
Operands: B0759 = brick_000759, B0672 = brick_000672, B0669 = brick_000669, B0418 = brick_000418, B0668 = brick_000668
Public evidence summary: 2 unique undirected pairs; 5 nodes
Reference output: {"value": 4}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: pending required human review.
28. **Id2-10213-neighbors-5.** Select every candidate directly adjacent to B0007 in the supplied connector graph.
 neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0007 = brick_000007, B0023 = brick_000023, B0001 = brick_000001, B0946 = brick_000946, B1163 = brick_001163, B0025 = brick_000025
Public evidence summary: 3 unique undirected pairs
Options: A: B0946; B: B0023; C: B1163; D: B0001; E: B0025
Reference output: {"choiceIds": ["B", "D", "E"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: pending required human review.
29. **Id2-10213-graph-removal-5.** Delete B0007 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
 graph-removal; connector-graph; graph-internal; integer.
Operands: B0946 = brick_000946, B0007 = brick_000007, B0023 = brick_000023, B0001 = brick_000001, B0025 = brick_000025, B1163 = brick_001163
Public evidence summary: 3 unique undirected pairs; 6 nodes
Reference output: {"value": 5}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: pending required human review.
30. **Id2-10213-coverage-1.** Which numbered instance lacks a connector definition in the coverage record, so this audit cannot establish that it is floating?
 coverage; source-data; metacognitive; single-choice.
Operands: B0396 = brick_000396, B0957 = brick_000957, B0706 = brick_000706, B0228 = brick_000228
Public evidence summary: {"coverage": [{"label": "B0228", "supported": true}, {"label": "B0396", "supported": true}, {"label": "B0957", "supported": true}, {"label": "B0706", "supported": false}]}
Options: A: B0957; B: B0228; C: B0396; D: B0706
Reference output: {"choiceId": "D"}

Derivation: public supported=false record

Equivalence: exact declared response

Review: pending required human review.

31. **ld2-10213-evidence-limit-1.** Only source geometry, connector matching and mesh intersection audits are available. Without mass, friction, material or dynamic tests, is gravitational stability established?

evidence-limit; source-data; metacognitive; single-choice.

Operands: none

Public evidence summary: {"available": ["Original CAD", "Connector inference", "Inset-mesh intersection candidates"]}

Options: A: Not established by this evidence; B: Established

Reference output: {"choiceId": "A"}

Derivation: fixed epistemic control label

Equivalence: exact declared response

Review: machine checks passed; no human certification.

32. **ld2-10213-restore-instance-1.** The display copy hides B0957. Restore only that numbered instance, preserving all other instances and source poses.

restore-instance; scene-edit; procedural; actions.

Operands: B0001 = brick_000001, B0957 = brick_000957

Public evidence summary: 2 public actions; budget 1; {"initialFacts": ["missing-brick_000957", "present-brick_000001"], "goalFacts": ["present-brick_000957", "present-brick_000001"], "absentFacts": ["missing-brick_000957"]}

Reference output: {"actionIds": ["restore-B0957"]}

Derivation: public precondition solver

Equivalence: any legal sequence reaching all goals within budget; display-only semantics

Review: machine checks passed; no human certification.

33. **ld2-10213-restore-instance-2.** The display copy hides B0396. Restore only that numbered instance, preserving all other instances and source poses.

restore-instance; scene-edit; procedural; actions.

Operands: B0396 = brick_000396, B0001 = brick_000001

Public evidence summary: 2 public actions; budget 1; {"initialFacts": ["missing-brick_000396", "present-brick_000001"], "goalFacts": ["present-brick_000396", "present-brick_000001"], "absentFacts": ["missing-brick_000396"]}

Reference output: {"actionIds": ["restore-B0396"]}

Derivation: public precondition solver

Equivalence: any legal sequence reaching all goals within budget; display-only semantics

Review: machine checks passed; no human certification.

34. **ld2-10213-restore-instance-3.** The display copy hides B0228. Restore only that numbered instance, preserving all other instances and source poses.

restore-instance; scene-edit; procedural; actions.

Operands: B0001 = brick_000001, B0228 = brick_000228

Public evidence summary: 2 public actions; budget 1; {"initialFacts": ["missing-brick_000228", "present-brick_000001"], "goalFacts": ["present-brick_000228", "present-brick_000001"], "absentFacts": ["missing-brick_000228"]}

Reference output: {"actionIds": ["restore-B0228"]}

Derivation: public precondition solver

Equivalence: any legal sequence reaching all goals within budget; display-only semantics

Review: machine checks passed; no human certification.

31 10265: Ford Mustang

1357 source instances; 40 tasks; D4 size band. 254 type strings; 19 color codes; 10 unsupported instances; 109 intersection candidates. 82 expanded author steps. Independent reparse: passed.



Attribution. Ulrich Röder [UR]. CC BY 2.0.

<https://library.ldraw.org/library/omr/10265-1.mpd>

Source SHA256. e9a2eb1b84d343a6faa9861d186159cf5d1fa1e202366eae1e76287df114ea95

Connector records. stud: 3166; axle: 344; hinge: 12; fixed: 0; ball: 4

Public input lock. 40c89a87d4ba0d068bcf063872a92bbaf1f1b5105d03b94158402fda60aa38c2

Status. Source preservation does not certify stability. Reference outputs below are internal review material. Full model inputs are the versioned public JSON and numbered PNGs.

1. **ld2-omr-10265-color-1.** Identify the original color of B0886; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B0886 = brick_000886

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-10265-color-1.png; structured input = {}.

Options: A: Black; B: Dark Tan; C: Dark Orange; D: Light Bluish Grey

Reference output: {"choiceId": "D"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

2. **ld2-omr-10265-shape-match-1.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0886?

shape-match; visual; atomic; single-choice.

Operands: B0882 = brick_000882, B0392 = brick_000392, B0205 = brick_000205, B0886 = brick_000886, B0060 = brick_000060

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-10265-shape-match-1.png; structured input = {}.

Options: A: B0392; B: B0060; C: B0205; D: B0882

Reference output: {"choiceId": "D"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

3. **ld2-omr-10265-distance-1.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0886?

distance; source-data; atomic; single-choice.

Operands: B0886 = brick_000886, B0599 = brick_000599, B0275 = brick_000275, B0105 = brick_000105

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0886": [59.654664, 31.2, 177.179945], "B0275": [-56, 37.6, 84], "B0105": [56, 50.4, -4], "B0599": [12.1, 21.6, -80]}}

Options: A: B0275; B: B0599; C: B0105

Reference output: {"choiceId": "A"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

4. **ld2-omr-10265-color-2.** Identify the original color of B1090; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B1090 = brick_001090

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-10265-color-2.png; structured input = {}.

Options: A: Tan; B: Trans Red; C: Yellow; D: Aqua

Reference output: {"choiceId": "A"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

5. **ld2-omr-10265-shape-match-2.** Ignoring color and pose, which numbered instance has the same LDraw part type as B1090?

shape-match; visual; atomic; single-choice.

Operands: B0527 = brick_000527, B1209 = brick_001209, B1090 = brick_001090, B0425 = brick_000425, B1089 = brick_001089

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-10265-shape-match-2.png; structured input = {}.

Options: A: B0425; B: B1089; C: B0527; D: B1209

Reference output: {"choiceId": "B"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

6. **ld2-omr-10265-distance-2.** Using the supplied geometry bounding-box centers, which candidate is nearest to B1090?

distance; source-data; atomic; single-choice.

Operands: B1090 = brick_001090, B0398 = brick_000398, B0057 = brick_000057, B1008 = brick_001008

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B1090": [48, 56.8, -40], "B0398": [58.32, 21.6, 180], "B0057": [53.6, 24, 24], "B1008": [-27.2, 29.453016, 10.473528]}}

Options: A: B0057; B: B1008; C: B0398

Reference output: {"choiceId": "A"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

7. **ld2-omr-10265-color-3.** Identify the original color of B0079; numbered isolation is allowed.
 color; visual; atomic; single-choice.
Operands: B0079 = brick_000079
Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-10265-color-3.png; structured input = {}.
Options: A: Aqua; B: Dark Orange; C: Tan; D: Green
Reference output: {"choiceId": "C"}
Derivation: source colorName by rendered B-label; source oracle only
Equivalence: exact declared response
Review: pending required human review.
8. **ld2-omr-10265-shape-match-3.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0079?
 shape-match; visual; atomic; single-choice.
Operands: B0269 = brick_000269, B0311 = brick_000311, B0759 = brick_000759, B0079 = brick_000079, B0283 = brick_000283
Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-10265-shape-match-3.png; structured input = {}.
Options: A: B0759; B: B0283; C: B0311; D: B0269
Reference output: {"choiceId": "D"}
Derivation: source partNumber equality across labeled candidates; source oracle only
Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending
Review: pending required human review.
9. **ld2-omr-10265-distance-3.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0079?
 distance; source-data; atomic; single-choice.
Operands: B1326 = brick_001326, B0079 = brick_000079, B0940 = brick_000940, B1261 = brick_001261
Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0079": [52, 21.6, 40], "B0940": [20, 32.49576, 70.615832], "B1326": [-48, 34.4, 165.8], "B1261": [-33.524999, 50.4, 116.479116]}}
Options: A: B1261; B: B1326; C: B0940
Reference output: {"choiceId": "C"}
Derivation: Euclidean norm from public rounded centers
Equivalence: exact declared response
Review: machine checks passed; no human certification.
10. **ld2-omr-10265-color-4.** Identify the original color of B0783; numbered isolation is allowed.
 color; visual; atomic; single-choice.
Operands: B0783 = brick_000783
Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-10265-color-4.png; structured input = {}.
Options: A: Light Bluish Grey; B: Blue; C: Brown; D: Yellow
Reference output: {"choiceId": "A"}
Derivation: source colorName by rendered B-label; source oracle only
Equivalence: exact declared response
Review: pending required human review.
11. **ld2-omr-10265-shape-match-4.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0783?
 shape-match; visual; atomic; single-choice.
Operands: B0813 = brick_000813, B0758 = brick_000758, B0783 = brick_000783, B0327 = brick_000327, B1206 = brick_001206
Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-10265-shape-match-4.png; structured input = {}.
Options: A: B0813; B: B0327; C: B1206; D: B0758

Reference output: {"choiceId": "D"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

12. **ld2-omr-10265-distance-4.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0783?

distance; source-data; atomic; single-choice.

Operands: B0157 = brick_000157, B0783 = brick_000783, B0683 = brick_000683, B1240 = brick_001240

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0783": [-40, 49.6, 92], "B1240": [40, 52.303984, 68.446956], "B0157": [56, 50.4, 128], "B0683": [-55.92, 12, 40]}}

Options: A: B0157; B: B1240; C: B0683

Reference output: {"choiceId": "C"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

13. **ld2-omr-10265-color-5.** Identify the original color of B1081; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B1081 = brick_001081

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-10265-color-5.png; structured input = {}.

Options: A: Light Grey; B: Dark Blue; C: Dark Bluish Grey; D: Tan

Reference output: {"choiceId": "C"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

14. **ld2-omr-10265-shape-match-5.** Ignoring color and pose, which numbered instance has the same LDraw part type as B1081?

shape-match; visual; atomic; single-choice.

Operands: B0528 = brick_000528, B1065 = brick_001065, B1265 = brick_001265, B1081 = brick_001081, B0546 = brick_000546

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-10265-shape-match-5.png; structured input = {}.

Options: A: B1265; B: B1065; C: B0546; D: B0528

Reference output: {"choiceId": "B"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

15. **ld2-omr-10265-distance-5.** Using the supplied geometry bounding-box centers, which candidate is nearest to B1081?

distance; source-data; atomic; single-choice.

Operands: B0977 = brick_000977, B0422 = brick_000422, B0379 = brick_000379, B1081 = brick_001081

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B1081": [-44, 53.6, -80], "B0422": [-24, 85.6, 28], "B0379": [-60.72, 18.4, 168], "B0977": [-24, 54.433218, 54.679544]}}

Options: A: B0422; B: B0977; C: B0379

Reference output: {"choiceId": "A"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

16. **ld2-omr-10265-interface-1.** Classify the supplied connector record for B0013 and B0010. This does not certify load capacity.
- interface; connector-graph; metacognitive; single-choice.
- Operands:* B0010 = brick_000010, B0013 = brick_000013
- Public evidence summary:* {"connectorRecord": {"family": "axle", "aConnector": 14, "bConnector": 1}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
- Options:* A: hinge; B: stud; C: axle; D: ball; E: fixed
- Reference output:* {"choiceId": "C"}
- Derivation:* public connectorRecord.family equality
- Equivalence:* exact declared response
- Review:* pending required human review.
17. **ld2-omr-10265-interface-2.** Classify the supplied connector record for B0548 and B0556. This does not certify load capacity.
- interface; connector-graph; metacognitive; single-choice.
- Operands:* B0548 = brick_000548, B0556 = brick_000556
- Public evidence summary:* {"connectorRecord": {"family": "ball", "aConnector": 3, "bConnector": 7}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
- Options:* A: axle; B: stud; C: fixed; D: hinge; E: ball
- Reference output:* {"choiceId": "E"}
- Derivation:* public connectorRecord.family equality
- Equivalence:* exact declared response
- Review:* machine checks passed; no human certification.
18. **ld2-omr-10265-interface-3.** Classify the supplied connector record for B0064 and B0065. This does not certify load capacity.
- interface; connector-graph; metacognitive; single-choice.
- Operands:* B0064 = brick_000064, B0065 = brick_000065
- Public evidence summary:* {"connectorRecord": {"family": "hinge", "aConnector": 1, "bConnector": 2}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
- Options:* A: ball; B: hinge; C: axle; D: stud; E: fixed
- Reference output:* {"choiceId": "B"}
- Derivation:* public connectorRecord.family equality
- Equivalence:* exact declared response
- Review:* machine checks passed; no human certification.
19. **ld2-omr-10265-interface-4.** Classify the supplied connector record for B0164 and B0165. This does not certify load capacity.
- interface; connector-graph; metacognitive; single-choice.
- Operands:* B0164 = brick_000164, B0165 = brick_000165
- Public evidence summary:* {"connectorRecord": {"family": "stud", "aConnector": 12, "bConnector": 7}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
- Options:* A: stud; B: axle; C: hinge; D: fixed; E: ball
- Reference output:* {"choiceId": "A"}
- Derivation:* public connectorRecord.family equality
- Equivalence:* exact declared response
- Review:* machine checks passed; no human certification.

20. **ld2-omr-10265-neighbors-1.** Select every candidate directly adjacent to B0145 in the supplied connector graph.
 neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0296 = brick_000296, B0148 = brick_000148, B1255 = brick_001255, B0145 = brick_000145, B0147 = brick_000147
Public evidence summary: 2 unique undirected pairs
Options: A: B0296; B: B0147; C: B0148; D: B1255
Reference output: {"choiceIds": ["B", "C"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: pending required human review.
21. **ld2-omr-10265-graph-removal-1.** Delete B0145 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
 graph-removal; connector-graph; graph-internal; integer.
Operands: B0148 = brick_000148, B0147 = brick_000147, B0296 = brick_000296, B0145 = brick_000145, B1255 = brick_001255
Public evidence summary: 2 unique undirected pairs; 5 nodes
Reference output: {"value": 4}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: pending required human review.
22. **ld2-omr-10265-neighbors-2.** Select every candidate directly adjacent to B0745 in the supplied connector graph.
 neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0740 = brick_000740, B0367 = brick_000367, B0751 = brick_000751, B0772 = brick_000772, B0745 = brick_000745
Public evidence summary: 2 unique undirected pairs
Options: A: B0740; B: B0367; C: B0772; D: B0751
Reference output: {"choiceIds": ["A", "D"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: machine checks passed; no human certification.
23. **ld2-omr-10265-graph-removal-2.** Delete B0745 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
 graph-removal; connector-graph; graph-internal; integer.
Operands: B0751 = brick_000751, B0367 = brick_000367, B0745 = brick_000745, B0740 = brick_000740, B0772 = brick_000772
Public evidence summary: 2 unique undirected pairs; 5 nodes
Reference output: {"value": 4}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: machine checks passed; no human certification.
24. **ld2-omr-10265-neighbors-3.** Select every candidate directly adjacent to B1252 in the supplied connector graph.
 neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B1258 = brick_001258, B0609 = brick_000609, B1257 = brick_001257, B1252 = brick_001252, B0805 = brick_000805
Public evidence summary: 2 unique undirected pairs
Options: A: B1258; B: B0805; C: B0609; D: B1257
Reference output: {"choiceIds": ["A", "D"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: pending required human review.

25. **ld2-omr-10265-graph-removal-3.** Delete B1252 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
graph-removal; connector-graph; graph-internal; integer.
Operands: B1252 = brick_001252, B0805 = brick_000805, B1258 = brick_001258, B1257 = brick_001257, B0609 = brick_000609
Public evidence summary: 2 unique undirected pairs; 5 nodes
Reference output: {"value": 4}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: pending required human review.
26. **ld2-omr-10265-neighbors-4.** Select every candidate directly adjacent to B1248 in the supplied connector graph.
neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B1342 = brick_001342, B1294 = brick_001294, B1324 = brick_001324, B1248 = brick_001248, B1250 = brick_001250
Public evidence summary: 2 unique undirected pairs
Options: A: B1342; B: B1250; C: B1294; D: B1324
Reference output: {"choiceIds": ["A", "B"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: machine checks passed; no human certification.
27. **ld2-omr-10265-graph-removal-4.** Delete B1248 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
graph-removal; connector-graph; graph-internal; integer.
Operands: B1324 = brick_001324, B1250 = brick_001250, B1248 = brick_001248, B1342 = brick_001342, B1294 = brick_001294
Public evidence summary: 2 unique undirected pairs; 5 nodes
Reference output: {"value": 4}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: machine checks passed; no human certification.
28. **ld2-omr-10265-neighbors-5.** Select every candidate directly adjacent to B0979 in the supplied connector graph.
neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0979 = brick_000979, B0980 = brick_000980, B0719 = brick_000719, B1133 = brick_001133, B0964 = brick_000964
Public evidence summary: 2 unique undirected pairs
Options: A: B0980; B: B0964; C: B0719; D: B1133
Reference output: {"choiceIds": ["A", "B"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: machine checks passed; no human certification.
29. **ld2-omr-10265-graph-removal-5.** Delete B0979 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
graph-removal; connector-graph; graph-internal; integer.
Operands: B1133 = brick_001133, B0980 = brick_000980, B0979 = brick_000979, B0719 = brick_000719, B0964 = brick_000964
Public evidence summary: 2 unique undirected pairs; 5 nodes
Reference output: {"value": 4}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: machine checks passed; no human certification.

30. **ld2-omr-10265-coverage-1.** Which numbered instance lacks a connector definition in the coverage record, so this audit cannot establish that it is floating?
 coverage; source-data; metacognitive; single-choice.
Operands: B0712 = brick_000712, B0886 = brick_000886, B0079 = brick_000079, B1090 = brick_001090
Public evidence summary: {"coverage": [{"label": "B0886", "supported": true}, {"label": "B1090", "supported": true}, {"label": "B0079", "supported": true}, {"label": "B0712", "supported": false}]}
Options: A: B0079; B: B0712; C: B0886; D: B1090
Reference output: {"choiceId": "B"}
Derivation: public supported=false record
Equivalence: exact declared response
Review: pending required human review.
31. **ld2-omr-10265-evidence-limit-1.** Only source geometry, connector matching and mesh intersection audits are available. Without mass, friction, material or dynamic tests, is gravitational stability established?
 evidence-limit; source-data; metacognitive; single-choice.
Operands: none
Public evidence summary: {"available": ["Original CAD", "Connector inference", "Inset-mesh intersection candidates"]}
Options: A: Not established by this evidence; B: Established
Reference output: {"choiceId": "A"}
Derivation: fixed epistemic control label
Equivalence: exact declared response
Review: machine checks passed; no human certification.
32. **ld2-omr-10265-source-step-1.** At which expanded author step does B0884 first appear in the supplied source-step table? Source order is not a collision-free assembly proof.
 source-step; source-data; procedural; single-choice.
Operands: B0884 = brick_000884, B0001 = brick_000001
Public evidence summary: 2 supplied author-step records; exact table in public JSON
Options: A: 54; B: 61; C: 66; D: 74
Reference output: {"choiceId": "A"}
Derivation: minimum public step index containing target
Equivalence: exact declared response
Review: pending required human review.
33. **ld2-omr-10265-source-step-2.** At which expanded author step does B0719 first appear in the supplied source-step table? Source order is not a collision-free assembly proof.
 source-step; source-data; procedural; single-choice.
Operands: B0719 = brick_000719, B0001 = brick_000001
Public evidence summary: 2 supplied author-step records; exact table in public JSON
Options: A: 52; B: 46; C: 34; D: 72
Reference output: {"choiceId": "B"}
Derivation: minimum public step index containing target
Equivalence: exact declared response
Review: pending required human review.
34. **ld2-omr-10265-source-step-3.** At which expanded author step does B0531 first appear in the supplied source-step table? Source order is not a collision-free assembly proof.
 source-step; source-data; procedural; single-choice.
Operands: B0531 = brick_000531, B0001 = brick_000001
Public evidence summary: 2 supplied author-step records; exact table in public JSON

Options: A: 27; B: 23; C: 7; D: 31

Reference output: {"choiceId": "D"}

Derivation: minimum public step index containing target

Equivalence: exact declared response

Review: pending required human review.

35. **ld2-omr-10265-source-step-4.** At which expanded author step does B0710 first appear in the supplied source-step table?

Source order is not a collision-free assembly proof.

source-step; source-data; procedural; single-choice.

Operands: B0001 = brick_000001, B0710 = brick_000710

Public evidence summary: 2 supplied author-step records; exact table in public JSON

Options: A: 60; B: 80; C: 45; D: 52

Reference output: {"choiceId": "C"}

Derivation: minimum public step index containing target

Equivalence: exact declared response

Review: pending required human review.

36. **ld2-omr-10265-source-step-5.** At which expanded author step does B0028 first appear in the supplied source-step table?

Source order is not a collision-free assembly proof.

source-step; source-data; procedural; single-choice.

Operands: B0028 = brick_000028, B0001 = brick_000001

Public evidence summary: 2 supplied author-step records; exact table in public JSON

Options: A: 10; B: 1; C: 6; D: 9

Reference output: {"choiceId": "A"}

Derivation: minimum public step index containing target

Equivalence: exact declared response

Review: pending required human review.

37. **ld2-omr-10265-source-sequence-1.** Reveal the supplied source-step window in author order. Actions change visibility only, without simulating insertion trajectories.

source-sequence; scene-edit; procedural; actions.

Operands: B0018 = brick_000018, B0005 = brick_000005, B0008 = brick_000008, B0001 = brick_000001, B0017 = brick_000017, B0006 = brick_000006, B0011 = brick_000011, B0010 = brick_000010, B0009 = brick_000009, B0012 = brick_000012, B0003 = brick_000003, B0015 = brick_000015, B0004 = brick_000004, B0014 = brick_000014, B0002 = brick_000002, B0007 = brick_000007, B0013 = brick_000013, B0016 = brick_000016

Public evidence summary: 6 public actions; budget 6; {"initialFacts": ["start"], "goalFacts": ["done-6"], "absentFacts": []}

Reference output: {"actionIds": ["step-1", "step-2", "step-3", "step-4", "step-5", "step-6"]}

Derivation: public precondition solver

Equivalence: any legal sequence reaching all goals within budget; display-only semantics

Review: pending required human review.

38. **ld2-omr-10265-restore-instance-1.** The display copy hides B0886. Restore only that numbered instance, preserving all other instances and source poses.

restore-instance; scene-edit; procedural; actions.

Operands: B0001 = brick_000001, B0886 = brick_000886

Public evidence summary: 2 public actions; budget 1; {"initialFacts": ["missing-brick_000886", "present-brick_000001"], "goalFacts": ["present-brick_000886", "present-brick_000001"], "absentFacts": ["missing-brick_000886"]}

Reference output: {"actionIds": ["restore-B0886"]}

Derivation: public precondition solver

Equivalence: any legal sequence reaching all goals within budget; display-only semantics

Review: pending required human review.

39. **ld2-omr-10265-restore-instance-2.** The display copy hides B1090. Restore only that numbered instance, preserving all other instances and source poses.
- restore-instance; scene-edit; procedural; actions.
- Operands:* B0001 = brick_000001, B1090 = brick_001090
- Public evidence summary:* 2 public actions; budget 1; {"initialFacts": ["missing-brick_001090", "present-brick_000001"], "goalFacts": ["present-brick_001090", "present-brick_000001"], "absentFacts": ["missing-brick_001090"]}
- Reference output:* {"actionIds": ["restore-B1090"]}
- Derivation:* public precondition solver
- Equivalence:* any legal sequence reaching all goals within budget; display-only semantics
- Review:* pending required human review.
40. **ld2-omr-10265-restore-instance-3.** The display copy hides B0079. Restore only that numbered instance, preserving all other instances and source poses.
- restore-instance; scene-edit; procedural; actions.
- Operands:* B0079 = brick_000079, B0001 = brick_000001
- Public evidence summary:* 2 public actions; budget 1; {"initialFacts": ["missing-brick_000079", "present-brick_000001"], "goalFacts": ["present-brick_000079", "present-brick_000001"], "absentFacts": ["missing-brick_000079"]}
- Reference output:* {"actionIds": ["restore-B0079"]}
- Derivation:* public precondition solver
- Equivalence:* any legal sequence reaching all goals within budget; display-only semantics
- Review:* pending required human review.

32 10220: Volkswagen T1 Camper Van

1428 source instances; 39 tasks; D4 size band. 177 type strings; 24 color codes; 100 unsupported instances; 7 intersection candidates. 433 expanded author steps. Independent reparse: passed.



Attribution. Stan Isachenko [angmarec]. CC BY 2.0.

<https://library.ldraw.org/omr/sets/1294>

Source SHA256. ded9a68fc1b569e2965aa67febd225ab9d4b496a5d944247488d64c5d1619373

Connector records. stud: 3438; axle: 142; hinge: 23; fixed: 16; ball: 0

Public input lock. 2068e99d6cb011f7c63a5b633d1a7f10b20f87b836e39c304ac89a84092e5159

Status. Source preservation does not certify stability. Reference outputs below are internal review material. Full model inputs are the versioned public JSON and numbered PNGs.

1. **ld2-10220-color-1.** Identify the original color of B0614; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B0614 = brick_000614

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10220-color-1.png; structured input = {}.

Options: A: Reddish Brown; B: White; C: Pearl Gold; D: Dark Red

Reference output: {"choiceId": "A"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

2. **ld2-10220-distance-1.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0614?

distance; source-data; atomic; single-choice.

Operands: B0977 = brick_000977, B1232 = brick_001232, B0847 = brick_000847, B0614 = brick_000614

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0614": [-24, 53.6, 24], "B0847": [48, 63.2, 216], "B1232": [32, 118.4, 152], "B0977": [36, 76, 128]}}

Options: A: B0977; B: B0847; C: B1232

Reference output: {"choiceId": "A"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

3. **ld2-10220-color-2.** Identify the original color of B1293; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B1293 = brick_001293

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10220-color-2.png; structured input = {}.

Options: A: Bright Light Orange; B: Trans Clear; C: Light Bluish Grey; D: White

Reference output: {"choiceId": "C"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

4. **ld2-10220-shape-match-1.** Ignoring color and pose, which numbered instance has the same LDraw part type as B1293?

shape-match; visual; atomic; single-choice.

Operands: B1423 = brick_001423, B0952 = brick_000952, B1293 = brick_001293, B1161 = brick_001161, B0657 = brick_000657

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10220-shape-match-1.png; structured input = {}.

Options: A: B0952; B: B1161; C: B0657; D: B1423

Reference output: {"choiceId": "B"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

5. **ld2-10220-distance-2.** Using the supplied geometry bounding-box centers, which candidate is nearest to B1293?

distance; source-data; atomic; single-choice.

Operands: B1293 = brick_001293, B1263 = brick_001263, B0550 = brick_000550, B0572 = brick_000572

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B1293": [-12, 122, 76], "B1263": [0, 114.4, 76], "B0550": [-52, 65.6, 156], "B0572": [0, 18.4, 244]}}

Options: A: B0550; B: B0572; C: B1263

Reference output: {"choiceId": "C"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

6. **ld2-10220-color-3.** Identify the original color of B0285; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B0285 = brick_000285

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10220-color-3.png; structured input = {}.

Options: A: Trans Green; B: Light Bluish Grey; C: Yellow; D: Red

Reference output: {"choiceId": "D"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

7. **ld2-10220-shape-match-2.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0285?
 shape-match; visual; atomic; single-choice.
Operands: B0780 = brick_000780, B0415 = brick_000415, B0874 = brick_000874, B0479 = brick_000479, B0285 = brick_000285
Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10220-shape-match-2.png; structured input = {}.
Options: A: B0415; B: B0479; C: B0780; D: B0874
Reference output: {"choiceId": "C"}
Derivation: source partNumber equality across labeled candidates; source oracle only
Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending
Review: pending required human review.
8. **ld2-10220-distance-3.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0285?
 distance; source-data; atomic; single-choice.
Operands: B0285 = brick_000285, B0284 = brick_000284, B0625 = brick_000625, B0043 = brick_000043
Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0285": [-36, 2.4, 244], "B0284": [36, 2.4, 244], "B0043": [25.2, 15.2, 12], "B0625": [8, 45.6, 80.4]}}
Options: A: B0625; B: B0043; C: B0284
Reference output: {"choiceId": "C"}
Derivation: Euclidean norm from public rounded centers
Equivalence: exact declared response
Review: machine checks passed; no human certification.
9. **ld2-10220-color-4.** Identify the original color of B0231; numbered isolation is allowed.
 color; visual; atomic; single-choice.
Operands: B0231 = brick_000231
Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10220-color-4.png; structured input = {}.
Options: A: Red; B: Dark Tan; C: Reddish Brown; D: Bright Light Orange
Reference output: {"choiceId": "A"}
Derivation: source colorName by rendered B-label; source oracle only
Equivalence: exact declared response
Review: pending required human review.
10. **ld2-10220-shape-match-3.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0231?
 shape-match; visual; atomic; single-choice.
Operands: B0743 = brick_000743, B0231 = brick_000231, B0485 = brick_000485, B0255 = brick_000255, B0747 = brick_000747
Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10220-shape-match-3.png; structured input = {}.
Options: A: B0255; B: B0485; C: B0747; D: B0743
Reference output: {"choiceId": "A"}
Derivation: source partNumber equality across labeled candidates; source oracle only
Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending
Review: pending required human review.
11. **ld2-10220-distance-4.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0231?
 distance; source-data; atomic; single-choice.
Operands: B1148 = brick_001148, B0231 = brick_000231, B0875 = brick_000875, B0889 = brick_000889
Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0231": [-28, 2.4, 120], "B0875": [-32, 72.8, 85.4], "B0889": [-20, 95.2, 68], "B1148": [0, 113.6, 192]}}
Options: A: B1148; B: B0889; C: B0875
Reference output: {"choiceId": "C"}
Derivation: Euclidean norm from public rounded centers
Equivalence: exact declared response
Review: pending required human review.

12. **ld2-10220-color-5.** Identify the original color of B0135; numbered isolation is allowed.
 color; visual; atomic; single-choice.
Operands: B0135 = brick_000135
Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10220-color-5.png; structured input = {}.
Options: A: Trans Red; B: Tan; C: Light Bluish Grey; D: Yellow
Reference output: {"choiceId": "C"}
Derivation: source colorName by rendered B-label; source oracle only
Equivalence: exact declared response
Review: pending required human review.
13. **ld2-10220-shape-match-4.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0135?
 shape-match; visual; atomic; single-choice.
Operands: B0439 = brick_000439, B0411 = brick_000411, B0501 = brick_000501, B0702 = brick_000702, B0135 = brick_000135
Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-10220-shape-match-4.png; structured input = {}.
Options: A: B0439; B: B0702; C: B0411; D: B0501
Reference output: {"choiceId": "D"}
Derivation: source partNumber equality across labeled candidates; source oracle only
Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending
Review: pending required human review.
14. **ld2-10220-distance-5.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0135?
 distance; source-data; atomic; single-choice.
Operands: B0377 = brick_000377, B0135 = brick_000135, B0460 = brick_000460, B0543 = brick_000543
Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0135": [20, 21.6, 44], "B0543": [-58.4, 56.8, 136], "B0377": [-36, 40.8, 4], "B0460": [-36, 31.2, 84]}}
Options: A: B0460; B: B0543; C: B0377
Reference output: {"choiceId": "A"}
Derivation: Euclidean norm from public rounded centers
Equivalence: exact declared response
Review: machine checks passed; no human certification.
15. **ld2-10220-interface-1.** Classify the supplied connector record for B1302 and B1270. This does not certify load capacity.
 interface; connector-graph; metacognitive; single-choice.
Operands: B1302 = brick_001302, B1270 = brick_001270
Public evidence summary: {"connectorRecord": {"family": "axle", "aConnector": 1, "bConnector": 0}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
Options: A: hinge; B: axle; C: stud; D: ball; E: fixed
Reference output: {"choiceId": "B"}
Derivation: public connectorRecord.family equality
Equivalence: exact declared response
Review: machine checks passed; no human certification.
16. **ld2-10220-interface-2.** Classify the supplied connector record for B1400 and B1404. This does not certify load capacity.
 interface; connector-graph; metacognitive; single-choice.
Operands: B1400 = brick_001400, B1404 = brick_001404
Public evidence summary: {"connectorRecord": {"family": "fixed", "aConnector": 1, "bConnector": 0}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
Options: A: fixed; B: hinge; C: ball; D: stud; E: axle

Reference output: {"choiceId": "A"}

Derivation: public connectorRecord.family equality

Equivalence: exact declared response

Review: machine checks passed; no human certification.

17. **ld2-10220-interface-3.** Classify the supplied connector record for B0492 and B0491. This does not certify load capacity.
interface; connector-graph; metacognitive; single-choice.

Operands: B0491 = brick_000491, B0492 = brick_000492

Public evidence summary: {"connectorRecord": {"family": "hinge", "aConnector": 1, "bConnector": 2}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}

Options: A: fixed; B: axle; C: ball; D: stud; E: hinge

Reference output: {"choiceId": "E"}

Derivation: public connectorRecord.family equality

Equivalence: exact declared response

Review: machine checks passed; no human certification.

18. **ld2-10220-interface-4.** Classify the supplied connector record for B0529 and B0542. This does not certify load capacity.
interface; connector-graph; metacognitive; single-choice.

Operands: B0542 = brick_000542, B0529 = brick_000529

Public evidence summary: {"connectorRecord": {"family": "stud", "aConnector": 4, "bConnector": 0}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}

Options: A: stud; B: hinge; C: fixed; D: ball; E: axle

Reference output: {"choiceId": "A"}

Derivation: public connectorRecord.family equality

Equivalence: exact declared response

Review: machine checks passed; no human certification.

19. **ld2-10220-neighbors-1.** Select every candidate directly adjacent to B1336 in the supplied connector graph.
neighbors; connector-graph; metacognitive; multiple-choice.

Operands: B1338 = brick_001338, B1337 = brick_001337, B0669 = brick_000669, B0824 = brick_000824, B1339 = brick_001339, B1336 = brick_001336

Public evidence summary: 3 unique undirected pairs

Options: A: B1337; B: B0669; C: B0824; D: B1338; E: B1339

Reference output: {"choiceIds": ["A", "D", "E"]}

Derivation: public incident edge endpoint set

Equivalence: unordered exact set; duplicate IDs invalid

Review: machine checks passed; no human certification.

20. **ld2-10220-graph-removal-1.** Delete B1336 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
graph-removal; connector-graph; graph-internal; integer.

Operands: B0824 = brick_000824, B1339 = brick_001339, B0669 = brick_000669, B1338 = brick_001338, B1337 = brick_001337, B1336 = brick_001336

Public evidence summary: 3 unique undirected pairs; 6 nodes

Reference output: {"value": 5}

Derivation: union-find component count after public target deletion

Equivalence: exact declared response

Review: machine checks passed; no human certification.

21. **ld2-10220-neighbors-2.** Select every candidate directly adjacent to B0616 in the supplied connector graph.
 neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0601 = brick_000601, B0120 = brick_000120, B0871 = brick_000871, B0616 = brick_000616, B0501 = brick_000501
Public evidence summary: 2 unique undirected pairs
Options: A: B0120; B: B0871; C: B0501; D: B0601
Reference output: {"choiceIds": ["B", "D"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: pending required human review.
22. **ld2-10220-graph-removal-2.** Delete B0616 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
 graph-removal; connector-graph; graph-internal; integer.
Operands: B0501 = brick_000501, B0871 = brick_000871, B0601 = brick_000601, B0120 = brick_000120, B0616 = brick_000616
Public evidence summary: 2 unique undirected pairs; 5 nodes
Reference output: {"value": 4}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: pending required human review.
23. **ld2-10220-neighbors-3.** Select every candidate directly adjacent to B0563 in the supplied connector graph.
 neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0563 = brick_000563, B0312 = brick_000312, B0643 = brick_000643, B0116 = brick_000116, B0566 = brick_000566
Public evidence summary: 2 unique undirected pairs
Options: A: B0312; B: B0116; C: B0643; D: B0566
Reference output: {"choiceIds": ["A", "D"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: pending required human review.
24. **ld2-10220-graph-removal-3.** Delete B0563 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
 graph-removal; connector-graph; graph-internal; integer.
Operands: B0563 = brick_000563, B0312 = brick_000312, B0116 = brick_000116, B0643 = brick_000643, B0566 = brick_000566
Public evidence summary: 2 unique undirected pairs; 5 nodes
Reference output: {"value": 4}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: pending required human review.
25. **ld2-10220-neighbors-4.** Select every candidate directly adjacent to B0211 in the supplied connector graph.
 neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0208 = brick_000208, B0211 = brick_000211, B0365 = brick_000365, B0213 = brick_000213, B1378 = brick_001378
Public evidence summary: 2 unique undirected pairs
Options: A: B1378; B: B0365; C: B0208; D: B0213
Reference output: {"choiceIds": ["C", "D"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: machine checks passed; no human certification.

26. **Id2-10220-graph-removal-4.** Delete B0211 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
graph-removal; connector-graph; graph-internal; integer.
Operands: B0211 = brick_000211, B0208 = brick_000208, B0213 = brick_000213, B0365 = brick_000365, B1378 = brick_001378
Public evidence summary: 2 unique undirected pairs; 5 nodes
Reference output: {"value": 4}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: machine checks passed; no human certification.
27. **Id2-10220-neighbors-5.** Select every candidate directly adjacent to B0347 in the supplied connector graph.
neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0338 = brick_000338, B0368 = brick_000368, B0347 = brick_000347, B0356 = brick_000356, B0344 = brick_000344, B0357 = brick_000357
Public evidence summary: 3 unique undirected pairs
Options: A: B0357; B: B0356; C: B0338; D: B0368; E: B0344
Reference output: {"choiceIds": ["B", "C", "D"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: machine checks passed; no human certification.
28. **Id2-10220-graph-removal-5.** Delete B0347 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
graph-removal; connector-graph; graph-internal; integer.
Operands: B0338 = brick_000338, B0357 = brick_000357, B0347 = brick_000347, B0344 = brick_000344, B0368 = brick_000368, B0356 = brick_000356
Public evidence summary: 3 unique undirected pairs; 6 nodes
Reference output: {"value": 5}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: machine checks passed; no human certification.
29. **Id2-10220-coverage-1.** Which numbered instance lacks a connector definition in the coverage record, so this audit cannot establish that it is floating?
coverage; source-data; metacognitive; single-choice.
Operands: B0285 = brick_000285, B0045 = brick_000045, B1293 = brick_001293, B0614 = brick_000614
Public evidence summary: {"coverage": [{"label": "B0285", "supported": true}, {"label": "B0614", "supported": true}, {"label": "B1293", "supported": true}, {"label": "B0045", "supported": false}]}
Options: A: B0045; B: B1293; C: B0285; D: B0614
Reference output: {"choiceId": "A"}
Derivation: public supported=false record
Equivalence: exact declared response
Review: pending required human review.
30. **Id2-10220-evidence-limit-1.** Only source geometry, connector matching and mesh intersection audits are available. Without mass, friction, material or dynamic tests, is gravitational stability established?
evidence-limit; source-data; metacognitive; single-choice.
Operands: none
Public evidence summary: {"available": ["Original CAD", "Connector inference", "Inset-mesh intersection candidates"]}
Options: A: Not established by this evidence; B: Established

Reference output: {"choiceId": "A"}

Derivation: fixed epistemic control label

Equivalence: exact declared response

Review: machine checks passed; no human certification.

31. **ld2-10220-source-step-1.** At which expanded author step does B0301 first appear in the supplied source-step table?
Source order is not a collision-free assembly proof.

source-step; source-data; procedural; single-choice.

Operands: B0001 = brick_000001, B0301 = brick_000301

Public evidence summary: 2 supplied author-step records; exact table in public JSON

Options: A: 2; B: 13; C: 300; D: 71

Reference output: {"choiceId": "D"}

Derivation: minimum public step index containing target

Equivalence: exact declared response

Review: machine checks passed; no human certification.

32. **ld2-10220-source-step-2.** At which expanded author step does B1307 first appear in the supplied source-step table?
Source order is not a collision-free assembly proof.

source-step; source-data; procedural; single-choice.

Operands: B0001 = brick_000001, B1307 = brick_001307

Public evidence summary: 2 supplied author-step records; exact table in public JSON

Options: A: 122; B: 382; C: 234; D: 149

Reference output: {"choiceId": "B"}

Derivation: minimum public step index containing target

Equivalence: exact declared response

Review: machine checks passed; no human certification.

33. **ld2-10220-source-step-3.** At which expanded author step does B1400 first appear in the supplied source-step table?
Source order is not a collision-free assembly proof.

source-step; source-data; procedural; single-choice.

Operands: B0001 = brick_000001, B1400 = brick_001400

Public evidence summary: 2 supplied author-step records; exact table in public JSON

Options: A: 418; B: 414; C: 423; D: 4

Reference output: {"choiceId": "B"}

Derivation: minimum public step index containing target

Equivalence: exact declared response

Review: machine checks passed; no human certification.

34. **ld2-10220-source-step-4.** At which expanded author step does B0973 first appear in the supplied source-step table?
Source order is not a collision-free assembly proof.

source-step; source-data; procedural; single-choice.

Operands: B0001 = brick_000001, B0973 = brick_000973

Public evidence summary: 2 supplied author-step records; exact table in public JSON

Options: A: 153; B: 391; C: 266; D: 286

Reference output: {"choiceId": "C"}

Derivation: minimum public step index containing target

Equivalence: exact declared response

Review: machine checks passed; no human certification.

35. **ld2-10220-source-step-5.** At which expanded author step does B1420 first appear in the supplied source-step table?
Source order is not a collision-free assembly proof.
source-step; source-data; procedural; single-choice.
Operands: B0001 = brick_000001, B1420 = brick_001420
Public evidence summary: 2 supplied author-step records; exact table in public JSON
Options: A: 43; B: 152; C: 430; D: 432
Reference output: {"choiceId": "C"}
Derivation: minimum public step index containing target
Equivalence: exact declared response
Review: machine checks passed; no human certification.
36. **ld2-10220-source-sequence-1.** Reveal the supplied source-step window in author order. Actions change visibility only, without simulating insertion trajectories.
source-sequence; scene-edit; procedural; actions.
Operands: B0016 = brick_000016, B0012 = brick_000012, B0022 = brick_000022, B0011 = brick_000011, B0001 = brick_000001, B0009 = brick_000009, B0007 = brick_000007, B0008 = brick_000008, B0002 = brick_000002, B0006 = brick_000006, B0017 = brick_000017, B0010 = brick_000010, B0023 = brick_000023, B0003 = brick_000003, B0015 = brick_000015, B0020 = brick_000020, B0005 = brick_000005, B0004 = brick_000004, B0018 = brick_000018, B0021 = brick_000021, B0013 = brick_000013, B0014 = brick_000014, B0019 = brick_000019
Public evidence summary: 6 public actions; budget 6; {"initialFacts": ["start"], "goalFacts": ["done-6"], "absentFacts": []}
Reference output: {"actionIds": ["step-1", "step-2", "step-3", "step-4", "step-5", "step-6"]}
Derivation: public precondition solver
Equivalence: any legal sequence reaching all goals within budget; display-only semantics
Review: machine checks passed; no human certification.
37. **ld2-10220-restore-instance-1.** The display copy hides B0614. Restore only that numbered instance, preserving all other instances and source poses.
restore-instance; scene-edit; procedural; actions.
Operands: B0614 = brick_000614, B0001 = brick_000001
Public evidence summary: 2 public actions; budget 1; {"initialFacts": ["missing-brick_000614", "present-brick_000001"], "goalFacts": ["present-brick_000614", "present-brick_000001"], "absentFacts": ["missing-brick_000614"]}
Reference output: {"actionIds": ["restore-B0614"]}
Derivation: public precondition solver
Equivalence: any legal sequence reaching all goals within budget; display-only semantics
Review: machine checks passed; no human certification.
38. **ld2-10220-restore-instance-2.** The display copy hides B1293. Restore only that numbered instance, preserving all other instances and source poses.
restore-instance; scene-edit; procedural; actions.
Operands: B0001 = brick_000001, B1293 = brick_001293
Public evidence summary: 2 public actions; budget 1; {"initialFacts": ["missing-brick_001293", "present-brick_000001"], "goalFacts": ["present-brick_001293", "present-brick_000001"], "absentFacts": ["missing-brick_001293"]}
Reference output: {"actionIds": ["restore-B1293"]}
Derivation: public precondition solver
Equivalence: any legal sequence reaching all goals within budget; display-only semantics
Review: machine checks passed; no human certification.
39. **ld2-10220-restore-instance-3.** The display copy hides B0285. Restore only that numbered instance, preserving all other instances and source poses.
restore-instance; scene-edit; procedural; actions.
Operands: B0285 = brick_000285, B0001 = brick_000001

Public evidence summary: 2 public actions; budget 1; {"initialFacts": ["missing-brick_000285", "present-brick_000001"], "goalFacts": ["present-brick_000285", "present-brick_000001"], "absentFacts": ["missing-brick_000285"]}

Reference output: {"actionIds": ["restore-B0285"]}

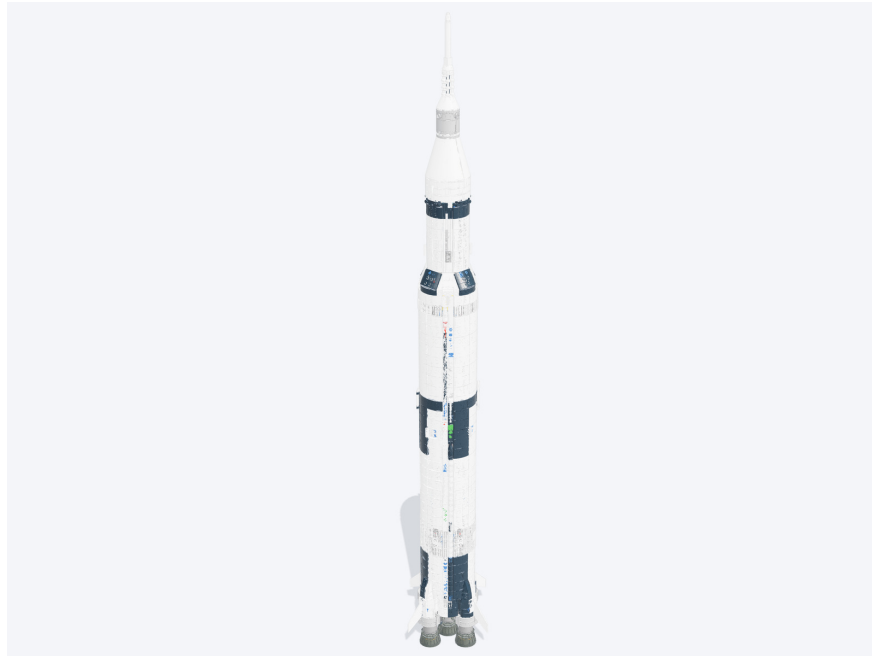
Derivation: public precondition solver

Equivalence: any legal sequence reaching all goals within budget; display-only semantics

Review: machine checks passed; no human certification.

33 21309: NASA Apollo Saturn V

1845 source instances; 39 tasks; D4 size band. 141 type strings; 14 color codes; 0 unsupported instances; 67 intersection candidates. 940 expanded author steps. Independent reparse: passed.



Attribution. Jaco van der Molen [Jaco]. CC BY 2.0.

<https://library.ldraw.org/library/omr/21309-1.mpd>

Source SHA256. eb56995265201179e0ecbad2865fb62fced524c44f3a4179badc964f844e5ba1

Connector records. stud: 5100; axle: 216; hinge: 16; fixed: 0; ball: 8

Public input lock. ae3142670b3b9f1b513a31473a59d676926e2f17657b32026a5d28cbe49e947b

Status. Source preservation does not certify stability. Reference outputs below are internal review material. Full model inputs are the versioned public JSON and numbered PNGs.

1. **ld2-omr-21309-color-1.** Identify the original color of B0496; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B0496 = brick_000496

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-21309-color-1.png; structured input = {}.

Options: A: Pearl Light Grey; B: White; C: Trans Orange; D: Yellow

Reference output: {"choiceId": "B"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

2. **ld2-omr-21309-shape-match-1.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0496?

shape-match; visual; atomic; single-choice.

Operands: B0496 = brick_000496, B1676 = brick_001676, B0498 = brick_000498, B0171 = brick_000171, B1478 = brick_001478

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-21309-shape-match-1.png; structured input = {}.

Options: A: B1676; B: B0171; C: B0498; D: B1478

Reference output: {"choiceId": "C"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

3. **ld2-omr-21309-distance-1.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0496?

distance; source-data; atomic; single-choice.

Operands: B1211 = brick_001211, B1130 = brick_001130, B0496 = brick_000496, B0093 = brick_000093

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0496": [-32.5858, 222.4, 26.9286], "B1211": [-33.01, 454.4, 27.354], "B1130": [-20, 532, 0], "B0093": [4, 122.4, -16.4]}}

Options: A: B1130; B: B0093; C: B1211

Reference output: {"choiceId": "B"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

4. **ld2-omr-21309-color-2.** Identify the original color of B0439; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B0439 = brick_000439

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-21309-color-2.png; structured input = {}.

Options: A: Blue; B: Red; C: White; D: Dark Bluish Grey

Reference output: {"choiceId": "B"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

5. **ld2-omr-21309-shape-match-2.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0439?

shape-match; visual; atomic; single-choice.

Operands: B0454 = brick_000454, B0536 = brick_000536, B0843 = brick_000843, B0439 = brick_000439, B1793 = brick_001793

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-21309-shape-match-2.png; structured input = {}.

Options: A: B1793; B: B0843; C: B0454; D: B0536

Reference output: {"choiceId": "C"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

6. **ld2-omr-21309-distance-2.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0439?

distance; source-data; atomic; single-choice.

Operands: B0642 = brick_000642, B0439 = brick_000439, B1371 = brick_001371, B0194 = brick_000194

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0439": [22.2624, 329.20596, 27.9184], "B0194": [40, 154.4, 12], "B0642": [-20.4486, 142.4, -38.3838], "B1371": [-40, 366.4, 12]}}

Options: A: B0642; B: B0194; C: B1371

Reference output: {"choiceId": "C"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

7. **ld2-omr-21309-color-3.** Identify the original color of B1134; numbered isolation is allowed.
 color; visual; atomic; single-choice.
Operands: B1134 = brick_001134
Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-21309-color-3.png; structured input = {}.
Options: A: Black; B: Yellow; C: Trans Orange; D: Green
Reference output: {"choiceId": "A"}
Derivation: source colorName by rendered B-label; source oracle only
Equivalence: exact declared response
Review: pending required human review.
8. **ld2-omr-21309-shape-match-3.** Ignoring color and pose, which numbered instance has the same LDraw part type as B1134?
 shape-match; visual; atomic; single-choice.
Operands: B1633 = brick_001633, B1136 = brick_001136, B0274 = brick_000274, B0809 = brick_000809, B1134 = brick_001134
Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-21309-shape-match-3.png; structured input = {}.
Options: A: B1633; B: B1136; C: B0809; D: B0274
Reference output: {"choiceId": "B"}
Derivation: source partNumber equality across labeled candidates; source oracle only
Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending
Review: pending required human review.
9. **ld2-omr-21309-distance-3.** Using the supplied geometry bounding-box centers, which candidate is nearest to B1134?
 distance; source-data; atomic; single-choice.
Operands: B1800 = brick_001800, B1134 = brick_001134, B1704 = brick_001704, B1365 = brick_001365
Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B1134": [-20, 538.4, 16.8], "B1365": [-34.4, 534.4, 0], "B1800": [0, 770.39994, 0], "B1704": [0, 570.4, 19.200032]}}
Options: A: B1365; B: B1704; C: B1800
Reference output: {"choiceId": "A"}
Derivation: Euclidean norm from public rounded centers
Equivalence: exact declared response
Review: machine checks passed; no human certification.
10. **ld2-omr-21309-color-4.** Identify the original color of B0754; numbered isolation is allowed.
 color; visual; atomic; single-choice.
Operands: B0754 = brick_000754
Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-21309-color-4.png; structured input = {}.
Options: A: Pearl Gold; B: Black; C: Light Bluish Grey; D: Green
Reference output: {"choiceId": "C"}
Derivation: source colorName by rendered B-label; source oracle only
Equivalence: exact declared response
Review: pending required human review.
11. **ld2-omr-21309-shape-match-4.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0754?
 shape-match; visual; atomic; single-choice.
Operands: B0212 = brick_000212, B0004 = brick_000004, B0036 = brick_000036, B0684 = brick_000684, B0754 = brick_000754
Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-21309-shape-match-4.png; structured input = {}.
Options: A: B0036; B: B0212; C: B0004; D: B0684

Reference output: {"choiceId": "D"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

12. **ld2-omr-21309-distance-4.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0754?

distance; source-data; atomic; single-choice.

Operands: B1554 = brick_001554, B0068 = brick_000068, B1390 = brick_001390, B0754 = brick_000754

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0754": [17.3568, 142.4, -33.8352], "B1390": [-40, 486.4, -12], "B0068": [27.3536, 54.4, 27.3536], "B1554": [8, 566.264406, -34.264011]}}

Options: A: B0068; B: B1390; C: B1554

Reference output: {"choiceId": "A"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: pending required human review.

13. **ld2-omr-21309-color-5.** Identify the original color of B1147; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B1147 = brick_001147

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-21309-color-5.png; structured input = {}.

Options: A: Black; B: Trans Orange; C: Red; D: Blue

Reference output: {"choiceId": "D"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

14. **ld2-omr-21309-shape-match-5.** Ignoring color and pose, which numbered instance has the same LDraw part type as B1147?

shape-match; visual; atomic; single-choice.

Operands: B1175 = brick_001175, B1740 = brick_001740, B1154 = brick_001154, B1410 = brick_001410, B1147 = brick_001147

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-21309-shape-match-5.png; structured input = {}.

Options: A: B1740; B: B1175; C: B1410; D: B1154

Reference output: {"choiceId": "A"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

15. **ld2-omr-21309-distance-5.** Using the supplied geometry bounding-box centers, which candidate is nearest to B1147?

distance; source-data; atomic; single-choice.

Operands: B1147 = brick_001147, B0920 = brick_000920, B1765 = brick_001765, B1750 = brick_001750

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B1147": [0, 340, 12], "B1750": [17.7936, 602.4, -18.7872], "B0920": [-26.2216, 6.4, 26.2216], "B1765": [8, 646.4, 28.800012]}}

Options: A: B1750; B: B0920; C: B1765

Reference output: {"choiceId": "A"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

16. **ld2-omr-21309-interface-1.** Classify the supplied connector record for B0478 and B0477. This does not certify load capacity.
- interface; connector-graph; metacognitive; single-choice.
- Operands:* B0478 = brick_000478, B0477 = brick_000477
- Public evidence summary:* {"connectorRecord": {"family": "axle", "aConnector": 1, "bConnector": 0}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
- Options:* A: axle; B: fixed; C: ball; D: stud; E: hinge
- Reference output:* {"choiceId": "A"}
- Derivation:* public connectorRecord.family equality
- Equivalence:* exact declared response
- Review:* machine checks passed; no human certification.
17. **ld2-omr-21309-interface-2.** Classify the supplied connector record for B1615 and B1737. This does not certify load capacity.
- interface; connector-graph; metacognitive; single-choice.
- Operands:* B1737 = brick_001737, B1615 = brick_001615
- Public evidence summary:* {"connectorRecord": {"family": "ball", "aConnector": 3, "bConnector": 3}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
- Options:* A: axle; B: stud; C: fixed; D: ball; E: hinge
- Reference output:* {"choiceId": "D"}
- Derivation:* public connectorRecord.family equality
- Equivalence:* exact declared response
- Review:* machine checks passed; no human certification.
18. **ld2-omr-21309-interface-3.** Classify the supplied connector record for B0204 and B0868. This does not certify load capacity.
- interface; connector-graph; metacognitive; single-choice.
- Operands:* B0204 = brick_000204, B0868 = brick_000868
- Public evidence summary:* {"connectorRecord": {"family": "hinge", "aConnector": 1, "bConnector": 1}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
- Options:* A: fixed; B: axle; C: ball; D: stud; E: hinge
- Reference output:* {"choiceId": "E"}
- Derivation:* public connectorRecord.family equality
- Equivalence:* exact declared response
- Review:* pending required human review.
19. **ld2-omr-21309-interface-4.** Classify the supplied connector record for B0965 and B0967. This does not certify load capacity.
- interface; connector-graph; metacognitive; single-choice.
- Operands:* B0965 = brick_000965, B0967 = brick_000967
- Public evidence summary:* {"connectorRecord": {"family": "stud", "aConnector": 13, "bConnector": 3}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
- Options:* A: axle; B: hinge; C: stud; D: ball; E: fixed
- Reference output:* {"choiceId": "C"}
- Derivation:* public connectorRecord.family equality
- Equivalence:* exact declared response
- Review:* machine checks passed; no human certification.

20. **ld2-omr-21309-neighbors-1.** Select every candidate directly adjacent to B1441 in the supplied connector graph.
 neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B1435 = brick_001435, B1441 = brick_001441, B1549 = brick_001549, B0432 = brick_000432, B1437 = brick_001437
Public evidence summary: 3 unique undirected pairs
Options: A: B0432; B: B1549; C: B1437; D: B1435
Reference output: {"choiceIds": ["C", "D"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: pending required human review.
21. **ld2-omr-21309-graph-removal-1.** Delete B1441 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
 graph-removal; connector-graph; graph-internal; integer.
Operands: B1437 = brick_001437, B1435 = brick_001435, B1441 = brick_001441, B1549 = brick_001549, B0432 = brick_000432
Public evidence summary: 3 unique undirected pairs; 5 nodes
Reference output: {"value": 3}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: pending required human review.
22. **ld2-omr-21309-neighbors-2.** Select every candidate directly adjacent to B1250 in the supplied connector graph.
 neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0989 = brick_000989, B1251 = brick_001251, B1250 = brick_001250, B1231 = brick_001231, B1019 = brick_001019, B1252 = brick_001252, B1253 = brick_001253, B1831 = brick_001831
Public evidence summary: 5 unique undirected pairs
Options: A: B0989; B: B1231; C: B1019; D: B1252; E: B1251; F: B1831; G: B1253
Reference output: {"choiceIds": ["A", "C", "D", "E", "G"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: machine checks passed; no human certification.
23. **ld2-omr-21309-graph-removal-2.** Delete B1250 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
 graph-removal; connector-graph; graph-internal; integer.
Operands: B1231 = brick_001231, B1831 = brick_001831, B1251 = brick_001251, B1019 = brick_001019, B1252 = brick_001252, B1253 = brick_001253, B0989 = brick_000989, B1250 = brick_001250
Public evidence summary: 5 unique undirected pairs; 8 nodes
Reference output: {"value": 7}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: machine checks passed; no human certification.
24. **ld2-omr-21309-neighbors-3.** Select every candidate directly adjacent to B1039 in the supplied connector graph.
 neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0670 = brick_000670, B1422 = brick_001422, B1040 = brick_001040, B1039 = brick_001039, B1038 = brick_001038
Public evidence summary: 2 unique undirected pairs
Options: A: B0670; B: B1422; C: B1040; D: B1038
Reference output: {"choiceIds": ["C", "D"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: machine checks passed; no human certification.

25. **ld2-omr-21309-graph-removal-3.** Delete B1039 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
graph-removal; connector-graph; graph-internal; integer.
Operands: B1039 = brick_001039, B1038 = brick_001038, B1422 = brick_001422, B1040 = brick_001040, B0670 = brick_000670
Public evidence summary: 2 unique undirected pairs; 5 nodes
Reference output: {"value": 4}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: machine checks passed; no human certification.
26. **ld2-omr-21309-neighbors-4.** Select every candidate directly adjacent to B0435 in the supplied connector graph.
neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0424 = brick_000424, B0657 = brick_000657, B0435 = brick_000435, B0484 = brick_000484, B0499 = brick_000499, B0485 = brick_000485, B0193 = brick_000193
Public evidence summary: 4 unique undirected pairs
Options: A: B0424; B: B0657; C: B0499; D: B0485; E: B0484; F: B0193
Reference output: {"choiceIds": ["A", "C", "D", "E"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: machine checks passed; no human certification.
27. **ld2-omr-21309-graph-removal-4.** Delete B0435 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
graph-removal; connector-graph; graph-internal; integer.
Operands: B0193 = brick_000193, B0435 = brick_000435, B0485 = brick_000485, B0484 = brick_000484, B0657 = brick_000657, B0499 = brick_000499, B0424 = brick_000424
Public evidence summary: 4 unique undirected pairs; 7 nodes
Reference output: {"value": 6}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: machine checks passed; no human certification.
28. **ld2-omr-21309-neighbors-5.** Select every candidate directly adjacent to B1053 in the supplied connector graph.
neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0416 = brick_000416, B1054 = brick_001054, B0538 = brick_000538, B1053 = brick_001053, B1055 = brick_001055
Public evidence summary: 2 unique undirected pairs
Options: A: B0416; B: B1055; C: B0538; D: B1054
Reference output: {"choiceIds": ["B", "D"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: machine checks passed; no human certification.
29. **ld2-omr-21309-graph-removal-5.** Delete B1053 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
graph-removal; connector-graph; graph-internal; integer.
Operands: B1054 = brick_001054, B1055 = brick_001055, B0538 = brick_000538, B1053 = brick_001053, B0416 = brick_000416
Public evidence summary: 2 unique undirected pairs; 5 nodes
Reference output: {"value": 4}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: machine checks passed; no human certification.

30. **Id2-omr-21309-evidence-limit-1.** Only source geometry, connector matching and mesh intersection audits are available. Without mass, friction, material or dynamic tests, is gravitational stability established?
evidence-limit; source-data; metacognitive; single-choice.
Operands: none
Public evidence summary: {"available": ["Original CAD", "Connector inference", "Inset-mesh intersection candidates"]}
Options: A: Not established by this evidence; B: Established
Reference output: {"choiceId": "A"}
Derivation: fixed epistemic control label
Equivalence: exact declared response
Review: machine checks passed; no human certification.
31. **Id2-omr-21309-source-step-1.** At which expanded author step does B1797 first appear in the supplied source-step table? Source order is not a collision-free assembly proof.
source-step; source-data; procedural; single-choice.
Operands: B0001 = brick_000001, B1797 = brick_001797
Public evidence summary: 2 supplied author-step records; exact table in public JSON
Options: A: 926; B: 933; C: 915; D: 676
Reference output: {"choiceId": "C"}
Derivation: minimum public step index containing target
Equivalence: exact declared response
Review: machine checks passed; no human certification.
32. **Id2-omr-21309-source-step-2.** At which expanded author step does B1075 first appear in the supplied source-step table? Source order is not a collision-free assembly proof.
source-step; source-data; procedural; single-choice.
Operands: B1075 = brick_001075, B0001 = brick_000001
Public evidence summary: 2 supplied author-step records; exact table in public JSON
Options: A: 526; B: 129; C: 647; D: 913
Reference output: {"choiceId": "A"}
Derivation: minimum public step index containing target
Equivalence: exact declared response
Review: machine checks passed; no human certification.
33. **Id2-omr-21309-source-step-3.** At which expanded author step does B0649 first appear in the supplied source-step table? Source order is not a collision-free assembly proof.
source-step; source-data; procedural; single-choice.
Operands: B0001 = brick_000001, B0649 = brick_000649
Public evidence summary: 2 supplied author-step records; exact table in public JSON
Options: A: 718; B: 78; C: 497; D: 285
Reference output: {"choiceId": "D"}
Derivation: minimum public step index containing target
Equivalence: exact declared response
Review: machine checks passed; no human certification.
34. **Id2-omr-21309-source-step-4.** At which expanded author step does B1156 first appear in the supplied source-step table? Source order is not a collision-free assembly proof.
source-step; source-data; procedural; single-choice.
Operands: B0001 = brick_000001, B1156 = brick_001156
Public evidence summary: 2 supplied author-step records; exact table in public JSON
Options: A: 781; B: 924; C: 912; D: 576

Reference output: {"choiceId": "D"}

Derivation: minimum public step index containing target

Equivalence: exact declared response

Review: machine checks passed; no human certification.

35. **ld2-omr-21309-source-step-5.** At which expanded author step does B0131 first appear in the supplied source-step table? Source order is not a collision-free assembly proof.

source-step; source-data; procedural; single-choice.

Operands: B0131 = brick_000131, B0001 = brick_000001

Public evidence summary: 2 supplied author-step records; exact table in public JSON

Options: A: 30; B: 74; C: 766; D: 20

Reference output: {"choiceId": "B"}

Derivation: minimum public step index containing target

Equivalence: exact declared response

Review: machine checks passed; no human certification.

36. **ld2-omr-21309-source-sequence-1.** Reveal the supplied source-step window in author order. Actions change visibility only, without simulating insertion trajectories.

source-sequence; scene-edit; procedural; actions.

Operands: B0007 = brick_000007, B0001 = brick_000001, B0002 = brick_000002, B0005 = brick_000005, B0004 = brick_000004, B0003 = brick_000003, B0006 = brick_000006

Public evidence summary: 6 public actions; budget 6; {"initialFacts": ["start"], "goalFacts": ["done-6"], "absentFacts": []}

Reference output: {"actionIds": ["step-1", "step-2", "step-3", "step-4", "step-5", "step-6"]}

Derivation: public precondition solver

Equivalence: any legal sequence reaching all goals within budget; display-only semantics

Review: machine checks passed; no human certification.

37. **ld2-omr-21309-restore-instance-1.** The display copy hides B0496. Restore only that numbered instance, preserving all other instances and source poses.

restore-instance; scene-edit; procedural; actions.

Operands: B0001 = brick_000001, B0496 = brick_000496

Public evidence summary: 2 public actions; budget 1; {"initialFacts": ["missing-brick_000496", "present-brick_000001"], "goalFacts": ["present-brick_000496", "present-brick_000001"], "absentFacts": ["missing-brick_000496"]}

Reference output: {"actionIds": ["restore-B0496"]}

Derivation: public precondition solver

Equivalence: any legal sequence reaching all goals within budget; display-only semantics

Review: machine checks passed; no human certification.

38. **ld2-omr-21309-restore-instance-2.** The display copy hides B0439. Restore only that numbered instance, preserving all other instances and source poses.

restore-instance; scene-edit; procedural; actions.

Operands: B0001 = brick_000001, B0439 = brick_000439

Public evidence summary: 2 public actions; budget 1; {"initialFacts": ["missing-brick_000439", "present-brick_000001"], "goalFacts": ["present-brick_000439", "present-brick_000001"], "absentFacts": ["missing-brick_000439"]}

Reference output: {"actionIds": ["restore-B0439"]}

Derivation: public precondition solver

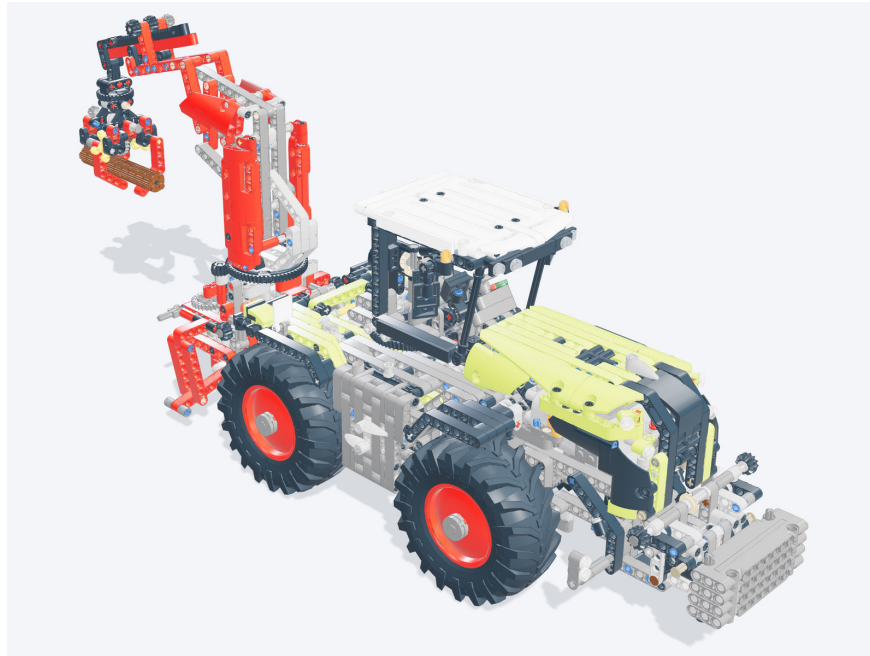
Equivalence: any legal sequence reaching all goals within budget; display-only semantics

Review: machine checks passed; no human certification.

39. **ld2-omr-21309-restore-instance-3.** The display copy hides B1134. Restore only that numbered instance, preserving all other instances and source poses.
- restore-instance; scene-edit; procedural; actions.
- Operands:* B1134 = brick_001134, B0001 = brick_000001
- Public evidence summary:* 2 public actions; budget 1; {"initialFacts": ["missing-brick_001134", "present-brick_000001"], "goalFacts": ["present-brick_001134", "present-brick_000001"], "absentFacts": ["missing-brick_001134"]}
- Reference output:* {"actionIds": ["restore-B1134"]}
- Derivation:* public precondition solver
- Equivalence:* any legal sequence reaching all goals within budget; display-only semantics
- Review:* machine checks passed; no human certification.

34 42054: CLAAS XERION 5000 TRAC VC

1978 source instances; 39 tasks; D4 size band. 160 type strings; 18 color codes; 28 unsupported instances; 54 intersection candidates. 229 expanded author steps. Independent reparse: passed.



Attribution. Philippe Hurbain [Philo]. CC BY 2.0.

<https://library.ldraw.org/library/omr/42054-1.mpd>

Source SHA256. aa2856b77726aaaa74145ba1e40dbd89d1897ff3577ac2eab83237bd6d65b04c

Connector records. stud: 168; axle: 2620; hinge: 2; fixed: 0; ball: 0

Public input lock. 0dec5b93310c7aaf6330dcfaf6e0a0474c1c4f7ad138c1d87013f90de85fa35f

Status. Source preservation does not certify stability. Reference outputs below are internal review material. Full model inputs are the versioned public JSON and numbered PNGs.

1. **ld2-omr-42054-color-1.** Identify the original color of B1103; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B1103 = brick_001103

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-42054-color-1.png; structured input = {}.

Options: A: Light Bluish Grey; B: Orange; C: Black; D: Rubber Black

Reference output: {"choiceId": "C"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

2. **ld2-omr-42054-shape-match-1.** Ignoring color and pose, which numbered instance has the same LDraw part type as B1103?

shape-match; visual; atomic; single-choice.

Operands: B1536 = brick_001536, B1103 = brick_001103, B1754 = brick_001754, B0911 = brick_000911, B0158 = brick_000158

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-42054-shape-match-1.png; structured input = {}.

Options: A: B1754; B: B1536; C: B0911; D: B0158

Reference output: {"choiceId": "A"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

3. **ld2-omr-42054-distance-1.** Using the supplied geometry bounding-box centers, which candidate is nearest to B1103?

distance; source-data; atomic; single-choice.

Operands: B0468 = brick_000468, B1103 = brick_001103, B1535 = brick_001535, B0455 = brick_000455

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B1103": [40, 144, 144], "B1535": [8, 32, -216], "B0468": [-32, 48, 106], "B0455": [-20, 68, 88]}}

Options: A: B1535; B: B0468; C: B0455

Reference output: {"choiceId": "C"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

4. **ld2-omr-42054-color-2.** Identify the original color of B1016; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B1016 = brick_001016

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-42054-color-2.png; structured input = {}.

Options: A: Red; B: Black; C: Orange; D: Pearl Silver

Reference output: {"choiceId": "B"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

5. **ld2-omr-42054-shape-match-2.** Ignoring color and pose, which numbered instance has the same LDraw part type as B1016?

shape-match; visual; atomic; single-choice.

Operands: B0191 = brick_000191, B0439 = brick_000439, B1357 = brick_001357, B1016 = brick_001016, B1766 = brick_001766

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-42054-shape-match-2.png; structured input = {}.

Options: A: B1357; B: B0191; C: B1766; D: B0439

Reference output: {"choiceId": "D"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

6. **ld2-omr-42054-distance-2.** Using the supplied geometry bounding-box centers, which candidate is nearest to B1016?

distance; source-data; atomic; single-choice.

Operands: B1016 = brick_001016, B1325 = brick_001325, B1356 = brick_001356, B0125 = brick_000125

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B1016": [8, 120, 72], "B1325": [0, 132, -40], "B1356": [24, 128, -32], "B0125": [0, 88, -104]}}

Options: A: B1356; B: B0125; C: B1325

Reference output: {"choiceId": "A"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: machine checks passed; no human certification.

7. **ld2-omr-42054-color-3.** Identify the original color of B0570; numbered isolation is allowed.
 color; visual; atomic; single-choice.
Operands: B0570 = brick_000570
Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-42054-color-3.png; structured input = {}.
Options: A: Trans Red; B: Rubber Black; C: Trans Orange; D: Dark Bluish Grey
Reference output: {"choiceId": "D"}
Derivation: source colorName by rendered B-label; source oracle only
Equivalence: exact declared response
Review: pending required human review.
8. **ld2-omr-42054-shape-match-3.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0570?
 shape-match; visual; atomic; single-choice.
Operands: B1867 = brick_001867, B0570 = brick_000570, B1885 = brick_001885, B0803 = brick_000803, B1394 = brick_001394
Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-42054-shape-match-3.png; structured input = {}.
Options: A: B1867; B: B1394; C: B0803; D: B1885
Reference output: {"choiceId": "B"}
Derivation: source partNumber equality across labeled candidates; source oracle only
Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending
Review: pending required human review.
9. **ld2-omr-42054-distance-3.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0570?
 distance; source-data; atomic; single-choice.
Operands: B1586 = brick_001586, B0016 = brick_000016, B1159 = brick_001159, B0570 = brick_000570
Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0570": [16.1, 88, 176], "B1586": [0, 62, -208], "B0016": [0, 96, -24], "B1159": [30.6176, 132.6864, 73.3612]}}
Options: A: B1586; B: B1159; C: B0016
Reference output: {"choiceId": "B"}
Derivation: Euclidean norm from public rounded centers
Equivalence: exact declared response
Review: pending required human review.
10. **ld2-omr-42054-color-4.** Identify the original color of B0724; numbered isolation is allowed.
 color; visual; atomic; single-choice.
Operands: B0724 = brick_000724
Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-42054-color-4.png; structured input = {}.
Options: A: Pearl Silver; B: White; C: Black; D: Blue
Reference output: {"choiceId": "C"}
Derivation: source colorName by rendered B-label; source oracle only
Equivalence: exact declared response
Review: pending required human review.
11. **ld2-omr-42054-shape-match-4.** Ignoring color and pose, which numbered instance has the same LDraw part type as B0724?
 shape-match; visual; atomic; single-choice.
Operands: B0485 = brick_000485, B0349 = brick_000349, B1765 = brick_001765, B0724 = brick_000724, B0529 = brick_000529
Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-42054-shape-match-4.png; structured input = {}.
Options: A: B1765; B: B0349; C: B0485; D: B0529

Reference output: {"choiceId": "D"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

12. **ld2-omr-42054-distance-4.** Using the supplied geometry bounding-box centers, which candidate is nearest to B0724?

distance; source-data; atomic; single-choice.

Operands: B0724 = brick_000724, B1307 = brick_001307, B0771 = brick_000771, B0366 = brick_000366

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B0724": [-24, 100, 72], "B0771": [-40, 112, -12], "B1307": [-24, 128, -32], "B0366": [0, 86, 8]}}

Options: A: B1307; B: B0771; C: B0366

Reference output: {"choiceId": "C"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: pending required human review.

13. **ld2-omr-42054-color-5.** Identify the original color of B1396; numbered isolation is allowed.

color; visual; atomic; single-choice.

Operands: B1396 = brick_001396

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-42054-color-5.png; structured input = {}.

Options: A: Tan; B: Orange; C: Dark Bluish Grey; D: Dark Tan

Reference output: {"choiceId": "C"}

Derivation: source colorName by rendered B-label; source oracle only

Equivalence: exact declared response

Review: pending required human review.

14. **ld2-omr-42054-shape-match-5.** Ignoring color and pose, which numbered instance has the same LDraw part type as B1396?

shape-match; visual; atomic; single-choice.

Operands: B0793 = brick_000793, B1082 = brick_001082, B0492 = brick_000492, B1763 = brick_001763, B1396 = brick_001396

Public evidence summary: Numbered PNG: benchmark/ldraw-v2/inputs/views/ld2-omr-42054-shape-match-5.png; structured input = {}.

Options: A: B0793; B: B0492; C: B1082; D: B1763

Reference output: {"choiceId": "C"}

Derivation: source partNumber equality across labeled candidates; source oracle only

Equivalence: same source partNumber; ignore color and pose; perceptual equivalence pending

Review: pending required human review.

15. **ld2-omr-42054-distance-5.** Using the supplied geometry bounding-box centers, which candidate is nearest to B1396?

distance; source-data; atomic; single-choice.

Operands: B0365 = brick_000365, B1396 = brick_001396, B0049 = brick_000049, B0827 = brick_000827

Public evidence summary: {"coordinateFrame": "Viewer world, millimeters, Euclidean distance", "centers": {"B1396": [-0.1, 154.0704, 7.7276], "B0365": [0, 74, 8], "B0827": [68, 100, -20], "B0049": [16, 40, -128]}}

Options: A: B0827; B: B0049; C: B0365

Reference output: {"choiceId": "C"}

Derivation: Euclidean norm from public rounded centers

Equivalence: exact declared response

Review: pending required human review.

16. **ld2-omr-42054-interface-1.** Classify the supplied connector record for B1852 and B1848. This does not certify load capacity.
- interface; connector-graph; metacognitive; single-choice.
- Operands:* B1848 = brick_001848, B1852 = brick_001852
- Public evidence summary:* {"connectorRecord": {"family": "axle", "aConnector": 0, "bConnector": 0}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
- Options:* A: stud; B: fixed; C: ball; D: axle; E: hinge
- Reference output:* {"choiceId": "D"}
- Derivation:* public connectorRecord.family equality
- Equivalence:* exact declared response
- Review:* machine checks passed; no human certification.
17. **ld2-omr-42054-interface-2.** Classify the supplied connector record for B1825 and B1824. This does not certify load capacity.
- interface; connector-graph; metacognitive; single-choice.
- Operands:* B1824 = brick_001824, B1825 = brick_001825
- Public evidence summary:* {"connectorRecord": {"family": "hinge", "aConnector": 1, "bConnector": 1}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
- Options:* A: ball; B: hinge; C: fixed; D: axle; E: stud
- Reference output:* {"choiceId": "B"}
- Derivation:* public connectorRecord.family equality
- Equivalence:* exact declared response
- Review:* machine checks passed; no human certification.
18. **ld2-omr-42054-interface-3.** Classify the supplied connector record for B0940 and B0942. This does not certify load capacity.
- interface; connector-graph; metacognitive; single-choice.
- Operands:* B0942 = brick_000942, B0940 = brick_000940
- Public evidence summary:* {"connectorRecord": {"family": "stud", "aConnector": 5, "bConnector": 0}, "scope": "Upstream connector tolerances and inset mesh intersections; unknown parts remain explicit, separate scene objects are not glued together; no load/force validation."}
- Options:* A: stud; B: hinge; C: ball; D: fixed; E: axle
- Reference output:* {"choiceId": "A"}
- Derivation:* public connectorRecord.family equality
- Equivalence:* exact declared response
- Review:* machine checks passed; no human certification.
19. **ld2-omr-42054-neighbors-1.** Select every candidate directly adjacent to B1563 in the supplied connector graph.
- neighbors; connector-graph; metacognitive; multiple-choice.
- Operands:* B1563 = brick_001563, B1037 = brick_001037, B0192 = brick_000192, B1559 = brick_001559, B1565 = brick_001565
- Public evidence summary:* 2 unique undirected pairs
- Options:* A: B1565; B: B0192; C: B1559; D: B1037
- Reference output:* {"choiceIds": ["A", "C"]}
- Derivation:* public incident edge endpoint set
- Equivalence:* unordered exact set; duplicate IDs invalid
- Review:* machine checks passed; no human certification.

20. **ld2-omr-42054-graph-removal-1.** Delete B1563 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
graph-removal; connector-graph; graph-internal; integer.
Operands: B1559 = brick_001559, B1565 = brick_001565, B1037 = brick_001037, B0192 = brick_000192, B1563 = brick_001563
Public evidence summary: 2 unique undirected pairs; 5 nodes
Reference output: {"value": 4}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: machine checks passed; no human certification.
21. **ld2-omr-42054-neighbors-2.** Select every candidate directly adjacent to B0710 in the supplied connector graph.
neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0711 = brick_000711, B1619 = brick_001619, B0644 = brick_000644, B0710 = brick_000710, B1207 = brick_001207
Public evidence summary: 2 unique undirected pairs
Options: A: B0711; B: B0644; C: B1207; D: B1619
Reference output: {"choiceIds": ["A", "B"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: machine checks passed; no human certification.
22. **ld2-omr-42054-graph-removal-2.** Delete B0710 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
graph-removal; connector-graph; graph-internal; integer.
Operands: B1619 = brick_001619, B0711 = brick_000711, B0710 = brick_000710, B1207 = brick_001207, B0644 = brick_000644
Public evidence summary: 2 unique undirected pairs; 5 nodes
Reference output: {"value": 4}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: machine checks passed; no human certification.
23. **ld2-omr-42054-neighbors-3.** Select every candidate directly adjacent to B1218 in the supplied connector graph.
neighbors; connector-graph; metacognitive; multiple-choice.
Operands: B0876 = brick_000876, B0027 = brick_000027, B1218 = brick_001218, B1217 = brick_001217, B1219 = brick_001219, B1216 = brick_001216
Public evidence summary: 3 unique undirected pairs
Options: A: B0876; B: B1216; C: B1217; D: B0027; E: B1219
Reference output: {"choiceIds": ["B", "C", "E"]}
Derivation: public incident edge endpoint set
Equivalence: unordered exact set; duplicate IDs invalid
Review: machine checks passed; no human certification.
24. **ld2-omr-42054-graph-removal-3.** Delete B1218 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
graph-removal; connector-graph; graph-internal; integer.
Operands: B0027 = brick_000027, B1219 = brick_001219, B0876 = brick_000876, B1216 = brick_001216, B1217 = brick_001217, B1218 = brick_001218
Public evidence summary: 3 unique undirected pairs; 6 nodes
Reference output: {"value": 5}
Derivation: union-find component count after public target deletion
Equivalence: exact declared response
Review: machine checks passed; no human certification.

25. **ld2-omr-42054-neighbors-4.** Select every candidate directly adjacent to B0478 in the supplied connector graph.
- neighbors; connector-graph; metacognitive; multiple-choice.
- Operands:* B0505 = brick_000505, B0459 = brick_000459, B0463 = brick_000463, B1616 = brick_001616, B0462 = brick_000462, B0468 = brick_000468, B0469 = brick_000469, B0125 = brick_000125, B0478 = brick_000478
- Public evidence summary:* 6 unique undirected pairs
- Options:* A: B0468; B: B0469; C: B0505; D: B0459; E: B0462; F: B0125; G: B0463; H: B1616
- Reference output:* {"choiceIds": ["A", "B", "C", "D", "E", "G"]}
- Derivation:* public incident edge endpoint set
- Equivalence:* unordered exact set; duplicate IDs invalid
- Review:* machine checks passed; no human certification.
26. **ld2-omr-42054-graph-removal-4.** Delete B0478 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
- graph-removal; connector-graph; graph-internal; integer.
- Operands:* B0478 = brick_000478, B0469 = brick_000469, B0459 = brick_000459, B1616 = brick_001616, B0462 = brick_000462, B0125 = brick_000125, B0505 = brick_000505, B0468 = brick_000468, B0463 = brick_000463
- Public evidence summary:* 6 unique undirected pairs; 9 nodes
- Reference output:* {"value": 8}
- Derivation:* union-find component count after public target deletion
- Equivalence:* exact declared response
- Review:* machine checks passed; no human certification.
27. **ld2-omr-42054-neighbors-5.** Select every candidate directly adjacent to B1973 in the supplied connector graph.
- neighbors; connector-graph; metacognitive; multiple-choice.
- Operands:* B1973 = brick_001973, B1967 = brick_001967, B0748 = brick_000748, B1547 = brick_001547, B1972 = brick_001972, B1974 = brick_001974
- Public evidence summary:* 5 unique undirected pairs
- Options:* A: B1967; B: B1547; C: B1974; D: B1972; E: B0748
- Reference output:* {"choiceIds": ["A", "C", "D"]}
- Derivation:* public incident edge endpoint set
- Equivalence:* unordered exact set; duplicate IDs invalid
- Review:* machine checks passed; no human certification.
28. **ld2-omr-42054-graph-removal-5.** Delete B1973 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.
- graph-removal; connector-graph; graph-internal; integer.
- Operands:* B1547 = brick_001547, B1973 = brick_001973, B1974 = brick_001974, B1967 = brick_001967, B1972 = brick_001972, B0748 = brick_000748
- Public evidence summary:* 5 unique undirected pairs; 6 nodes
- Reference output:* {"value": 3}
- Derivation:* union-find component count after public target deletion
- Equivalence:* exact declared response
- Review:* machine checks passed; no human certification.
29. **ld2-omr-42054-coverage-1.** Which numbered instance lacks a connector definition in the coverage record, so this audit cannot establish that it is floating?
- coverage; source-data; metacognitive; single-choice.
- Operands:* B1016 = brick_001016, B0570 = brick_000570, B1103 = brick_001103, B0146 = brick_000146
- Public evidence summary:* {"coverage": [{"label": "B0146", "supported": false}, {"label": "B0570", "supported": true}, {"label": "B1103", "supported": true}, {"label": "B1016", "supported": true}]}

Options: A: B0570; B: B0146; C: B1103; D: B1016

Reference output: {"choiceId": "B"}

Derivation: public supported=false record

Equivalence: exact declared response

Review: pending required human review.

30. **ld2-omr-42054-evidence-limit-1.** Only source geometry, connector matching and mesh intersection audits are available. Without mass, friction, material or dynamic tests, is gravitational stability established?

evidence-limit; source-data; metacognitive; single-choice.

Operands: none

Public evidence summary: {"available": ["Original CAD", "Connector inference", "Inset-mesh intersection candidates"]}

Options: A: Not established by this evidence; B: Established

Reference output: {"choiceId": "A"}

Derivation: fixed epistemic control label

Equivalence: exact declared response

Review: machine checks passed; no human certification.

31. **ld2-omr-42054-source-step-1.** At which expanded author step does B1230 first appear in the supplied source-step table? Source order is not a collision-free assembly proof.

source-step; source-data; procedural; single-choice.

Operands: B0001 = brick_000001, B1230 = brick_001230

Public evidence summary: 2 supplied author-step records; exact table in public JSON

Options: A: 48; B: 95; C: 135; D: 109

Reference output: {"choiceId": "C"}

Derivation: minimum public step index containing target

Equivalence: exact declared response

Review: machine checks passed; no human certification.

32. **ld2-omr-42054-source-step-2.** At which expanded author step does B1177 first appear in the supplied source-step table? Source order is not a collision-free assembly proof.

source-step; source-data; procedural; single-choice.

Operands: B1177 = brick_001177, B0001 = brick_000001

Public evidence summary: 2 supplied author-step records; exact table in public JSON

Options: A: 131; B: 221; C: 132; D: 206

Reference output: {"choiceId": "A"}

Derivation: minimum public step index containing target

Equivalence: exact declared response

Review: machine checks passed; no human certification.

33. **ld2-omr-42054-source-step-3.** At which expanded author step does B0711 first appear in the supplied source-step table? Source order is not a collision-free assembly proof.

source-step; source-data; procedural; single-choice.

Operands: B0711 = brick_000711, B0001 = brick_000001

Public evidence summary: 2 supplied author-step records; exact table in public JSON

Options: A: 65; B: 175; C: 85; D: 136

Reference output: {"choiceId": "C"}

Derivation: minimum public step index containing target

Equivalence: exact declared response

Review: machine checks passed; no human certification.

34. **ld2-omr-42054-source-step-4.** At which expanded author step does B0180 first appear in the supplied source-step table?
Source order is not a collision-free assembly proof.
source-step; source-data; procedural; single-choice.
Operands: B0001 = brick_000001, B0180 = brick_000180
Public evidence summary: 2 supplied author-step records; exact table in public JSON
Options: A: 136; B: 30; C: 19; D: 82
Reference output: {"choiceId": "C"}
Derivation: minimum public step index containing target
Equivalence: exact declared response
Review: machine checks passed; no human certification.
35. **ld2-omr-42054-source-step-5.** At which expanded author step does B0363 first appear in the supplied source-step table?
Source order is not a collision-free assembly proof.
source-step; source-data; procedural; single-choice.
Operands: B0001 = brick_000001, B0363 = brick_000363
Public evidence summary: 2 supplied author-step records; exact table in public JSON
Options: A: 128; B: 19; C: 43; D: 11
Reference output: {"choiceId": "C"}
Derivation: minimum public step index containing target
Equivalence: exact declared response
Review: machine checks passed; no human certification.
36. **ld2-omr-42054-source-sequence-1.** Reveal the supplied source-step window in author order. Actions change visibility only, without simulating insertion trajectories.
source-sequence; scene-edit; procedural; actions.
Operands: B0027 = brick_000027, B0021 = brick_000021, B0047 = brick_000047, B0044 = brick_000044, B0001 = brick_000001, B0035 = brick_000035, B0025 = brick_000025, B0042 = brick_000042, B0005 = brick_000005, B0020 = brick_000020, B0058 = brick_000058, B0050 = brick_000050, B0037 = brick_000037, B0003 = brick_000003, B0039 = brick_000039, B0030 = brick_000030, B0029 = brick_000029, B0034 = brick_000034, B0009 = brick_000009, B0011 = brick_000011, B0016 = brick_000016, B0012 = brick_000012, B0051 = brick_000051, B0022 = brick_000022, B0004 = brick_000004, B0048 = brick_000048, B0006 = brick_000006, B0046 = brick_000046, B0023 = brick_000023, B0032 = brick_000032, B0013 = brick_000013, B0019 = brick_000019, B0026 = brick_000026, B0014 = brick_000014, B0031 = brick_000031, B0024 = brick_000024, B0056 = brick_000056, B0017 = brick_000017, B0008 = brick_000008, B0007 = brick_000007, B0045 = brick_000045, B0053 = brick_000053, B0018 = brick_000018, B0041 = brick_000041, B0038 = brick_000038, B0036 = brick_000036, B0057 = brick_000057, B0010 = brick_000010, B0043 = brick_000043, B0052 = brick_000052, B0054 = brick_000054, B0033 = brick_000033, B0055 = brick_000055, B0028 = brick_000028, B0002 = brick_000002, B0049 = brick_000049, B0040 = brick_000040, B0015 = brick_000015
Public evidence summary: 6 public actions; budget 6; {"initialFacts": ["start"], "goalFacts": ["done-6"], "absentFacts": []}
Reference output: {"actionIds": ["step-1", "step-2", "step-3", "step-4", "step-5", "step-6"]}
Derivation: public precondition solver
Equivalence: any legal sequence reaching all goals within budget; display-only semantics
Review: pending required human review.
37. **ld2-omr-42054-restore-instance-1.** The display copy hides B1103. Restore only that numbered instance, preserving all other instances and source poses.
restore-instance; scene-edit; procedural; actions.
Operands: B1103 = brick_001103, B0001 = brick_000001
Public evidence summary: 2 public actions; budget 1; {"initialFacts": ["missing-brick_001103", "present-brick_000001"], "goalFacts": ["present-brick_001103", "present-brick_000001"], "absentFacts": ["missing-brick_001103"]}
Reference output: {"actionIds": ["restore-B1103"]}
Derivation: public precondition solver
Equivalence: any legal sequence reaching all goals within budget; display-only semantics
Review: machine checks passed; no human certification.

38. **ld2-omr-42054-restore-instance-2.** The display copy hides B1016. Restore only that numbered instance, preserving all other instances and source poses.
- restore-instance; scene-edit; procedural; actions.
- Operands:* B0001 = brick_000001, B1016 = brick_001016
- Public evidence summary:* 2 public actions; budget 1; {"initialFacts": ["missing-brick_001016", "present-brick_000001"], "goalFacts": ["present-brick_001016", "present-brick_000001"], "absentFacts": ["missing-brick_001016"]}
- Reference output:* {"actionIds": ["restore-B1016"]}
- Derivation:* public precondition solver
- Equivalence:* any legal sequence reaching all goals within budget; display-only semantics
- Review:* machine checks passed; no human certification.
39. **ld2-omr-42054-restore-instance-3.** The display copy hides B0570. Restore only that numbered instance, preserving all other instances and source poses.
- restore-instance; scene-edit; procedural; actions.
- Operands:* B0570 = brick_000570, B0001 = brick_000001
- Public evidence summary:* 2 public actions; budget 1; {"initialFacts": ["missing-brick_000570", "present-brick_000001"], "goalFacts": ["present-brick_000570", "present-brick_000001"], "absentFacts": ["missing-brick_000570"]}
- Reference output:* {"actionIds": ["restore-B0570"]}
- Derivation:* public precondition solver
- Equivalence:* any legal sequence reaching all goals within budget; display-only semantics
- Review:* machine checks passed; no human certification.